On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

Xiang Huang

July 2026

Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences (a_n), (b_n) with $d_n = \mathrm{lcm}(1,\dots,n)^3$. We prove *unconditionally* that $G_n = e^{o(n)}$ for a set of positive integers of natural density 1, with the quantitative bound $\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon\big((\log N)^2\big)$ for every $\varepsilon > 0$; in particular, the exceptional set has upper Banach density zero and finite harmonic weight. The key ingredient is a new bound $Z(p) = O(p^{2/3})$ on the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$, derived from the Apéry recurrence via a gap polynomial argument. A companion centered-coefficient analysis of the bordered value-return certificates proves the *uniform* fiber bound $\max_a \#\{j < p : b_j \equiv a\} = O(p^{3/4})$ and the collision-energy bound $\sum_a \#\{j : b_j \equiv a\}^2 = O(p^{7/4})$, both unconditional, with the underlying certificate nonvanishing proved in the sharp range $1 \le h < k < p$. Computation for all 78,496 primes $5 \le p \le 10^6$ gives $Z(p) \le 12$ with mean ≈ 1. The pair count $Z(p)/2$ fits Poisson(1/2). For each canonical gap polynomial N_h whose splitting field has full hyperoctahedral Galois group, we prove via Chebotarev that its root-pair count follows the signed fixed-point distribution, tending to Poisson(1/2) as $h \to \infty$; the corresponding law for the actual coefficient zeros remains conjectural. The Poisson tail predicts $\max_{p \le N} Z(p) \sim 2 \log\log N$, so $Z(p)$ is almost certainly unbounded. Under the weaker Cesàro hypothesis $\sum_{p \le x} Z(p) \ll \pi(x)$ (Hypothesis $\bar{Z}$), which the Poisson model supports, we prove $\log G_n = O(\sqrt{n})$ for density 1, matching the unconditional small-prime floor, with the conditional exceptional set shrinking to $O(N^{1/3})$ and having upper Banach density zero. The proof combines a dyadic divisor-incidence bound with a sub-interval zero-count argument to reduce the leading-digit bottleneck from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ unconditionally, and to $O(N^{4/3}/\log N)$ under Hypothesis $\bar{Z}$. A prime-wise decomposition of the residual L^2-norm $\|\eta\|^2 = D + O$ (diagonal + off-diagonal) shows that $D = T + O(1/\ln N)$ unconditionally, reducing the all-n conjecture $G_n = e^{o(n)}$ to a two-prime dispersion input: the cross-prime interaction O is $o(T)$. Computation through $N = 262{,}144$ confirms $|O/T| < 2 \times 10^{-3}$. Two structural reductions further localize the gap: prime counting alone disposes of every prime $p \le n/(f(n) \log n)$ for any $f \to \infty$, so only the quotient classes $\lfloor n/p \rfloor < f(n) \log n$ matter; and a single fixed factorial moment $(\mathrm{HM})_k$ with $k > 6$—at the independence scale, compatible with $Z(p) \ll p^{2/3}$, and proved here unconditionally for $k = 2$—would give the full conjecture with the power saving $\log G_n \ll n^{2/3+2/k+o(1)}$.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5)\, u_n - n^3\, u_{n-1}, \quad n \ge 1, \tag{1}$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n-th Apéry number. Set $d_n = \mathrm{lcm}(1,\dots,n)^3$; Apéry [3] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$. Write $P(n) = 34n^3 + 51n^2 + 27n + 5$ for the middle coefficient.

2 The Wronskian identity

Lemma 1. *For all $n \geq 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. □

Corollary 2. *For every prime $p \geq 5$,*

$$v_p(G_n) \leq 3\lfloor \log_p n \rfloor. \tag{2}$$

Proof. G_n divides $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$. For $p \geq 5$, $v_p(6) = 0$ and $v_p(d_n/n^3) = 3\lfloor \log_p n \rfloor - 3v_p(n) \leq 3\lfloor \log_p n \rfloor$. □

3 Small primes: unconditional bound

Proposition 3. $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n})$.

Proof. By (2), $v_p(G_n) \log p \leq 3 \log n$ for every prime $p \geq 5$. Primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. □

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [18])**.** *For every prime p and $n = \sum_{i=0}^{s} n_i p^i$ in base p,*

$$b_n \equiv \prod_{i=0}^{s} b_{n_i} \pmod{p}. \tag{3}$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [22] and McIntosh [21].

5 The zero-count function

Definition 5. For a prime $p \geq 5$, define $Z(p) = \#\{j \in \{0, \ldots, p-1\} : b_j \equiv 0 \pmod{p}\}$.

5.1 Unconditional bound

Lemma 6 (No consecutive zeros)**.** *For every prime $p \geq 5$ and every $j \in \{0, \ldots, p-2\}$, b_j and b_{j+1} cannot both vanish modulo p.*

Proof. Suppose $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$ with $0 \leq m \leq p-2$. The recurrence (1) at $n = m$ gives $(m+1)^3 b_{m+1} = P(m)\, b_m - m^3\, b_{m-1}$, so $m^3 b_{m-1} \equiv 0 \pmod{p}$. Since $1 \leq m \leq p-2$ implies $p \nmid m$, we get $b_{m-1} \equiv 0$. Iterating gives $b_0 \equiv 0 \pmod{p}$, contradicting $b_0 = 1$. □

Lemma 7 (Gap polynomial)**.** *Fix $h \geq 2$. Suppose $b_m \equiv 0 \pmod p$ and $m + h \leq p - 1$. Then*

$$b_{m+h} \equiv C_h(m)\, b_{m+1} \pmod p, \tag{4}$$

where $C_h(m)$ is a rational function of m whose numerator (after clearing the denominator $\prod_{j=1}^{h-1}(m+j+1)^3$) is a polynomial of degree exactly $3(h-1)$ with positive leading coefficient.

In particular, if $b_m \equiv b_{m+h} \equiv 0 \pmod p$, then by Lemma 6 $b_{m+1} \not\equiv 0$, so $C_h(m) \equiv 0 \pmod p$, and m is constrained to at most $3(h-1)$ residue classes modulo p, provided C_h is not identically zero over $\mathbb{F}_p$.

Proof. We iterate the recurrence from the initial condition $(b_m, b_{m+1}) = (0, b_{m+1})$. At step $n = m+1$: $b_{m+2} = P(m+1)\, b_{m+1}/(m+2)^3$, since $b_m = 0$ kills the second term. At step $n = m+j$ for $j \geq 2$: b_{m+j+1} is a $\mathbb{Q}(m)$-linear combination of b_{m+j} and b_{m+j-1}, both already rational multiples of b_{m+1}. The leading coefficient of $P(n)$ is 34 and the subtracted term has leading coefficient 1, so at each step the degree in m increases by 3 and the leading coefficient remains positive. After $h-1$ steps the numerator has degree $3(h-1)$.

For $h = 2$: $C_2(m) = P(m+1)/(m+2)^3$, with numerator $P(m+1) = (2m+3)(17m^2+51m+39)$, a cubic with leading coefficient 34. □

Lemma 8 (Nonvanishing over $\mathbb{F}_p$)**.** *For every prime $p \geq 7$ and every $2 \leq h \leq p$, the numerator of C_h is not the zero polynomial modulo p.*

Proof. Write N_h for the numerator of C_h (a polynomial of degree $3(h-1)$), satisfying $N_{h+1}(m) = P(m+h)\, N_h(m) - (m+h)^6\, N_{h-1}(m)$, with $N_1 = 1$ and $N_2 = P(m+1)$.

First evaluation. Equation (4) at $m = -1$ (where $b_{-1} = 0$, $b_0 = 1$) gives $C_h(-1) = b_{h-1}$, hence $N_h(-1) = b_{h-1} \cdot \big((h-1)!\big)^3$.

Second evaluation. Setting $m = -2$ in the recurrence gives $N_{h+1}(-2) = P(h-2)\, N_h(-2) - (h-2)^6\, N_{h-1}(-2)$, with $N_1(-2) = 1$ and $N_2(-2) = P(-1) = -5$. By induction using the Apéry recurrence at $n = h-2$:

$$N_h(-2) \;=\; -5\, b_{h-2}\, \big((h-2)!\big)^3 \qquad (h \geq 2). \tag{5}$$

Conclusion. If $N_h \equiv 0 \pmod p$ as a polynomial ($2 \leq h \leq p$, $p \geq 7$), then $N_h(-1) \equiv N_h(-2) \equiv 0$, forcing $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod p$ (since $p \nmid 5$ and $p \nmid (h-1)!$, $(h-2)!$). These are consecutive zeros, contradicting Lemma 6. □

Proposition 9. *For every prime $p \geq 7$,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \tfrac{3}{2}(H-1)(H-2) + 1, \tag{6}$$

where H is any integer with $2 \leq H \leq p$. Optimizing $H = \lfloor (p/3)^{1/3} \rfloor + 1$ gives $Z(p) \leq \frac{3^{4/3}}{2}\, p^{2/3} + O(p^{1/3}) = O(p^{2/3})$ unconditionally for all primes. In particular, $Z(p) = o(p)$.

Proof. Partition $\{0, \dots, p-1\}$ into $\lceil p/H \rceil$ consecutive blocks of length at most H. In each block, mark the first zero; all other zeros have gap $h \in \{2, \dots, H-1\}$ to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap-h pair constrains m to at most $3(h-1)$ residue classes mod p (the denominator factors $(m+j+1)^3$ are nonzero since $0 \leq m+j+1 \leq p-1$). By Lemma 8, $C_h \not\equiv 0$ over $\mathbb{F}_p$, so the numerator roots are at most $3(h-1)$. Summing:

$$Z(p) \leq \left\lceil \frac{p}{H} \right\rceil + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \tfrac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real $H > 0$ occurs at $H = (p/3)^{1/3}$, giving the bound. For $p = 2, 3, 5$: $Z(p) = 0$ (since b_j for $j < 5$ is coprime to $2, 3, 5$). $\square$

Corollary 10 (Sub-interval zero count)**.** *For every prime $p \geq 7$ and every interval $I \subseteq \{0, \ldots, p-1\}$,*

$$\#\{j \in I : p \mid b_j\} = O(|I|^{2/3}).$$

Proof. Apply the proof of Proposition 9 to the sub-interval I in place of $\{0, \ldots, p-1\}$: partition I into $\lceil |I|/H \rceil$ blocks of length H, bound the non-first zeros in each block by the same gap polynomial argument (Lemma 7 applies regardless of the interval), and optimize. $\square$

Remark 11. The leading coefficient of the numerator of C_h is $U_{h-1}(17)$, the Chebyshev polynomial of the second kind evaluated at 17, satisfying $\ell_1 = 1$, $\ell_{h+1} = 34\ell_h - \ell_{h-1}$. Let $N_h(m)$ denote the cleared numerator of C_h. The identity $P(-u-1) = -P(u)$ (evident from $P(n) = (2n+1)(17n^2 + 17n + 5)$) propagates through the recurrence to give $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$, verified computationally for $h \leq 200$. For even h, N_h is an *odd* polynomial in $(m + (h+1)/2)$, forcing the structural factor $(2m+h+1)$. At $h = 2$: factor $(2m+3)$ in $P(m+1) = (2m+3)(17m^2+51m+39)$. The computational data (Section 10) suggests $Z(p) = O(1)$, far stronger than $O(p^{2/3})$.

Remark 12 (Content restriction)*.* A stronger nonvanishing holds: if $p \geq 7$ divides the content of N_h (the gcd of all its coefficients), then $h \geq 2p$. Indeed, $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} \big((r-1)!\big)^3 \big((h-r)!\big)^3$ for $1 \leq r \leq h$ (by induction from the Apéry recurrence); choosing $r = 1, 2$ and using $p \nmid (h-1)!, (h-2)!, 5$ for $h \leq p$ recovers Lemma 8, while the range $p < h < 2p$ is excluded similarly via $r = h - p + 1$. Setting $\beta_n = (n!)^3 b_n$ and $F(z) = \sum_{n \geq 0} b_n z^n$, the endpoint factorization packages into the double generating function $\sum_{r,s \geq 0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u)\, F(v)$, showing that the boundary array is a rank-one tensor.

Remark 13 (Adjacent resultant formula)*.* The adjacent resultants satisfy the exact formula

$$\big|\mathrm{Res}(N_h, N_{h+1})\big| = \prod_{j=1}^{h-1} \big(j!^3\, b_j\big)^6. \tag{7}$$

This follows from the recursion $\mathrm{Res}(N_h, N_{h+1}) = N_h(-h)^6\ \mathrm{Res}(N_{h-1}, N_h)$ (since $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$), combined with $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1}((h-1)!)^3 b_{h-1}$ (the reflection identity and the first evaluation from Lemma 8). Verified for $h = 2, 3, 4, 5$; the cases $h = 2, 3, 4$ give $\mathrm{Res}(N_2, N_3) = -5^6$, $\mathrm{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$, $\mathrm{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$.

A key consequence: $p \mid \mathrm{Res}(N_h, N_{h+1})$ if and only if $p \mid b_j$ for some $1 \leq j \leq h-1$ (when $p > h$, so factorials are coprime to p). In particular, N_2 and N_3 have disjoint root sets modulo p for all $p \geq 7$ (since $5^6 \neq 0$).

Remark 14 (Sharpness of the bound)*.* The exponent $2/3$ in Proposition 9 is optimal for any argument using only the degree bounds $\deg N_h = 3(h-1)$. The exact extremal problem $\max\big\{\sum r_h : r_h \leq 3(h-1),\ \sum h\, r_h \leq p\big\}$ has value $\frac{3}{2} p^{2/3} + O(p^{1/3})$, with leading constant $3/2$ achieved by filling gap sizes $h = 2, 3, \ldots, H$ to capacity and using the remaining space at $h = H+1$, where $H \sim p^{1/3}$. Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield $O(p^{1/2})$: an abstract polynomial model on $\sim p^{2/3}$ points can satisfy $\deg F_h \leq 3(h-1)$ for all relevant h. Improving the exponent requires genuinely arithmetic input about the family (N_h).

Remark 15 (Dodgson condensation)*.* The Desnanot–Jacobi identity (Lewis Carroll's condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m)\, N_h(m+1) - N_{h+1}(m)\, N_{h+1}(m+1) = -\prod_{j=2}^{h+1} (m+j)^6. \tag{8}$$

This is an identity in $\mathbb{Z}[m]$, verified for $h \le 7$. A consequence: for $m \not\equiv -j \pmod p$ $(2 \le j \le h{+}1)$, the pairs $(m, h{+}2)$ and $(m{+}1, h)$ cannot both lie in the zero relation $\mathcal{Z} = \{(m,h) : N_h(m) \equiv 0\}$, since $N_{h+1}(m)\, N_{h+1}(m{+}1)$ would equal a nonzero product of sixth powers. After a meromorphic gauge $\tau_h(x) = N_h(x)/d_h(x)$ satisfying $d_{h+2}(x)\, d_h(x{+}1) = \prod_{j=2}^{h+1}(x{+}j)^6$, this becomes an exact SL_2-tiling: $\tau_{h+1}(x)\,\tau_{h+1}(x{+}1) - \tau_{h+2}(x)\,\tau_h(x{+}1) = 1$, equivalently an A_1 T-system (discrete Liouville equation). A zero of τ forces its transverse neighbors to be units (*singularity confinement*), but abstract SL_2-tilings can have zeros of density 1/2 (e.g., $q_n = 0, 1, 0, -1, \ldots$ with $q_n^2 - q_{n-1}q_{n+1} = 1$), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 21), and polynomial degree bounds—is essential.

Remark 16 (Projective orbit reformulation). Let $B_n = n!^3\, b_n$ and D_n be the second solution of the renormalized recurrence $w_{n+1} = P(n)\, w_n - n^6\, w_{n-1}$ with $(D_0, D_1) = (0, 1)$. Their Casoratian is $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$, nonzero for $n < p$. Define the *projective Apéry orbit* $\pi\colon \{0, \ldots, p{-}1\} \to \mathbb{P}^1(\mathbb{F}_p)$ by $\pi_n = [B_n : D_n]$. For $0 \le m < m + h < p$, $N_h(m) \equiv 0 \pmod p$ if and only if $\pi_m = \pi_{m+h}$. Thus the zero-count $Z(p)$ equals the number of distinct collisions in the orbit π. The reflection symmetry $P(-1{-}n) = -P(n)$ implies $b_{p-1-n} \equiv b_n \pmod p$, making the orbit a palindrome: $\pi_n = \pi_{p-1-n}$. In particular, $Z(p)$ is even except at the non-ordinary primes $p \mid a_p(f)$ (at most two primes $p \le 10^5$).

Remark 17 (Galois structure). For each fixed h, the Galois group of the splitting field of N_h over $\mathbb{Q}$ is not the full symmetric group: the reflection $N_h(-m{-}h{-}1) = (-1)^{h-1} N_h(m)$ forces roots into pairs $\{r, -r{-}h{-}1\}$, and $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ (hyperoctahedral group) with $m = \lfloor 3(h{-}1)/2 \rfloor$ (confirmed exactly for $h \le 5$, $2 \le h \le 9$ by obstruction tests). By the Chebotarev density theorem, the average number of $\mathbb{F}_p$-roots of N_h is 1 for odd h and 2 for even h. This prediction is verified: for 262 primes in $[1000, 3000]$, the measured averages are $0.98, 2.03, 1.11, 2.07, 1.11, 1.88$ for $h = 3, \ldots, 8$.

Proposition 18 (Hyperoctahedral Poisson law)**.** *Assume* $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ *with* $m = \lfloor 3(h{-}1)/2 \rfloor$. *Let* $Y_h(p)$ *be the number of non-central reflection-pairs of roots of* N_h *that split completely over* $\mathbb{F}_p$. *Then for each fixed* $k \le m$, *the factorial moments in the prime-aspect satisfy*

$$\lim_{x \to \infty} \frac{1}{\pi(x)} \sum_{p \le x} (Y_h(p))_k = \Big(\frac{1}{2}\Big)^k,$$

where $(Y)_k = Y(Y{-}1)\cdots(Y{-}k{+}1)$. *Since these match the factorial moments of* Poisson(1/2) *for all* k, *the prime-aspect distribution of* $Y_h(p)$ *converges to* Poisson(1/2) *as* $h \to \infty$.

Proof. In $G = C_2 \wr S_m = \{(\sigma, \varepsilon) : \sigma \in S_m,\ \varepsilon \in \{\pm 1\}^m\}$, a *positive fixed point* of $g = (\sigma, \varepsilon)$ is an index i with $\sigma(i) = i$ and $\varepsilon_i = +1$. The Frobenius at p acts on the m reflection-pairs, and $Y_h(p)$ equals the number of positive fixed points. For any k distinct indices $i_1, \ldots, i_k$, the fraction of $g \in G$ fixing all k positively is

$$\frac{2^{m-k}\,(m-k)!}{2^m\, m!} = \frac{1}{2^k\,(m)_k}.$$

Summing over all $(m)_k$ ordered k-tuples, $\mathbf{E}_{g \in G}[(Y(g))_k] = (1/2)^k$, which matches the factorial moments of Poisson(1/2). The Chebotarev density theorem then gives the prime-aspect convergence (Remark 20 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all $k \le m$, $Y_h(p) \to \text{Poisson}(1/2)$ as $m \to \infty$. □

Remark 19 (Prime-aspect vs. orbit-aspect). Proposition 18 is a *prime-aspect* statement: for fixed h and varying p, the root-pair count $Y_h(p)$ converges to Poisson(1/2). The zero-count $Z(p)$ aggregates contributions from all gaps $h = 1, \ldots, p{-}1$ at a *fixed* prime p, so the orbit-aspect independence needed to deduce $Z(p)/2 \sim \text{Poisson}(1/2)$ is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

Remark 20 (GRH-effective error). For each fixed k, the k-th factorial moment does not require the full splitting field K_h (of degree $2^m m!$, double-exponential in h). Let $E_{h,k}$ be the fixed field of the stabilizer of an ordered signed k-tuple with distinct underlying reflection pairs; then $[E_{h,k} : \mathbb{Q}] = 2^k\,(m)_k = O_k(h^k)$. Under GRH for $\zeta_{E_{h,k}}$,

$$\frac{1}{\pi(x)} \sum_{\substack{p \le x \\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\!\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where $D_{h,k} = |\operatorname{Disc}(E_{h,k})|$. The Apéry recurrence gives the coefficient-height bound $\log \|N_h\|_1 = O(h \log h)$, from which $\log D_{h,k} \ll_k h^{k+2} \log h$. Thus an $O(h^{-C})$ error holds in the polynomial range $x \ge h^{2k+4+2C+\varepsilon}$, for each fixed k.

At $x = 10^6$ with $h = 10$ ($m = 13$), the degree contributions alone for $k = 1, 2, 3$ are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

5.2 Dual bounds: the restart identity and column estimate

The bound $Z(p) = O(p^{2/3})$ controls zeros along a *fixed row* (varying m for a given gap h). A dual bound controls zeros along a *fixed column* (varying h for a given starting point x_0).

Lemma 21 (Restart identity)**.** *Suppose $N_r(x_0) = 0$ and $N_{r+1}(x_0) \neq 0$ for some $x_0 \in \mathbb{F}_p$ and $r \ge 1$. Then for every $d \ge 0$,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \tag{9}$$

In particular, if $N_{r+d}(x_0) = 0$ as well, then $N_d(x_0 + r) = 0$.

Proof. The sequence $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$ has $Y_0 = 0$, $Y_1 = 1$, and satisfies $Y_{d+1} = P((x_0+r)+d)\, Y_d - ((x_0+r)+d)^6\, Y_{d-1}$, which is the defining recurrence for $N_d(x_0 + r)$. □

Proposition 22 (Column bound)**.** *For every $x_0 \in \mathbb{F}_p$ and every interval I of H consecutive h-values with $H \le p$,*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod p\} \le \tfrac{3^{4/3}}{2}\, H^{2/3} + O(H^{1/3} + 1). \tag{10}$$

Proof. Split I at the unique transition h where $x_0 + h \equiv 0 \pmod p$ (if it exists in I), creating at most two nonsingular pieces plus an $O(1)$ additive cost.

On a nonsingular piece, order the zero levels. For threshold $L \ge 2$: a gap of length $d \le L$ between consecutive zeros at levels $r < r + d$ gives $N_d(x_0 + r) = 0$ by Lemma 21. For fixed d, there are at most $3(d-1)$ values of $x_0 + r$ satisfying this (by Lemma 8 and $\deg N_d = 3(d-1)$). Gaps longer than L contribute at most $H/(L+1)$ zeros. Summing:

$$\text{zero count} \le \frac{H}{L+1} + \sum_{d=2}^{L} 3(d-1) + O(1) = \frac{H}{L+1} + \tfrac{3}{2} L(L-1) + O(1).$$

Optimizing $L \asymp (H/3)^{1/3}$ gives the stated bound. □

Computationally, the column count at mesoscopic scale $H = \lfloor\sqrt{p}\rfloor$ is far below the $O(H^{2/3})$ bound: for all primes $p \le 100{,}003$ and all x_0, $\#\{h \le H : N_h(x_0) \equiv 0\} \le 2$. For $p \ge 10{,}007$, the maximum is ≤ 1.

Remark 23 (Rectangular incidence estimate). The row bound $|R_h| \le 3(h{-}1)$ and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x,h) : x \in \mathbb{F}_p,\ 2 \le h \le H,\ N_h(x) \equiv 0\} \le \min\{H^2,\ pH^{2/3}\}.$$

The first bound sums row degrees; the second sums column estimates. They cross at $H \sim p^{3/4}$. Note that a uniform bound $R_p(H) \le CH$ for all primes is *false*: by the Chebotarev density theorem, for each fixed H, infinitely many primes $p > H^2$ have every N_h $(2 \le h \le H)$ splitting completely over $\mathbb{F}_p$, giving $R_p(H) = \frac{3}{2}H(H-1)$. The correct target is the *mesoscopic* estimate $R_p(\lfloor\sqrt{p}\rfloor) = O(\sqrt{p})$, where the split-prime obstruction does not apply because $H = \sqrt{p}$ grows with p. More generally, if $R_p(H) \ll H^\beta$ then $Z(p) \le p/H + R_p(H)$ is optimized at $H \sim p^{1/(\beta+1)}$, giving $Z(p) \ll p^{\beta/(\beta+1)}$: the current $\beta = 2$ gives $2/3$ and $\beta = 1$ yields $1/2$. Reflection forces $R_p(H) \ge H/2$ from central roots, so $\beta = 1$ is the intrinsic limit of this first-moment method.

5.3 The continuant Bezout identity

Write $S_d(x) = \prod_{j=2}^{d}(x+j)$.

Proposition 24 (Bezout identity)**.** *For integers* $2 \le d < e$*,*

$$N_{e-1}(x{+}1)\,N_d(x) - N_{d-1}(x{+}1)\,N_e(x) = S_d(x)^6\,N_{e-d}(x+d). \tag{11}$$

In particular, after saturating the cut-edge divisor S_d*,*

$$(N_d, N_e) : S_d^\infty = (N_d,\ N_{e-d}(x+d)) : S_d^\infty.$$

Proof. The continuant addition formula (splitting the interval $[1, e{-}1]$ at position d) gives $N_e = N_d\,N_{e-d+1}(x{+}d{-}1) - (x{+}d)^6\,N_{d-1}\,N_{e-d}(x{+}d)$. The Dodgson identity at index $d{-}2$ gives $N_{d-1}(x{+}1)\,N_{d-1} - N_{d-2}(x{+}1)\,N_d = \prod_{j=2}^{d-1}(x{+}j)^6$. Combining these two identities yields (11). Verified computationally for all $2 \le d < e \le 9$. □

Corollary 25 (Separated-block criterion)**.** *Away from the cut-edge divisor* $S_d(x) = 0$*,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod p \quad\Longleftrightarrow\quad N_d(\alpha) \equiv N_{e-d}(\alpha+d) \equiv 0 \pmod p.$$

In particular, $\deg\gcd_{\mathbb{F}_p}(N_d, N_e) \le 3(\min(d, e{-}d) - 1)$*, which improves the naive* $3(d{-}1)$ *when* $e - d < d$*.*

Remark 26 (Separated-block resultant). Define $\mathcal{R}_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x{+}d))$. The Bezout identity gives

$$|\,\mathrm{Res}(N_d, N_e)| = \prod_{j=1}^{d-1}\big((j!)^3 b_j\big)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ($e = d+1$), $N_1 = 1$ gives $\mathcal{R}_{d,1} = 1$, recovering the adjacent resultant formula. For nonadjacent indices, $\mathcal{R}_{d,r}$ is genuinely new. The reflection identity gives $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$. The subtraction step does *not iterate*: after one application, $N_d(x)$ and $N_{e-d}(x+d)$ live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For $d = 2$, writing $P(y) = (2y{+}1)Q(y)$ with $Q(y) = 17y^2{+}17y{+}5$ and θ a root of Q, the norm decomposition

$\mathcal{R}_{2,r} = 2^{3(r-1)}G_r(-\frac{1}{2}) \cdot 17^{3(r-1)}\,\mathrm{N}_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$ (where $G_r = N_r(x+1)$) gives $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$. At $p = 71$: $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$ with $-5/2 \notin \{-1,-2,-3,-4\}$, a nonboundary common root inside the mesoscopic range ($4^3 < 71$). This shows the boundary-only gcd statement is *false*.

Remark 27 (Collision energy and the $H^{1/2}$ barrier). The column exponent $2/3$ is the natural barrier for the consecutive-gap argument: even if each gap of length d has only d (rather than $3(d-1)$) admissible starts, the sum $\sum_{d\le L} d \asymp L^2$ still gives $H/L + L^2$, optimized at $H^{2/3}$. To achieve $H^{1/2}$, one needs the aggregate collision energy $\mathcal{E}_p(H) = \sum_{d+r\le H} \#\{x : N_d(x) \equiv N_r(x+d) \equiv 0\} \ll H^{3/2}$, which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The A_1 T-system $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$ admits tame solutions with zero density 1/2, e.g. $\tau_{2k} = 0$, $\tau_{2k+1} = (-1)^k$, so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing $M(n) = \left(\begin{smallmatrix} P(n) & -n^6 \\ 1 & 0 \end{smallmatrix}\right)$, so that $N_h(x) = 0$ iff a fixed matrix coefficient of the interval product $M(x+h-1)\cdots M(x+1)$ vanishes), the conjectured $\mathcal{E}_p(H) \ll H^{3/2}$ is analogous to a Rudnev-type point–plane incidence bound in $\mathrm{PGL}_2(\mathbb{F}_p)$, subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

p	71	503	2003	5003	7919	20011	50021	10^5+3
$H = \lfloor\sqrt{p}\rfloor$	8	22	44	70	88	141	223	316
$R_p(H)/H$	1.75	1.59	1.36	1.50	1.75	1.63	1.56	1.46
$\mathcal{E}_p(H)$	1	2	2	2	10	0	2	0
$\mathcal{E}_p(H)/H^{3/2}$	.044	.019	.007	.003	.012	0	.001	0

The ratio $R_p(H)/H$ clusters around 3/2 (the Chebotarev prediction from Remark 17), and $\mathcal{E}_p(H)/H^{3/2} \to 0$, consistent with $\mathcal{E}_p(H) \ll H^{3/2}$. The key arithmetic object is the shifted resultant $S_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$: for $p \nmid S_{d,r}$, the gap polynomials N_d and $N_r(\cdot+d)$ share no root over $\mathbb{F}_p$, so that gap pair contributes zero to $\mathcal{E}_p$. A Hadamard bound gives $\log|S_{d,r}| \ll (d+r)^2\log(d+r)$, so the total number of prime divisors across all pairs (d,r) with $d+r \le H$ is $O(H^4 \log H)$. Averaging over the $\sim X/\log X$ primes in $(X, 2X]$ gives $\mathcal{E}_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ primes, provided $H \le X^{2/7-\varepsilon}$—rigorous but far from the global target $H = p-1$.

A sharper route uses the bipartite incidence graph $\Gamma_p(H)$: left vertices are gap lengths d, right vertices are gap lengths r, with an edge when N_d and $N_r(\cdot + d)$ share a root over $\mathbb{F}_p$. Each edge certifies a triple collision, so $\mathcal{E}_p(H) = e(\Gamma_p(H))$. A uniform fiber bound $\deg_L(d) \le C$ (the number of r-partners for each d) and a uniform codegree bound (at most C common right neighbors for any two left vertices) would place $\Gamma_p(H)$ in the regime of the Kővári–Sós–Turán theorem, giving $\mathcal{E}_p(H) \ll H^{3/2}$—the collision-energy scale needed for the mesoscopic target $R_p(\sqrt{p}) = O(\sqrt{p})$. The Dodgson identity (Remark 15) restricts local configurations in Γ_p, but by itself controls $O(H)$ pairs out of $O(H^2)$; upgrading it to a global codegree bound is the key missing step.

5.4 Value fibers: the bordered certificate

The zero-count bound $Z(p) = O(p^{2/3})$ (Proposition 9) concerns the single fiber $a = 0$, where one return condition $b_t \equiv b_{t+h} \equiv 0$ already forces a gap polynomial to vanish. For a *general* fiber $\{t : b_t \equiv a \pmod p\}$ with $a \neq 0$, two return conditions are needed to eliminate the two unknowns (b_{t+1}, a), and the resulting certificate is a bordered 2×2 determinant $D_{h,k}$ built from the gap polynomials and the second solution of the cleared recurrence. The determinant factors exactly as $D_{h,k} = \Pi_h \cdot \Delta_{h,k}$, and the analogue of Lemma 8 for the reduced certificate $\Delta_{h,k}$ is subtler: the endpoint evaluations assert equalities of separated Apéry values rather than consecutive zeros, so the no-consecutive-zero mechanism does not apply (an explicit two-point obstruction occurs at $p = 131$, $(h, k) = (12, 55)$; see the proof below). Nevertheless, a centered-coefficient analysis—using the Pell–Chebyshev structure of the leading coefficients rather than endpoint values—proves the nonvanishing in the *full* range $1 \le h < k < p$ (Theorem 32 below), and the range $k < p$ is sharp: at $p = 7$ the certificates $\Delta_{h,21}$ vanish identically. Consequently the same block-partition count that proves $Z(p) = O(p^{2/3})$ yields an *unconditional uniform* fiber bound $O(p^{3/4})$. We first state the counting mechanism conditionally, then prove the nonvanishing.

Proposition 28 (Uniform fiber bound conditional on bordered nonvanishing)**.** *Let*

$$P(n) = 34n^3 + 51n^2 + 27n + 5$$

and let $(b_n)_{n\ge0}$ be the Apéry numbers, with $b_0 = 1$, $b_1 = 5$, and

$$(n + 1)^3 b_{n+1} = P(n)b_n - n^3 b_{n-1}.$$

For a prime $p \ge 7$ and $a \in \mathbb{F}_p$, put

$$N_p(a) = \#\{0 \le t < p : b_t \equiv a \pmod p\}.$$

Define $N_0(x) = 0$, $N_1(x) = 1$, and, for $m \ge 1$,

$$N_{m+1}(x) = P(x + m)N_m(x) - (x + m)^6 N_{m-1}(x),$$

and set

$$\Pi_m(x) = \prod_{j=1}^{m} (x + j)^3, \qquad \begin{aligned} \Delta_{h,k}(x) = N_h(x)\Pi_{k-h}(x + h) - N_k(x) \\ + \Pi_h(x)N_{k-h}(x + h). \end{aligned}$$

Let H be an integer with $2 \le H \le (p - 1)/2$, and assume

$$\Delta_{h,k} \not\equiv 0 \pmod p \quad \text{in } \mathbb{F}_p[x] \qquad (1 \le h < k \le H). \tag{NV$_{p,H}$}$$

Then, for every $a \in \mathbb{F}_p^\times$,

$$N_p(a) \le 2\left\lfloor \frac{p - 1}{H + 1} \right\rfloor + \frac{H(H - 1)(2H - 1)}{2} + 2. \tag{12}$$

In particular, if (NV$_{p,H}$) holds for

$$H = \left\lceil (2p/3)^{1/4} \right\rceil,$$

then

$$N_p(a) \le 4(2/3)^{3/4}p^{3/4} + \sqrt{6}\,p^{1/2} + 3(2/3)^{1/4}p^{1/4} + 3. \tag{13}$$

Thus the leading constant supplied by this argument is

$$4(2/3)^{3/4} = 2.951151\ldots.$$

Proof. We first establish the exact algebra behind the bordered certificate. For a base point r put

$$Y_m = \left(\frac{(r + m)!}{r!}\right)^3 b_{r+m} = \Pi_m(r)b_{r+m}.$$

As long as $0 \le r$ and $r+m \le p-1$, clearing the denominators in the Apéry recurrence gives

$$Y_{m+1} = P(r+m)Y_m - (r+m)^6 Y_{m-1}, \tag{14}$$

with $Y_0 = b_r$ and $Y_1 = (r+1)^3 b_{r+1}$. Let $B_0(x) = 1$, $B_1(x) = 0$, and let B_m satisfy the same recurrence in m as N_m. The two fundamental solutions of (14) give

$$Y_m = N_m(r)Y_1 + B_m(r)Y_0. \tag{15}$$

Moreover,

$$B_m(x) = -(x+1)^6 N_{m-1}(x+1) \qquad (m \ge 1). \tag{16}$$

Indeed, this has the correct value at $m = 1$, and its induction step is exactly the recurrence for $N_{m-1}(x+1)$.

Write $R_m(x) = \Pi_m(x) - B_m(x)$. If $b_t = b_{t+m} = a$, equation (15) becomes

$$N_m(t)(t+1)^3 b_{t+1} - R_m(t)a = 0. \tag{17}$$

Consequently, if $b_t = b_{t+h} = b_{t+k} = a$ with $1 \le h < k$ and $t+k \le p-1$, the nonzero vector $((t+1)^3 b_{t+1}, a)$ is killed by the two rows in (17). Hence

$$D_{h,k}(t) = 0, \qquad D_{h,k}(x) := N_h(x)R_k(x) - N_k(x)R_h(x). \tag{18}$$

The structural factors of this determinant are important. A continuant Casoratian calculation gives

$$N_h B_k - N_k B_h = -\Pi_h^2 N_{k-h}(x+h). \tag{19}$$

For completeness, both sides, as functions of k, satisfy the same second-order recurrence; at $k = h$ they vanish, while at $k = h+1$ they equal $-\Pi_h^2$. Substituting (19) in (18) yields the exact factorization

$$\boxed{D_{h,k}(x) = \Pi_h(x)\Delta_{h,k}(x).} \tag{20}$$

Let $\ell_0 = 0$, $\ell_1 = 1$, and $\ell_{m+1} = 34\ell_m - \ell_{m-1}$. Since ℓ_m is the leading coefficient of N_m, the leading coefficient of $\Delta_{h,k}$ is

$$\ell_h + \ell_{k-h} - \ell_k < 0.$$

The inequality follows, for example, from the strict increase of (ℓ_m) and $\ell_{m+1} > 2\ell_m$. Therefore, over $\mathbb{Z}$,

$$\deg \Delta_{h,k} = 3(k-1), \qquad \deg D_{h,k} = 3(h+k) - 3. \tag{21}$$

The leading coefficient can, however, vanish modulo p, so its characteristic-zero nonvanishing does not prove ($\mathrm{NV}_{p,H}$).

Besides the cubic boundary factor Π_h, interval reflection gives all three immediately forced midpoint factors:

$$\begin{aligned} 2 \mid h &\implies 2x+h+1 \mid \Delta_{h,k}, \\ 2 \mid k &\implies 2x+k+1 \mid \Delta_{h,k}, \\ 2 \mid (k-h) &\implies 2x+h+k+1 \mid \Delta_{h,k}. \end{aligned} \tag{22}$$

These follow by applying $N_m(-x-m-1) = (-1)^{m-1}N_m(x)$ to the first interval, the whole interval, and the translated final interval, respectively. Exactly one of the three lengths $h, k, k-h$ is even unless h and k are both even, in which case all three are even. These are the factors forced by the three reflections; in the exceptional smallest case $\Delta_{1,2} = -4(2x+3)^3$.

We record why the known endpoint facts do not establish the missing nonvanishing. Using Remark 12 in (20) gives, for $1 \le r \le h$,

$$\Delta_{h,k}(-r) = (-1)^{r-1} b_{r-1}\big(b_{h-r} - b_{k-r}\big)\big((r-1)!(k-r)!\big)^3, \tag{23}$$

whereas, for $h < r \le k$,

$$\Delta_{h,k}(-r) = (-1)^{r-1} b_{k-r}\big(b_{r-h-1} - b_{r-1}\big)\big((r-1)!(k-r)!\big)^3. \tag{24}$$

In particular, the raw determinant vanishes structurally at $-1, \dots, -h$. After removing Π_h, its first two evaluations (when $h \ge 2$) are

$$\Delta_{h,k}(-1) = (k-1)!^3(b_{h-1} - b_{k-1}), \quad \Delta_{h,k}(-2) = -5(k-2)!^3(b_{h-2} - b_{k-2}).$$

Thus their simultaneous vanishing gives two equalities, not two consecutive zeros, and Lemma 6 does not contradict it. For example, modulo 131, the recurrence gives

$$b_{10} = b_{53} = 15, \qquad b_{11} = b_{54} = 15;$$

hence both evaluations vanish for $(h,k) = (12,55)$, while $\Delta_{12,55}(-3) = 2$. Removing the forced midpoint factor $2x + 13$ does not change this example, since it is nonzero at -1 and -2. Lemma 8 proves the corresponding assertion for the individual N_m, but it does not imply $(\mathrm{NV}_{p,H})$ for this bordered combination.

We now use the explicitly stated hypothesis $(\mathrm{NV}_{p,H})$. List the a-fiber in increasing order as

$$t_1 < t_2 < \cdots < t_s.$$

For $1 \le i \le s-2$ put

$$h_i = t_{i+1} - t_i, \qquad k_i = t_{i+2} - t_i.$$

If $k_i \le H$, then (20) and $t_i + k_i \le p-1$ show that $\Pi_{h_i}(t_i)$ is a unit modulo p. Thus t_i is a root of the nonzero polynomial Δ_{h_i,k_i}. For a fixed pair (h,k) there are at most $3(k-1)$ such bases, by (21). Summing over all short patterns gives

$$\sum_{k=2}^{H}\sum_{h=1}^{k-1} 3(k-1) = 3\sum_{j=1}^{H-1} j^2 = \frac{H(H-1)(2H-1)}{2}. \tag{25}$$

For the remaining indices, $t_{i+2} - t_i > H$, hence this integer is at least $H+1$. Splitting the indices i according to parity, the corresponding increments telescope and have total at most $p-1$ in each parity class. The number of long indices is therefore at most

$$2\left\lfloor \frac{p-1}{H+1} \right\rfloor.$$

Together with the two final fiber elements, this proves (12).

Finally put $x = (2p/3)^{1/4}$ and $H = \lceil x \rceil$. This H lies in $[2, (p-1)/2]$ for $p \ge 7$. Using $H(H-1)(2H-1)/2 < H^3 \le (x+1)^3$ and $2(p-1)/(H+1) < 2p/x = 3x^3$, the right-hand side of (12) is at most

$$4x^3 + 3x^2 + 3x + 3,$$

which is exactly (13). □

Corollary 29 (Collision energy, conditional form)**.** *Under the bordered nonvanishing hypothesis of Proposition 28, let $U(p)$ denote the right-hand side of* (13)*. Then*

$$E(p) := \sum_{a\in\mathbb{F}_p} N_p(a)^2 \le p\max\{Z(p), U(p)\} = O(p^{7/4}).$$

Proof. By Proposition 9, $N_p(0) = Z(p) = O(p^{2/3})$, while the conditional proposition gives $N_p(a) \le U(p) = O(p^{3/4})$ for $a \ne 0$. Since $\sum_a N_p(a) = p$,

$$E(p) \le \big(\max_a N_p(a)\big)\sum_a N_p(a) \le p\max\{Z(p), U(p)\} = O(p^{7/4}).$$

□

Proposition 30 (An exact degeneration outside the nonsingular range)**.** *Over $\mathbb{F}_7[x]$ one has*

$$\Delta_{h,21}(x) = 0 \qquad (1 \le h < 21).$$

In particular, a restriction on the size of k relative to the characteristic is essential.

Proof. Put

$$A_j(x) = \begin{pmatrix} P(x+j) & -(x+j)^6 \\ 1 & 0 \end{pmatrix}, \qquad T = A_7A_6\cdots A_1, \qquad q = x^7 - x.$$

All matrices in this proof are over $\mathbb{F}_7[x]$. Expanding the seven factors, equivalently applying the continuant recurrence seven times, gives the exact identities

$$\operatorname{tr} T = -q^3, \qquad \det T = \prod_{j=1}^{7}(x+j)^6 = q^6.$$

The second identity also follows immediately by multiplying the determinants. Cayley–Hamilton now gives

$$T^2 + q^3T + q^6I = 0, \qquad \text{hence} \qquad T^3 = q^9 I. \tag{26}$$

Since $A_{j+7} = A_j$ in characteristic 7, the transfer matrix for twenty-one steps is T^3. Applying (26) to the initial vectors $(N_1, N_0)^\top = (1,0)^\top$ and $(B_1, B_0)^\top = (0,1)^\top$ yields

$$N_{21} = 0, \qquad B_{21} = q^9.$$

On the other hand,

$$\Pi_{21} = \left(\prod_{j=1}^{21}(x+j)\right)^3 = q^9.$$

Thus $R_{21} = \Pi_{21} - B_{21} = 0$, and consequently

$$D_{h,21} = N_hR_{21} - N_{21}R_h = 0.$$

The factorization $D_{h,21} = \Pi_h\Delta_{h,21}$ and the fact that $\mathbb{F}_7[x]$ is an integral domain finish the proof. □

We next compute the few leading coefficients that detect every possible degeneration before a singular step is reached. Recall that $\ell_0 = 0$, $\ell_1 = 1$, and $\ell_{m+1} = 34\ell_m - \ell_{m-1}$.

Lemma 31 (Centered continuant coefficients)**.** *For $m \geq 1$, set*

$$\widehat{N}_m(Y) = N_m\left(Y - \frac{m+1}{2}\right).$$

Let u_m and v_m be respectively the coefficients of Y^{3m-5} and Y^{3m-7} in $\widehat{N}_m$; a coefficient with a negative exponent is understood to be zero. Then

$$u_m = \frac{-(m-1)(32m^2 - 64m - 11)\ell_m + 5m\ell_{m-1}}{256}, \tag{27}$$

$$v_m = \frac{(m-1)(5120m^5 - 31744m^4 + 62016m^3 - 19264m^2 - 37024m - 2095)\ell_m}{655360} - \frac{m(320m^3 - 1600m^2 - 320m + 621)\ell_{m-1}}{131072}. \tag{28}$$

Moreover, if

$$\widehat{\Pi}_m(Y) = \Pi_m\left(Y - \frac{m+1}{2}\right),$$

then

$$\widehat{\Pi}_m(Y) = Y^{3m} - \frac{m(m^2-1)}{8}Y^{3m-2} + \text{terms of degree at most } 3m-4. \tag{29}$$

All these identities hold over $\mathbb{Z}[1/10]$.

Proof. The reflection identity $N_m(-x-m-1) = (-1)^{m-1}N_m(x)$ shows that $\widehat{N}_m$ contains only powers of the same parity as $3m-3$. To compare its first three possible powers, center the recurrence for N_{m+1} by putting $T = x + (m+2)/2$. The natural centered variables for N_m and N_{m-1} are then $T - 1/2$ and $T - 1$, respectively, and

$$P(x+m) = 34\left(T + \frac{m-1}{2}\right)^3 + \frac{3}{2}\left(T + \frac{m-1}{2}\right), \qquad (x+m)^6 = \left(T + \frac{m-2}{2}\right)^6.$$

Comparison of the coefficients two and four places below the leading one gives

$$u_{m+1} = 34u_m - u_{m-1} - \frac{3}{4}(17m^2 - 17m - 2)\ell_m + \frac{3}{4}m(m-2)\ell_{m-1}, \tag{30}$$

$$v_{m+1} = 34v_m - v_{m-1} - \frac{3}{4}(17m^2 - 17m - 36)u_m + \frac{3}{4}(m^2 - 2m - 4)u_{m-1} \tag{31}$$

$$+ \frac{3(m-1)(17m^3 + 51m^2 - 90m - 24)}{64}\ell_m - \frac{3m(m-2)(m^2-6)}{16}\ell_{m-1}. \tag{32}$$

For completeness, the coefficient extraction in the second line uses only

$$e_2 = \frac{S_1^2 - S_2}{2}, \qquad e_4 = \frac{S_1^4 - 6S_1^2S_2 + 3S_2^2 + 8S_1S_3 - 6S_4}{24}$$

for the appropriate repeated shifts. The cases $m = 1, 2, 3$ are checked directly from $N_1 = 1$ and $N_2 = P(x+1)$. Substitution of (27) and (28) into (30)–(32), followed by $\ell_{m+1} = 34\ell_m - \ell_{m-1}$, makes the coefficients of both ℓ_m and ℓ_{m-1} agree identically. This proves the two closed forms by induction.

Finally,

$$\widehat{\Pi}_m(Y) = \prod_{j=1}^{m}\left(Y + j - \frac{m+1}{2}\right)^3.$$

The repeated shifts have sum zero, while the coefficient two below the top is

$$-\frac{3}{2}\sum_{j=1}^{m}\left(j-\frac{m+1}{2}\right)^2=-\frac{m(m^2-1)}{8}.$$

This proves (29). ☐

Theorem 32 (Range nonvanishing of the bordered certificate)**.** *Let $p \geq 7$ be prime. For every pair of integers*

$$1 \leq h < k < p,$$

the reduced bordered certificate satisfies

$$\Delta_{h,k} \not\equiv 0 \pmod p \qquad \text{in } \mathbb{F}_p[x].$$

Consequently ($\mathrm{NV}_{p,H}$) *holds for every $H \leq p-1$; in particular it holds throughout the range required in Proposition 28.*

Proof. Work in $\mathbb{F}_p$. Put

$$d = k-h, \qquad n = 3(k-1), \qquad z = x + \frac{k+1}{2}.$$

The two centered variables in the first product defining $\Delta_{h,k}$ are $z - d/2$ and $z + h/2$; in the last product they occur in the opposite order. Write

$$\Delta_{h,k}(x) = Lz^n + C_1 z^{n-1} + C_2 z^{n-2} + \cdots .$$

Multiplying the centered expansions in Lemma 31 gives

$$L = \ell_h + \ell_d - \ell_k, \tag{33}$$
$$C_1 = \frac{3}{2}(d\ell_h - h\ell_d), \tag{34}$$
$$C_2 = u_h + u_d - u_k + \frac{d(-d^2+1+12d-3hk)\ell_h + h(-h^2+1+12h-3dk)\ell_d}{8}. \tag{35}$$

For example, in the first product the degree-two loss from the shifted leading powers is $3d(4d-hk)/8$; adding the second coefficient of $\widehat{\Pi}_d$ gives the coefficient displayed in (35).

Suppose for a contradiction that $\Delta_{h,k} = 0$. Then $L = C_1 = C_2 = 0$. Set

$$A = \ell_h, \quad B = \ell_{h-1}, \quad C = \ell_d, \quad D = \ell_{d-1}.$$

The recurrence for ℓ_m gives, by induction,

$$\ell_{h+d} = 34AC - AD - BC, \tag{36}$$
$$\ell_{h+d-1} = AC - BD, \tag{37}$$
$$B^2 - 34AB + A^2 = 1, \qquad D^2 - 34CD + C^2 = 1. \tag{38}$$

Since h, d, k are nonzero in $\mathbb{F}_p$, equation (34) gives a unique $\rho \in \mathbb{F}_p$ with

$$A = h\rho, \qquad C = d\rho. \tag{39}$$

We first show that $\rho = 0$. Otherwise put $U = h\rho$. From $L = 0$, (39), and (36), after division by ρ,

$$hD = d(34U - B) - k. \tag{40}$$

Substitute (39) and (40) into the second Cassini identity in (38), multiply it by h^2, and subtract d^2 times the first Cassini identity. Direct expansion leaves

$$2dk(B - 17U + 1) = 0.$$

Thus $B = 17U - 1$. The first Cassini identity now reduces to $-288U^2 = 0$, impossible because $p \geq 7$ and $U \neq 0$. Therefore

$$A = C = 0. \tag{41}$$

Equations (36)–(38) now give

$$\ell_k = 0, \qquad \ell_{k-1} = -BD, \qquad B, D \in \{1, -1\}.$$

Using (27) in (35), together with (41), we obtain

$$C_2 = \frac{5}{256}\big(hB + dD + kBD\big). \tag{42}$$

For $(B, D) = (1, 1), (1, -1), (-1, 1), (-1, -1)$, the parenthesis in (42) is respectively $2k, -2d, -2h, 0$. Since $h, d, k < p$, its only possible zero is therefore

$$B = D = -1, \qquad \ell_{k-1} = -1. \tag{43}$$

It remains to inspect one more even coefficient. Under (41) and (43), formulas (27) and (28) become, for $m = h, d, k$,

$$u_m = -\frac{5m}{256}, \qquad v_m = \frac{m(320m^3 - 1600m^2 - 320m + 621)}{131072}. \tag{44}$$

(The dangerous case cannot contain $m = 1$, since $\ell_1 = 1$.) Let C_4 denote the coefficient of z^{n-4} in $\Delta_{h,k}$. In the present case $A = C = 0$, the shifted products and (29) give

$$C_4 = v_h + v_d - v_k \tag{45}$$
$$+ \frac{1}{8}u_h d(-d^2 + 1 + 30d - 3hk) + \frac{1}{8}u_d h(-h^2 + 1 + 30h - 3dk). \tag{46}$$

Indeed, by the parity structure of $\widehat{N}_m$ and $\widehat{\Pi}_m$, every contribution to the z^{n-4} coefficient of the two shifted products other than those displayed carries the factor ℓ_h (in $\widehat{N}_h \cdot \widehat{\Pi}_d$) or ℓ_d (in $\widehat{\Pi}_h \cdot \widehat{N}_d$)—namely the pairings of the leading coefficient of $\widehat{N}$ with the levels $0, -2, -4$ of $\widehat{\Pi}$ and with the shift binomials—and these vanish here; equation (46) is *not* asserted for general (h, k). For the u_h term the degree-two loss from the shifted leading powers is $3d(10d - hk)/8$, and the contribution from $\widehat{\Pi}_d$ is $-d(d^2 - 1)/8$; the other term is symmetric. Substitution of (44) into (46) gives

$$v_h + v_d - v_k = -\frac{5hd(4d^2 + 6dh - 15d + 4h^2 - 15h - 2)}{2048},$$
$$\text{the two } u\text{-terms} = \frac{10hd(2d^2 + 3dh - 15d + 2h^2 - 15h - 1)}{2048}.$$

Consequently

$$C_4 = -\frac{75hdk}{2048}.$$

This is nonzero in $\mathbb{F}_p$, because $p \geq 7$ and $0 < h, d, k < p$. This final contradiction proves the theorem. $\square$

Corollary 33 (Unconditional uniform fiber and energy bounds)**.** *For every prime $p \geq 7$, every $a \in \mathbb{F}_p^\times$, and every $2 \leq H \leq (p-1)/2$,*

$$N_p(a) \leq 2\left\lfloor \frac{p-1}{H+1} \right\rfloor + \frac{H(H-1)(2H-1)}{2} + 2.$$

In particular, with $H = \lceil (2p/3)^{1/4} \rceil$,

$$N_p(a) \leq 4(2/3)^{3/4} p^{3/4} + \sqrt{6}\, p^{1/2} + 3(2/3)^{1/4} p^{1/4} + 3.$$

Moreover,

$$\sum_{a \in \mathbb{F}_p} N_p(a)^2 = O(p^{7/4}).$$

Proof. Theorem 32 supplies hypothesis ($\mathrm{NV}_{p,H}$) in Proposition 28. That proposition gives the first two assertions, and Corollary 29 gives the last one. □

Remark 34 (Verification and sharpness). The nonvanishing theorem and its boundary are machine-verified from several independent directions: (i) direct polynomial-level scans find no identically vanishing $\Delta_{h,k}$ for any prime $7 \leq p \leq 700$ and $1 \leq h < k \leq 40$, nor in 1000 random cases with $p \leq 997$; (ii) the centered coefficient formulas for L, C_1, C_2 were checked against exact expansions over $\mathbb{Q}$, and the danger-case pipeline ($L = C_1 = C_2 = 0$, $B = D = -1$, $C_3 = 0$, $C_4 = -75hdk/2048$) was confirmed on 309 constructed rank-of-apparition cases across all primes $p \leq 3000$ and on three genuine drop-four cases at $p \approx 10^7$, in every instance against the actual centered polynomial; (iii) the $p = 7$, $k = 21$ degeneration is exact: $T^3 = (x^7 - x)^9 I$ for the seven-step transfer matrix, so $N_{21} \equiv 0$ and $R_{21} \equiv 0$ identically. The theorem makes the range restriction $k < p$ explicit and sharp; for the fiber count only $k \leq H \asymp p^{1/4}$ is used.

Remark 35 (Why the uniform fiber bound matters). The energy bound $E(p) = \sum_a N_p(a)^2 = O(p^{7/4})$ (Corollary 33) is the first *unconditional* power saving over the trivial $E(p) \leq p \max_a N_p(a) \leq p^2$ scale for the value distribution of Apéry numbers modulo p; the random-map prediction is $E(p) \asymp p$. Any bound $E(p) \ll p^{2-\eta}$ gives $Z(p) \leq E(p)^{1/2} \ll p^{1-\eta/2}$, so energy savings are a strictly weaker—hence potentially more accessible—avenue than direct zero-set bounds, and the value-collision structure enters the two-prime dispersion analysis of Section 12 through the equal-value pattern counts. Note that the $p^{3/4}$ fiber bound also complements the zero-fiber bound: $Z(p) = O(p^{2/3})$ remains the stronger statement for $a = 0$, so $\max_a N_p(a) = O(p^{3/4})$ with the maximum over *all* a.

5.5 Computational evidence

Proposition 36. *For all* 78,496 *primes* $5 \leq p \leq 10^6$*:*

1. $Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$ *for all but two primes; the two exceptions ($p = 11$ and $p = 3137$) have $Z(p) = 1$ (non-ordinary primes, where $p \mid a_p(f)$).* $\max Z(p) = 12$ *at* $p = 159{,}977$.
2. *The exact histogram is:*

$Z(p)$	*0*	*1*	*2*	*4*	*6*	*8*	*10*	*12*
count	*47632*	*2*	*23730*	*6045*	*951*	*123*	*12*	*1*
fraction	*.607*	*.000*	*.302*	*.077*	*.012*	*.002*	*.000*	*.000*
Poi($\frac{1}{2}$)	*.607*	—	*.303*	*.076*	*.013*	*.002*	*.000*	*.000*

The mean $Z(p) = 1.000$, stable across all ranges ($\chi^2 = 3.42$ on 4 d.f. against Poisson(1/2) *for the pair count). The empirical factorial moments of $K(p) = Z(p)/2$ over the* 78,494 *ordinary primes are:*

r	*1*	*2*	*3*	*4*	*5*	*6*
$\frac{1}{\pi_{\text{ord}}} \sum (K)_r$	.4998	.2490	.1210	.0605	.0275	.0092
2^{-r}	.5000	.2500	.1250	.0625	.0312	.0156

The first four moments match Poisson(1/2) to better than 3%*; the higher-moment deviation is consistent with sampling variance ($K \geq 5$ has only* 13 *primes in this range).*

3. *The symmetry $b_j \equiv b_{p-1-j} \pmod p$ holds for all tested primes: zeros come in palindromic pairs $\{j, p-1-j\}$, so $Z(p)$ is even (except at non-ordinary primes). The pair count $Z(p)/2$ fits* Poisson(1/2) *to within sampling noise (table above). Proposition 18 proves* Poisson(1/2) *for the root-pair count $Y_h(p)$ of each fixed gap polynomial in the prime-aspect; the transfer to the actual coefficient zero count $K(p) = Z(p)/2$ (Remark 19) remains conjectural.*

Remark 37 (Mesoscopic root count). At the scale $H = \lfloor \sqrt{p} \rfloor$, the total root count $R_p(H) = \sum_{h=2}^{H} r_p(h)$ satisfies $R_p(H)/H \approx 3/2$ for all tested primes, consistent with the Chebotarev prediction from Remark 17:

p	$H = \lfloor \sqrt{p} \rfloor$	$R_p(H)$	$R_p(H)/H$
503	22	35	1.59
2003	44	60	1.36
5003	70	105	1.50
10007	100	154	1.54
20011	141	230	1.63
50021	223	347	1.56
100003	316	460	1.46

The ratio $R_p(H)/H$ stabilizes near 3/2 (range 1.32–1.75 for $p > 500$), supporting the mesoscopic conjecture in Remark 23.

5.6 Hypothesis Z

Hypothesis 38 (Hypothesis Z)**.** *There exists an absolute constant C such that $Z(p) \leq C$ for all primes p.*

Remark 39. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ has a fundamentally different structure. Stienstra–Beukers [25] showed that A_p is related to a *CM* modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod p$ for all $p \equiv 3 \pmod 4$ — a *positive density* of primes with guaranteed zeros. In contrast, the $\zeta(3)$ Apéry numbers are governed by the non-CM newform $f(z) = \eta(2z)^4 \eta(4z)^4 \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis $\bar{Z}$ should hold for the $\zeta(3)$ sequence, even though the pointwise bound $Z(p) \leq C$ (Hypothesis Z) likely fails.

Remark 40. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod p$ [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [19]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is

$\sim C_f/p$ (the Sato–Tate distribution on the normalized coefficients $a_p/p^{3/2}$ admits $\sim 4p^{1/2}$ multiples of p in $[-2p^{3/2}, 2p^{3/2}]$, each contributing mass $\sim p^{-3/2}$). The expected count up to x is thus $C_f \log\log x$ with $C_f \approx 1$. Finding 2 non-ordinary primes ($p = 11, 3137$) below 10^6 is consistent with $\log\log(10^6) \approx 2.9$. This is not a standard Lang–Trotter problem: the target set $\{a : p \mid a\}$ varies with p, and no analogue of Elkies' theorem (infinitely many supersingular primes) is known for weight ≥ 4.

Remark 41 (Evidence against Hypothesis Z). The $2 \cdot \text{Poisson}(1/2)$ distribution (Remark 10 below) has unbounded support: extreme-value theory predicts $\max_{p \leq N} Z(p) \sim 2\log\log N$. The current record $Z(p) = 12$ at $p = 159{,}977$ (with record progression $2, 4, 6, 8, 10, 12$) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with $Z(p) = 14$ should appear near $p \approx 2 \times 10^7$ (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to $p = 5 \times 10^7$ would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

Hypothesis 42 (Hypothesis $\bar{Z}$)**.** $\sum_{p \leq x} Z(p) \ll \pi(x)$*, equivalently, the average of* $Z(p)$ *over primes* $p \leq x$ *is bounded.*

Remark 43. Hypothesis $\bar{Z}$ is strictly weaker than Hypothesis Z and consistent with $Z(p) \sim 2 \cdot \text{Poisson}(1/2)$ (which gives $\mathbb{E}[Z(p)] = 1$, independent of p). It suffices for the conditional bound $\log G_n = O(\sqrt{n})$ (Theorem 66).

We state the full distributional prediction of the Poisson model (cf. Proposition 18):

Hypothesis 44 (Poisson conjecture)**.** *For every good prime* $p \geq 5$*, put*

$$\nu_p := \mathbf{1}_{\{p \mid a_p(f)\}}, \qquad K_*(p) := \frac{Z(p) - \nu_p}{2},$$

where f *is the weight-*4 *newform associated to the Apéry differential operator. Thus* $K_*(p)$ *counts noncentral reflected pairs of zeros of* $(b_j \bmod p)_{0 \leq j < p}$*. For every fixed* $r \geq 1$*,*

$$\frac{1}{\pi(X)} \sum_{p \leq X} \binom{K_*(p)}{r} \longrightarrow \frac{1}{2^r\, r!}.$$

On ordinary primes ($\nu_p = 0$)*, this reduces to* $K(p) = Z(p)/2 \to \text{Poisson}(1/2)$*, consistent with the data in Proposition 36.*

Remark 45. The Poisson conjecture implies Hypothesis $\bar{Z}$. Taking $r = 1$ gives $\sum_{p \leq X} K_*(p) = (\frac{1}{2} + o(1))\,\pi(X)$. Since $Z(p) = 2K_*(p) + \nu_p$ and $0 \leq \nu_p \leq 1$,

$$\sum_{p \leq X} Z(p) = 2\sum_{p \leq X} K_*(p) + \sum_{p \leq X} \nu_p \leq (2 + o(1))\,\pi(X),$$

which is Hypothesis $\bar{Z}$. A proof would require the fixed-order joint independence of reflection-pair zero events—the correlation theorem underlying the factorial-moment computation in Proposition 18—for the deterministic Apéry orbit, not just the hyperoctahedral random model.

Remark 46 (Computational evidence for $\bar{Z}$). Over dyadic blocks of primes, $(\sum_{P < p \leq 2P} Z(p))/|\{P < p \leq 2P\}|$ ranges from 0.86 to 1.07 for $P = 2^k$, $3 \leq k \leq 12$ (666 primes up to 4999), with overall average $\sum Z(p)/\pi = 0.91$. The Poisson prediction is 1.

6 The denominator connection lemma

Lemma 47. *Let $p \geq 7$ be prime with $n/2 < p \leq n$, and set $r = n - p$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_r$.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \dots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2 (variation of parameters). From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^{n} \frac{6}{m^3\, b_m\, b_{m-1}}. \tag{47}$$

For $p > n/2$, the only $m \in \{1, \dots, n\}$ with $p \mid m$ is $m = p$.

Step 3. By Theorem 4, $b_p \equiv b_0\, b_1 = 5 \pmod{p}$ (since p has base-p digits $(0, 1)$), and by the symmetry $b_j \equiv b_{p-1-j}$, $b_{p-1} \equiv b_0 = 1 \pmod{p}$. Since $p \geq 7$, both are nonzero mod p.

The $m = p$ term in (47) has $v_p = -3$; all other terms have $v_p \geq 0$. Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p\, b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using $b_n \equiv 5\, b_r \pmod{p}$ (Theorem 4, since $n = 1 \cdot p + r$):

$$p^3\, a_n \equiv \tfrac{6}{5} \cdot 5\, b_r = 6\, b_r \pmod{p}.$$

Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. $\square$

Verification. For all $n \leq 300$ and primes $p \in (n/2, n]$ with $p \geq 7$: $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases (100%).

Corollary 48 (Lucas reduction)**.** *For $p \geq 7$ and $p \in (n/2, n]$: $v_p(G_n) \geq 1$ iff $p \mid b_{n-p}$. By the Lucas congruence $b_n \equiv b_1\, b_{n-p} = 5\, b_{n-p} \pmod{p}$, this is equivalent (for $p \neq 5$) to $p \mid b_n$. Thus $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$ equals $\omega_{(n/2,n]}(b_n) + O(1)$, the number of* distinct prime factors of b_n lying between $n/2$ and n*, up to an absolute constant.*

The full conjecture $G_n = e^{o(n)}$ for all n is therefore equivalent to the *top-half radical estimate*:

$$\log \mathrm{rad}_{(n/2,n]}(b_n) = o(n). \tag{48}$$

Since $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52\, n$, the height bound gives $\omega_{(n/2,n]}(b_n) \leq \log b_n / \log(n/2) = O(n/\log n)$, precisely one $o(1)$ factor short of the target. Computation for $n \leq$ 2,000,000 gives $B(n) \leq 3$ (with exactly 53 maximizers and no value $B(n) \geq 4$) and $B(n) = 0$ for 94.8% of values (see Remark 49 for the Poisson analysis).

Remark 49 (Random-model prediction). Under a random-integer model, $\Pr(p \mid b_n) \approx 1/p$ for each prime p, so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\!\Big(\frac{\log n}{\log(n/2)}\Big) + O(1/\log n) \approx \frac{\log 2}{\log n} \;\to\; 0.$$

Computation for $n \leq$ 200,000 confirms: $\mathbb{E}[B] \approx 0.065$, $\max B(n) = 3$, and the histogram matches Poisson($\log 2/\log n$) to high accuracy: $B(n) = 0$ for 93.7%, $B(n) = 1$ for 6.0%, $B(n) = 2$ for 0.20%, $B(n) = 3$ for 0.006% (the Poisson predictions with $\lambda = 0.065$ are 93.7%, 6.1%, 0.20%, 0.004%).

The dispersion ratio $\mathrm{Var}(B)/\mathbb{E}[B] = 1.003$ is the exact Poisson signature. The extended scan to $n \le 2{,}000{,}000$ (with all 148,930 primes $7 \le p \le 2 \times 10^6$) confirms the picture at ten times the range: $\max B(n) = 3$ still, overall $\mathrm{Var}(B)/\mathbb{E}[B] = 0.9991$, per-shell $\mathrm{Var}/\mathbb{E}$ between 0.95 and 1.00, and windowed sums over 64 windows per dyadic shell show no localized pile-up (largest window sum 481 against a window mean of 429.5 at $N = 2^{19}$). By CRT-based approximate independence across primes, $B(n)$ should follow $\mathrm{Poisson}(\log 2/\log n)$ with $\lambda \to 0$. Extreme-value theory then predicts $\max_{n \le N} B(n) = O(\log N/\log\log N)$—trivially $o(n/\log n)$—so the random model *strongly* predicts the full conjecture. The obstruction is proving that b_n behaves "randomly" with respect to its prime factors in $(n/2, n]$, despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

7 The unconditional density-one theorem

Theorem 50. *For a set of positive integers n of natural density 1, $G_n = e^{o(n)}$.*

Proof. Fix N large and consider $n \in (N, 2N]$. Write $B(n) = \#\{\text{primes } p > \sqrt{n} \text{ with } v_p(G_n) \ge 1\}$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$ for every n. Each such prime contributes $v_p(G_n)\log p \le 3\log n$ (since $v_p(d_n) = 3$ for $p > \sqrt{n}$), so $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$. It suffices to show $B(n) = o(n/\log n)$ for all but $o(N)$ values of $n \in (N, 2N]$.

For $p > \sqrt{n}$, write $n = qp + r$ with $1 \le q \le \sqrt{2N}$ and $0 \le r < p$. By Lemma 55(iii), $v_p(G_n) \ge 1$ requires $D_q b_r \equiv 0 \pmod{p}$. By Lemma 56, leading-digit bad primes ($p \mid b_q$, $p \nmid b_r$) have $v_p(G_n) = 0$ and do not contribute to $B(n)$. The remaining contributions come from *lower-digit bad primes* ($r \in \mathcal{Z}_p$) and *companion-block primes* ($D_q \equiv 0 \pmod{p}$, $p \nmid b_r$).

Lower-digit incidence. For fixed $p \in (\sqrt{N}, 2N]$, an interval of N consecutive integers contains at most $N \cdot Z(p)/p + Z(p)$ values of n whose residue $r = n \bmod p$ lies in $\mathcal{Z}_p$. Set $\rho_p = Z(p)/p$. The total lower-digit incidence is

$$R_N \le \sum_{\sqrt{N} < p \le 2N} \bigl(N\rho_p + Z(p)\bigr).$$

By Proposition 9, $\rho_p \to 0$. Since $\pi(2N) = O(N/\log N)$:

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \le 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence $R_N = o(N^2/\log N)$.

Companion-block incidence. By Proposition 59, the companion-block contribution to $\log G_n$ is $O(n^{2/3}) = o(n)$ at every index. Hence companion-block primes do not affect the density-one statement, and we set $P_N = 0$ in the counting argument.

Conclusion. Write $\sum_{n \in (N,2N]} B(n) \le R_N + P_N = o(N^2/\log N)$. For any fixed $\delta > 0$, Markov's inequality gives

$$\#\bigl\{n \in (N, 2N] : B(n) > \delta n/\log n\bigr\} \le \frac{o(N^2/\log N)}{\delta N/\log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so $B(n) = o(n/\log n)$ for density 1. For all such n, $\log G_n = O(\sqrt{n}) + 3B(n)\log n = O(\sqrt{n}) + o(n) = o(n)$. □

Corollary 51 (Quantitative exceptional set)**.** *For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon(N^{2/3}).$$

Proof. We make the density-one argument quantitative using the explicit bound $Z(p) = O(p^{2/3})$ from Proposition 9. For $n \in (N, 2N]$, $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$, so $\log G_n > \varepsilon n$ requires $B(n) > \varepsilon N/(7\log N)$ for N large.

Partial summation with $Z(p) \le Cp^{2/3}$ gives

$$\sum_{\sqrt{N}<p\le 2N} \frac{Z(p)}{p} \le C \sum_{p\le 2N} p^{-1/3} = \tfrac{3}{2}\, C\, \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies $R_N = N\sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$ (the second sum, $\sum Z(p) \le C\sum p^{2/3}$, contributes the same order). The companion-block incidence $P_N = O(N^{3/2})$ is dominated by R_N. Hence $\sum_{n\in(N,2N]} B(n) = O(N^{5/3}/\log N)$.

By Markov's inequality,

$$\#\{n \in (N, 2N] : B(n) > \varepsilon N/(7\log N)\} \le \frac{O(N^{5/3}/\log N)}{\varepsilon N/(7\log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals $(N/2^{j+1}, N/2^j]$ gives the global bound, since $\sum_{j\ge 0}(N/2^j)^{2/3} = O(N^{2/3})$. □

In particular, the exceptional set has finite harmonic weight:

Corollary 52 (Finite harmonic weight)**.** *For every $\varepsilon > 0$,*

$$\sum_{\substack{n\ge 1\\ \log G_n > \varepsilon\, n}} \frac{1}{n} < \infty.$$

Proof. The contribution from $(2^k, 2^{k+1}]$ is $O(2^{-k/3})$ by Corollary 51, and the geometric series converges. □

Remark 53. The exponent $2/3$ in the exceptional set is inherited from the lower-digit incidence R_N, which uses $Z(p) = O(p^{2/3})$. More generally, if $Z_p(I) \le C|I|^\alpha$ uniformly for all primes and sub-intervals (as Corollary 10 provides with $\alpha = 2/3$), then $R_N = O(N^{1+\alpha}/\log N)$ and $L_N = O(N^{(4-\alpha)/(3-\alpha)}/\log N)$. The exceptional set is therefore $O(\max(N^\alpha,\ N^{1/(3-\alpha)})/\varepsilon)$, with the crossover at $\alpha = (3-\sqrt{5})/2 \approx 0.382$. Under Hypothesis $\bar{Z}$, the dyadic bound $D(P,Q) \le \sum Z(p) = O(P/\log P)$ is equivalent to $\alpha = 0$ in this formula, giving the exceptional set $O(N^{1/3}/\varepsilon)$ (Corollary 68). Theorem 66 shows the matching density-one bound $\log G_n = O(\sqrt{n})$, where $O(\sqrt{n})$ is now the small-prime floor, not the leading-digit contribution.

8 Codegree amplification

We begin with a structural rigidity lemma: consecutive integers cannot share a top-half bad prime.

Lemma 54 (Consecutive coprimality)**.** *Let $p > 5$ be prime and $n \ge p$. If $p \mid b_{n \bmod p}$ and $p \mid b_{(n+1) \bmod p}$, then $p \le n/2$. Equivalently, no prime $p > n/2$ with $p > 5$ can be lower-digit bad for both n and $n+1$.*

Proof. Since $p > n/2$, both n and $n+1$ have base-p representations $n = p+r$, $n+1 = p+(r+1)$ with $0 \le r < r+1 < p$ (provided $r < p-1$; the case $r = p-1$ gives $n+1 = 2p$ and $b_{n+1} \equiv b_2 b_0 = 73 \not\equiv 0$ for $p > 73$, a finite check).

Suppose $p \mid b_r$ and $p \mid b_{r+1}$ with $r, r+1 < p$. The Apéry recurrence gives

$$(k+1)^3 b_{k+1} = P(k) b_k - k^3 b_{k-1},$$

and since $k+1 < p$ implies $(k+1)^3$ is invertible mod p, the conditions $p \mid b_k$ and $p \mid b_{k+1}$ force $p \mid b_{k-1}$. Iterating downward from $k = r$ gives $p \mid b_j$ for all $0 \le j \le r+1$, contradicting $b_0 = 1$. □

A prime that divides b_n only through a leading base-p digit contributes nothing to G_n.

Lemma 55 (Block system for $p^3 a_n \bmod p$)**.** *Let $p \ge 7$, $p \le n < p^2$, and write $n = qp + r$ with $1 \le q < p$ and $0 \le r < p$. Define $\Delta_{q,r} \equiv p^3 a_{qp+r} \pmod p$ and $D_q = \Delta_{q,0}$. Then:*

- *(i) $\Delta_{q,r} \equiv D_q b_r \pmod p$ for every $0 \le r < p$;*
- *(ii) $D_q b_{q-1} - D_{q-1} b_q \equiv 6/q^3 \pmod p$;*
- *(iii) $p \mid G_n$ if and only if $D_q b_r \equiv 0 \pmod p$.*

Proof. For $p > \sqrt{n}$, $v_p(\mathrm{lcm}(1, \dots, n)) = 1$, so $v_p(d_n) = 3$. Since $d_n a_n \in \mathbf{Z}$, $p^3 a_n \in \mathbf{Z}_p$, and $v_p(d_n a_n) = v_p(p^3 a_n)$. Meanwhile $v_p(d_n b_n) = 3 + v_p(b_n) \ge 3$, so $p \mid G_n \Leftrightarrow p \mid p^3 a_n$.

(i) Multiply the Apéry recurrence at index $qp + r$ by p^3 and reduce modulo p. Since $qp + r \equiv r \pmod p$, the sequence $r \mapsto \Delta_{q,r}$ satisfies the same recurrence $(r+1)^3 \Delta_{q,r+1} = P(r)\Delta_{q,r} - r^3 \Delta_{q,r-1}$ as $r \mapsto D_q b_r$, with matching initial values $\Delta_{q,0} = D_q$ and $\Delta_{q,1} = 5D_q = D_q b_1$. Since $(r+1)^3$ is invertible mod p for $0 \le r \le p-2$, uniqueness gives the identity.

(ii) Multiply the Wronskian $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ at $n = qp$ by p^3 and reduce mod p, using $b_{qp} \equiv b_q$ and $b_{qp-1} \equiv b_{q-1} b_{p-1} \equiv b_{q-1}$ (Lucas), and $p^3 a_{qp-1} \equiv D_{q-1} b_{p-1} \equiv D_{q-1}$ (part (i)).

(iii) Immediate from (i) and the observation above. □

Lemma 56 (Leading-digit vanishing)**.** *Let $p \ge 7$, $p \le n < p^2$, $n = qp + r$ with $q \ge 1$. If $p \mid b_q$ and $p \nmid b_r$, then $v_p(G_n) = 0$.*

Proof. Since consecutive Apéry numbers below the p-th cannot both vanish mod p (the recurrence propagates to $b_0 = 1$), $p \nmid b_{q-1}$. Part (ii) of Lemma 55 gives $D_q b_{q-1} \equiv 6/q^3 \not\equiv 0 \pmod p$, so $D_q \not\equiv 0$. By part (iii), $p \mid G_n$ iff $D_q b_r \equiv 0$, but $D_q \not\equiv 0$ and $p \nmid b_r$, so $D_q b_r \not\equiv 0$ and $v_p(G_n) = 0$. □

Remark 57 (Block constants are Apéry numbers). The block constant D_q of Lemma 55 equals $a_q \bmod p$, where a_q is the Apéry rational companion. Indeed, a_q satisfies the same Wronskian recurrence $a_q b_{q-1} - a_{q-1} b_q = 6/q^3$ with $a_0 = 0$, $a_1 = 6$, so $D_q = a_q$ by uniqueness. Consequently, $D_q \equiv 0 \pmod p$ iff p divides the numerator of a_q (in lowest terms) and p does not divide its denominator.

Since $|a_q| \le C(1+\sqrt{2})^{4q}$ and $\mathrm{denom}(a_q) \mid \mathrm{lcm}(1, \dots, q)^3$, the numerator of a_q has $O(q)$ prime divisors. In particular, for fixed q, at most $O(q)$ primes p can be companion-block bad at quotient q. Computation through $n = 2000$ confirms that the companion-block count $\#\{p > \sqrt{n} : p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})\}$ is at most 3 for every n in this range.

Proposition 58 (Reflection of the companion solution)**.** *Let $p \ge 5$ be prime. Every p-integral solution (u_n) of the Apéry recurrence* (1) *on $\{0, \dots, p-1\}$ is palindromic modulo p:*

$$u_{p-1-j} \equiv u_j \pmod p \qquad (0 \le j \le p-1).$$

In particular, for the companion solution $a_0 = 0$, $a_1 = 6$,

$$a_{p-1-j} \equiv a_j \pmod p, \qquad a_{p-1} \equiv 0 \pmod p.$$

Proof. Let R denote the reversal operator, $(Ru)_j = u_{p-1-j}$. The coefficient polynomial $P(n) = 34n^3 + 51n^2 + 27n + 5$ satisfies $P(-1-x) = -P(x)$, so substituting $n \mapsto p-2-n$ in the recurrence shows that R maps p-integral solutions to p-integral solutions modulo p.

The integer Apéry solution (b_n) satisfies $b_{p-1} \equiv 1$ and $b_{p-2} \equiv 5 \pmod p$: in the binomial sum $b_n = \sum_k \binom{n}{k}^2\binom{n+k}{k}^2$, only the $k=0$ term survives modulo p for $n = p-1$, while for $n = p-2$ only $k = 0, 1$ survive, contributing $1+4 = 5$. Since Rb and b have the same initial values $(b_0, b_1) = (1, 5)$, uniqueness gives $Rb = b$.

For two solutions x, y, the discrete Wronskian $W_n(x,y) = x_n y_{n-1} - x_{n-1}y_n$ satisfies $(n+1)^3 W_{n+1} = n^3 W_n$, so $W_n(x,y) = C(x,y)/n^3$. Using $Rb = b$:

$$W_n(Ru, b) = -W_{p-n}(u, b) = W_n(u, b).$$

Hence $W_n(Ru - u, b) = 0$, so $Ru - u = cb$ for some $c \in \mathbf{F}_p$. Applying R: on one hand $R(Ru-u) = -(Ru-u)$, on the other $R(cb) = cb$. Since p is odd, $c = 0$ and $Ru = u$. □

Proposition 59 (Companion-block height bound)**.** *Assume $D_q \equiv a_q \pmod p$. For every $n \geq 1$, the total logarithmic weight of companion-block primes is at most $O(n^{2/3})$:*

$$\sum_{\substack{p > \sqrt{n} \\ p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})}} \log p \;\leq\; C\, n^{2/3}$$

for an absolute constant C. In particular, companion-block primes contribute $o(n)$ to $\log G_n$ at every index.

Proof. Write $a_q = A_q/C_q$ in lowest terms. From the telescoping identity $a_q = b_q \sum_{k=1}^{q} 6/(k^3 b_k b_{k-1})$ and $b_k \geq 1$, we obtain $|A_q| \leq |a_q| \cdot C_q \leq C_0(1+\sqrt{2})^{4q}$, hence each numerator A_q has at most $O(q)$ prime divisors.

For a fixed n, set $Q = \lfloor n^{1/3} \rfloor$. A companion-block prime $p > \sqrt{n}$ has quotient $q = \lfloor n/p \rfloor < \sqrt{n}$.

Small quotients ($q \leq Q$): for each q, the primes dividing A_q contribute $\sum \log p \leq \log |A_q| = O(q)$. Summing over $q \leq Q$: $\sum_{q=1}^{Q} O(q) = O(Q^2) = O(n^{2/3})$.

Large quotients ($q > Q$): each such prime satisfies $p < n/Q = n^{2/3}$. Since each prime determines its quotient uniquely, the total log-weight from this range is bounded by the Chebyshev sum $\sum_{p < n^{2/3}} \log p = \theta(n^{2/3}) = O(n^{2/3})$. □

Proposition 60 (Quotient reduction)**.** *Let f be any function with $f(n) \to \infty$ and $f(n) \leq \log n$. Then*

$$\sum_{\substack{\sqrt{n} < p \leq n/(f(n)\log n) \\ p \mid G_n}} v_p(G_n) \log p \;=\; O\!\left(\frac{n}{f(n)}\right) = o(n).$$

Consequently,

$$\log G_n = O(\sqrt{n}) + O(n^{2/3}) + O\!\left(\frac{n}{f(n)}\right) + \sum_{\substack{n/(f(n)\log n) < p \leq n \\ p \mid b_{n \bmod p}}} v_p(G_n) \log p,$$

and the full conjecture $G_n = e^{o(n)}$ is equivalent to the last sum being $o(n)$ for some (equivalently, every) such f. Every prime in that sum has quotient $q = \lfloor n/p \rfloor < f(n) \log n$.

Proof. For $p > \sqrt{n}$ we have $p^2 > n$, so $\lfloor \log_p n \rfloor = 1$ and $v_p(G_n) \le 3$ by (2); each bad prime therefore contributes at most $3 \log p \le 3 \log n$. The number of primes up to $Y = n/(f(n) \log n)$ is $\pi(Y) \ll Y/\log Y \ll n/(f(n) \log^2 n)$, using $\log Y \ge \frac{1}{2} \log n$ for large n (as $f(n) \le \log n$). Multiplying by $3 \log n$ gives the first display. For the second display, split $\log G_n$ into: primes $p \le \sqrt{n}$ (Proposition 3: $O(\sqrt{n})$); companion-block primes $p > \sqrt{n}$ with $p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})$ (Proposition 59: $O(n^{2/3})$); the remaining bad primes $p > \sqrt{n}$, which by Lemma 55(iii) and Lemma 56 satisfy the lower-digit condition $p \mid b_{n \bmod p}$, split at $n/(f(n) \log n)$ by the first display. For the equivalence: one direction is the display; the other is immediate since any sub-sum of $\sum_p v_p(G_n) \log p$ is at most $\log G_n$. $\square$

Remark 61 (The conjecture lives at bounded quotients). Proposition 60 localizes the entire remaining difficulty to the top $O(\log \log n)$ dyadic blocks of primes: only $p \in (n/(f(n) \log n), n]$ matters, i.e., the quotient classes $q = \lfloor n/p \rfloor < f(n) \log n$. For each fixed q, the bad primes with quotient q form the shifted-divisibility set

$$\{\, p = (n - r)/q \text{ prime} : \ 0 \le r < p, \quad p \mid b_r \,\},$$

in which both the modulus and the index move with r. The case $q = 1$ is precisely the top-half radical estimate (48); the bounded-quotient cases are its mild deformations. Thus any method that resolves the top-half channel uniformly in $q < f(n) \log n$ for a single slowly growing f proves the full conjecture. Conversely, prime counting alone is already sufficient below this range, so no further structural information about small bad primes is needed anywhere in the argument.

The first-moment bound in Corollary 51 can be sharpened to a polylogarithmic exceptional set by exploiting the gap polynomials a second time, now as a *codegree* bound between nearby integers.

Lemma 62 (Common bad-prime codegree)**.** *Let $m, n \in (N, 2N]$ with $h = n - m \ge 1$, and let $P_0 \ge 1$. The number of primes $p > P_0$ that are simultaneously lower-digit bad for m and for n (i.e., $p \mid b_{m \bmod p}$ and $p \mid b_{n \bmod p}$) is at most $O(h \log N / \log P_0)$. In particular, if $P_0 \ge N^\delta$ for any fixed $\delta > 0$, the codegree is $O(h)$.*

Proof. We split the common bad primes into three classes.

Small primes ($P_0 < p \le h$): counting all of them trivially, $\pi(h) \ll h \ll h \log N / \log P_0$ (as $P_0 \le 2N$ in any nonvacuous application).

No wrap ($p > h$ and $(m \bmod p) + h < p$): here $n \bmod p = (m \bmod p) + h$, so the two zeros of b modulo p are at distance exactly h, and the gap implication (Lemma 7) gives $p \mid N_h(m \bmod p)$; polynomial periodicity $N_h(m \bmod p) \equiv N_h(m) \pmod{p}$ then gives $p \mid N_h(m)$. The gap polynomial has degree $3(h-1)$ in m (the tridiagonal determinant representation gives exactly this), so for $m \in (N, 2N]$, $|N_h(m)| \le (CN^3)^h$. Since $N_h(m) > 0$ (the associated tridiagonal matrix is positive definite at positive integers), the number of prime factors exceeding P_0 is at most $3h \log(CN)/\log P_0$.

Wrap ($p > h$ and $(m \bmod p) + h \ge p$): writing $r = m \bmod p$, the wrap condition says $r = p - j$ for some $1 \le j \le h$, i.e. $p \mid m + j$. Hence every wrapped prime divides the nonzero integer $\prod_{j=1}^h (m + j) \le (3N)^h$, so there are at most $h \log(3N)/\log P_0$ of them—no zero-set information is needed in this case. (If $r + h = p$ exactly, then $n \equiv 0$ and $b_0 = 1$, so p is not lower-digit bad for n; this boundary case is vacuous but in any event covered by $j = h$.)

Summing the three classes gives $O(h \log N / \log P_0)$. $\square$

Remark 63 (Wrap is essential). The no-wrap case alone does not exhaust the lemma: at $p = 37$ one has $\mathcal{Z}_{37} = \{17, 19\}$, and $m = 93$, $n = 128$ (so $h = 35$) give residues 19 and 17—not differing by h. Here $p \mid m + 18$, and the wrap class accounts for it. In the top-half application ($p \in (N, 2N]$, quotient 1) wrap cannot occur, since $n \le 2N < 2p$.

Theorem 64 (Polylogarithmic exceptional set)**.** *For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon\big((\log N)^2\big).$$

Proof. Fix $\varepsilon > 0$ and consider $n \in (N, 2N]$ with $\log G_n > \varepsilon n$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$. Each prime $p > \sqrt{n}$ satisfies $v_p(G_n) \le 3$ (since $v_p(d_n) = 3$). By Lemma 56, leading-digit bad primes ($p \mid b_{\lfloor n/p \rfloor}$ with $n \bmod p \notin \mathcal{Z}_p$) satisfy $v_p(G_n) = 0$. The remaining primes with $v_p(G_n) \ge 1$ are the lower-digit bad primes ($n \bmod p \in \mathcal{Z}_p$) and the *companion-block primes* ($D_q \equiv 0 \pmod p$ with $p \nmid b_r$, cf. Lemma 55). By Proposition 59, the total logarithmic weight of companion-block primes is $O(n^{2/3})$ at every index, hence $o(n)$. Hence $\log G_n \le O(\sqrt{n}) + 3\, t(n) \log n$, and $\log G_n > \varepsilon n$ forces

$$t(n) := \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\} \ge \frac{\varepsilon n}{4 \log n} \tag{49}$$

for N sufficiently large.

Partition $(N, 2N]$ into intervals J of length $Y = c_1 \varepsilon^2 N / \log N$, where $c_1 > 0$ is a small constant to be chosen. Write $\mathcal{E}_J = \{n \in J : \log G_n > \varepsilon n\}$ and $M = |\mathcal{E}_J|$.

For each prime $p \in (\sqrt{N}, 2N]$, let d_p count the elements of $\mathcal{E}_J$ for which p is lower-digit bad. By (49),

$$I := \sum_p d_p \ge \frac{\varepsilon N M}{4 \log N}.$$

The prime universe has size $L = \pi(2N) \le 4N / \log N$.

Upper bound on pair incidence. For $m, n \in \mathcal{E}_J$ with $m < n$, Lemma 62 gives at most $C(n - m)$ common lower-digit bad primes (taking $P_0 = N^\delta$ in the codegree lemma). Hence

$$\sum_p \binom{d_p}{2} = \sum_{\substack{m<n \\ m,n \in \mathcal{E}_J}} |\mathcal{P}_m \cap \mathcal{P}_n| \le C \sum_{\substack{m<n \\ m,n \in \mathcal{E}_J}} (n - m) \le C Y M^2.$$

Lower bound on pair incidence. By Cauchy–Schwarz applied to the d_p,

$$\sum_p \binom{d_p}{2} \ge \frac{1}{2}\left(\frac{I^2}{L} - I\right) \ge \frac{1}{2}\left(\frac{\varepsilon^2 N M^2}{64 \log N} - I\right).$$

Combining gives $CYM^2 \ge \varepsilon^2 N M^2 / (128 \log N) - I/2$. Since $I \le LM \le 4NM/\log N$ and $Y = c_1 \varepsilon^2 N / \log N$, for c_1 chosen smaller than $1/(128C)$, the YM^2 term cannot absorb the quadratic lower bound once $M > C'/\varepsilon^2$. Hence $M = O(1/\varepsilon^2)$.

There are $N/Y = O(\log N / \varepsilon^2)$ intervals. The total on $(N, 2N]$ is therefore $O(\log N / \varepsilon^4)$. Summing over $O(\log N)$ dyadic blocks $(N/2^{j+1}, N/2^j]$ gives the result. $\square$

Theorem 64 supersedes Corollary 51. The proof's critical-window estimate—every interval of length $\asymp \varepsilon^2 N / \log N$ in $(N, 2N]$ contains $O_\varepsilon(1)$ exceptional integers—directly gives the uniform local bound

$$\sup_{M \ge 0} \#\{M < n \le M + L : \log G_n > \varepsilon n\} \ll_\varepsilon L^{2/3} + 1,$$

by applying the critical-window argument to any interval $[M, M + L]$ with $N = M$ and $Y = c\varepsilon^2 M / \log M$. Dividing by L and letting $L \to \infty$ yields *upper Banach density zero* unconditionally, strengthening the harmonic-weight corollary 52.

9 Counting bad primes: the conditional result

We first sharpen the leading-digit incidence from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ via a dyadic argument combining divisor-counting with the sub-interval zero bound; this is unconditional and also strengthens the conditional theorem below.

Proposition 65 (Dyadic leading-digit bound)**.** *The leading-digit incidence L_N (counting pairs (n, p) with $n \in (N, 2N]$, $p > \sqrt{n}$ prime, and $p \mid b_{\lfloor n/p \rfloor}$) satisfies*

$$L_N \;=\; O\!\left(\frac{N^{10/7}}{\log N}\right).$$

Proof. For dyadic blocks $P = 2^j$, $Q = 2^k$ with $Q < P$ and $PQ \in [N/4,\, 4N]$, define

$$D(P, Q) = \#\{(p, q) : P < p \le 2P,\ Q < q \le 2Q,\ p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Sub-interval zero counting:* By Corollary 10, each prime $p \in (P, 2P]$ has at most $O(Q^{2/3})$ zeros of b_j in $(Q, 2Q]$. Summing over $\pi(2P)$ primes gives $D(P, Q) = O(P\,Q^{2/3}/\log P)$.
- *Divisor counting:* Since $\log b_q \le (q{+}1)\log 64$, the number of prime factors of b_q exceeding P is at most $\log b_q/\log P = O(q/\log P)$. Summing over $q \in (Q, 2Q]$ gives $D(P, Q) \le \sum_q O(q/\log P) = O(Q^2/\log P)$.

Hence $D(P, Q) \le \min(P\,Q^{2/3},\, Q^2)/\log P$.

Each pair (p, q) in $D(P, Q)$ contributes to at most $p \le 2P$ values of n with $\lfloor n/p \rfloor = q$, so the block's contribution to L_N is at most $D(P, Q) \cdot 2P \le 2\min(P^{4/3}\,N^{2/3},\ N^2/P)/\log P$ (using $Q = \Theta(N/P)$). The crossover is at $P_0 = N^{4/7}$. Summing $O(\log N)$ dyadic blocks $P \in [\sqrt{N},\, 2N]$:

$$L_N \;\le\; \sum_{P \le N^{4/7}} \frac{2P^{4/3}\,N^{2/3}}{\log P} \;+\; \sum_{P > N^{4/7}} \frac{2N^2}{P\log P} \;=\; O\!\left(\frac{N^{10/7}}{\log N}\right),$$

since both sums are geometric series dominated by the $P = N^{4/7}$ boundary term. □

Theorem 66. *Assume Hypothesis $\bar{Z}$. Then for a set of positive integers n of natural density* 1*,*

$$\log G_n = O(\sqrt{n}).$$

Proof. Fix N and $\omega(N) \to \infty$ arbitrarily slowly. As in Theorem 50, decompose $\log G_n = \sum_p v_p(G_n)\log p$. The small-prime contribution is $O(\sqrt{n})$. Split $B(n) = B_{\text{low}}(n) + B_{\text{lead}}(n)$, where B_{low} counts bad primes $p > \sqrt{n}$ with $n \bmod p \in \mathcal{Z}_p$ and B_{lead} counts those with $\lfloor n/p \rfloor \in \mathcal{Z}_p$.

Lower digit. Under Hypothesis $\bar{Z}$, set $M(t) = \sum_{p \le t} Z(p) \ll t/\log t$. Abel summation gives $\sum_{\sqrt{N} < p \le 2N} Z(p)/p = O(1)$, so $R_N = \sum B_{\text{low}}(n) = O(N)$. By Markov, $B_{\text{low}}(n) \le \omega(N)$ for all but $O(N/\omega(N)) = o(N)$ values.

Leading digit. As in the proof of Corollary 68, Hypothesis $\bar{Z}$ gives $L_N = O(N^{4/3}/\log N)$. By Markov with threshold $\omega(N)\,N^{1/3}/\log N$:

$$\#\big\{n \in (N, 2N] : B_{\text{lead}}(n) > \omega(N)\,N^{1/3}/\log N\big\} \le \frac{O(N^{4/3}/\log N)}{\omega(N)\,N^{1/3}/\log N} = O(N/\omega(N)) = o(N).$$

Conclusion. For all remaining n:

$$\log G_n \le O(\sqrt{n}) + 3\big(\omega(N) + \omega(N)\, N^{1/3}/\log N\big)\log n.$$

Since $\omega(N)\cdot N^{1/3} \ll N^{1/2}$ for any $\omega(N) \to \infty$ slowly enough ($\omega \le N^{1/6}$ suffices), the second term is $O(N^{1/2}) = O(\sqrt{n})$. □

Remark 67 (The leading-digit bottleneck). The gap between the conditional bound $O(\sqrt{n})$ (Theorem 66) and the conjectured $O(\log n)$ comes from the small-prime contribution (Proposition 3), which gives an unconditional floor of $O(\sqrt{n})$. Under Hypothesis $\bar{Z}$, the big-prime contribution is now absorbed into this floor, thanks to Proposition 65.

Heuristically, $B_{\text{lead}}(n) = O(1)$ because the leading digit $\lfloor n/p \rfloor$ lands in $\mathcal{Z}_p$ with probability $\le C/p$ for each prime, giving expected $B_{\text{lead}} = \sum C/p = O(1)$. Making this rigorous would require $D(P,Q) \ll Q/\log P$ (the random-divisibility prediction), improving the proven $D(P,Q) \le \min(P\,Q^{2/3},\, Q^2)/\log P$. The small-prime bound $O(\sqrt{n})$ itself would need a finer analysis of $v_p(G_n)$ for small primes.

Corollary 68 (Conditional exceptional set)**.** *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon(N^{1/3}).$$

Proof. Under Hypothesis $\bar{Z}$, $R_N = O(N)$. For the leading-digit incidence, Hypothesis $\bar{Z}$ also strengthens the dyadic bound: since each prime $p \in (P, 2P]$ contributes at most $Z(p)$ zeros to any sub-interval, $D(P,Q) \le \sum_{P<p\le 2P} Z(p) = O(P/\log P)$ (using $\sum_{p\le x} Z(p) = O(\pi(x))$). The crossover with the divisor bound $D(P,Q) \le Q^2/\log P$ shifts from $P_0 = N^{4/7}$ to $P_0 = N^{2/3}$, giving $L_N = O(N^{4/3}/\log N)$. The threshold $B(n) > \varepsilon N/(7\log N)$ is exceeded by at most

$$\frac{O(N) + O(N^{4/3}/\log N)}{\varepsilon N/(7\log N)} = O(N^{1/3}/\varepsilon).$$

Dyadic summation over $(N/2^{j+1}, N/2^j]$ gives the global bound. □

Under Hypothesis $\bar{Z}$, the exceptional set is not only sparse globally but cannot cluster in any interval:

Corollary 69 (Upper Banach density zero)**.** *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$, the set $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$ has upper Banach density zero:*

$$\lim_{L\to\infty} \sup_{M\ge 0} \frac{|E_\varepsilon \cap (M, M+L]|}{L} = 0.$$

Proof. For $I = (M, M+L]$ with $M > 2L$, every prime $p > n/2$ with $n \in I$ lies in $(M/2,\, M+L]$. Under Hypothesis $\bar{Z}$, each such prime contributes at most $Z(p)$ candidate bad integers to I, giving $\sum_{n\in I} B(n) \le \sum_{M/2<p\le M+L} Z(p) \ll \pi(M+L) = O(M/\log M)$. For $n \in I \cap E_\varepsilon$, the bound $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$ and $\log G_n > \varepsilon n$ force $B(n) \gg \varepsilon M/\log M$ for M large. Therefore $|I \cap E_\varepsilon| \ll_\varepsilon 1 = o(L)$. The case $M \le 2L$ follows similarly by covering $(0, 3L]$ with $O(\log L)$ dyadic intervals. □

Corollary 70. *The condition $Z(p) = o(p)$ is* necessary *for the conjecture: if $Z(p) \ge \varepsilon p$ for a positive proportion of primes, double-counting gives $\log G_n \ge c\varepsilon n$ for a positive proportion of n. Conversely, $Z(p) = o(p)$ implies $G_n = e^{o(n)}$ for density 1 (Theorem 50).*

Proof. For each prime $p \in (n/2, n]$ with $Z(p) \geq \varepsilon p$, Lemma 47 gives $v_p(G_n) \geq 1$ whenever $n - p \in \mathcal{Z}_p$. Suppose a proportion $\delta > 0$ of primes satisfy $Z(p) \geq \varepsilon p$. Summing over $n \in (N, 2N]$ and bad primes $p \in (n/2, n]$, the total incidence count is $\geq \delta\, \pi(2N) \cdot \varepsilon N/2 = \Omega(N^2/\log N)$. Dividing by N and applying Markov's inequality: a proportion $\geq \varepsilon\delta/4$ of integers n have $B(n) \geq \varepsilon\delta N/(4 \log N)$, giving $\log G_n \geq 3B(n) \log(N/2) \geq c\varepsilon\delta\, n$. $\square$

Remark 71 (The pointwise gap). The full conjecture $G_n = e^{o(n)}$ for *every* n is strictly stronger than the density-one statement. By Corollary 48, it is equivalent to the top-half radical estimate (48). The height bound gives $B(n) = O(n/\log n)$, but for an *adversarial* family (Z_p) with $|Z_p| \leq 1$, one may choose $Z_p = \{N - p\}$ for primes in $(N/2, N]$, producing $B(N) \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$. Thus the needed pointwise bound $B(n) = o(n/\log n)$ requires arithmetic information specific to the Apéry sequence, not just the cardinalities $Z(p)$.

By Corollary 48, every bad prime $p > n/2$ satisfies $p \mid b_n$. Hence the conjecture implies $\log \operatorname{rad}_{(n/2,n]}(b_n) = o(n)$: the large prime radical of b_n must be subexponential, even though $|b_n| = e^{\Theta(n)}$.

The Poisson model predicts that $B(n) \geq 1$ occurs infinitely often: $\mathbb{P}(B(n) \geq 1) \sim \log 2/\log n$, and $\sum (\log n)^{-1}$ diverges. This is compatible with the conjecture, since one bad prime contributes only $O(\log n)$ to $\log G_n$. To violate $G_n = e^{o(n)}$, one needs $B(n) \geq \varepsilon n/(6 \log n)$; the Poisson tail gives $\mathbb{P}(B(n) \geq k_n) \leq e^{-cn}$ for $k_n = \lceil \varepsilon n/(6 \log n) \rceil$, and the sum converges. Thus the model predicts: bad-prime occurrences are inevitable, but catastrophic pile-ups are finitely many.

10 Computational verification

We computed G_n exactly for $n = 1, \ldots, 1000$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	0.93
51–100	0.108	0.26
101–200	0.086	0.18
201–300	0.048	0.10
301–400	0.038	0.08
401–500	0.024	0.05
501–600	0.029	0.06
601–700	0.017	0.04
701–800	0.014	0.04
801–900	0.016	0.04
901–1000	0.013	0.03

A power-law fit gives $\log(G_n)/n \sim 5\, n^{-0.87}$, i.e., $\log G_n \sim 5\, n^{0.13}$, far stronger than the density-one bound $o(n)$ of Theorem 50.

The exponent 0.13 should be interpreted with caution: if $\log G_n \sim \kappa \log n$, the effective log-log slope is $1/\log n = 0.145$ at $n = 1000$, close to the fitted 0.13. A random-gcd model—assigning each prime $p \leq n$ an independent probability $\sim 1/p$ of dividing G_n—predicts $\mathbb{E}[\log G_n] \sim \log n$ by the weighted Mertens theorem, with standard deviation $\sim (\log n)/\sqrt{2}$ of the same order. The data thus cannot yet distinguish $\kappa \log n$ from $n^{0.13}$; the logarithmic prediction is arithmetically more natural.

10.1 Extended modular verification

For each n, define $B(n) = \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\}$, so that $\log G_n \le O(\sqrt{n}) + 3\,B(n)\log n$ (Section 7). The ratio $B(n) \cdot \log n/n$ governs $\log G_n/n$: whenever it tends to zero, $G_n = e^{o(n)}$. We computed $B(n)$ for all $n \le 100{,}000$ by modular arithmetic (iterating the Apéry recurrence $\bmod\, p$ for each prime $p \le 200{,}001$ and checking whether $n \bmod p \in \mathcal{Z}_p$).

n range	max $B(n)$	avg $B/T(n)$	max $B/T(n)$
1–100	3	0.038	0.260
101–500	4	0.013	0.124
501–1,000	4	0.006	0.036
1,001–2,000	4	0.003	0.027
2,001–5,000	4	0.001	0.015
5,001–10,000	5	0.0008	0.007
10,001–20,000	5	0.0004	0.004
20,001–50,000	6	0.0002	0.002
50,001–100,000	6	0.0001	0.001

Here $T(n) = n/\log n$, the density-one threshold. The ratio $\max B/T$ decreases monotonically. The overall maximum is $B(n) = 6$ at $n = 20{,}546$, giving the explicit bound $\log G_n/n \le 3 \cdot 6 \cdot \log(20{,}546)/20{,}546 + O(1/\sqrt{20{,}546}) = 0.009$, which confirms $G_n = e^{o(n)}$ at every index $n \le 100{,}000$. The Poisson prediction for the maximum is $\max_{n \le N} B(n) \approx 2\log\log N \approx 5.6$ at $N = 10^5$, consistent with the observed 6.

11 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [4] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base-p digits. Quantitatively, since $b_n \equiv \prod b_{d_i} \pmod{p}$ (Theorem 4), we have $p \nmid b_n$ iff every base-p digit of n avoids $\mathcal{Z}_p$. Hence

$$\#\{n < p^r : p \mid b_n\} = p^r - (p - Z(p))^r.$$

2. For the gap-2 polynomial $C_2(m) = P(m+1)$, the factorization $P(x) = (2x+1)(17x^2+17x+5)$ shows that gap-2 pairs can only occur at $m = (p-3)/2$ (when $2(m+1)+1 = 2m+3 \equiv 0$) or at roots of the quadratic $17x^2 + 17x + 5$ (discriminant -51; exists over $\mathbb{F}_p$ iff $\left(\frac{-51}{p}\right) = 1$). Empirically, all observed gap-2 pairs arise from the linear factor at $m = (p-3)/2$.

3. For $1 \le j \le p-2$, the Apéry number $b_j \bmod p$ admits a character-sum representation as a Greene [14] Gaussian hypergeometric ${}_4F_3$ over $\mathbb{F}_p$ with parameter $\chi = \omega^j$ (Teichmüller character). However, the Weil bound controls the archimedean size $|b_j|_\infty = O(p^{3/2})$ but does *not* bound $Z(p)$, since zeros of b_j modulo p are a p-adic event. The gap polynomial argument above is specifically adapted to this p-adic zero-count problem.

4. Malik–Straub [20] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [23] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings.

5. In the standard toric coordinate, the truncated polynomial $H_p(t) = \sum_{j=0}^{p-1} b_j\, t^j \in \mathbb{F}_p[t]$ is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in $\mathbb{F}_p$ are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic $Z(p)$ counts vanishing *coefficients* of H_p, equivalently the vanishing finite Mellin coefficients $b_j = -\sum_{t\in\mathbb{F}_p^\times} H_p(t)\, t^{-j}$ for $1 \le j \le p-2$. Thus $Z(p)$ is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent F-crystal theory do not by themselves bound $Z(p)$; they control the root locus, not the Fourier support. A bounded-complexity geometric lift *does* exist: the lisse sheaf $\mathcal{V}_\ell = R^2 f_* \mathbb{Q}_\ell$ on the smooth locus of the K3 pencil (rank 22, conductor independent of p) satisfies $\mathrm{Tr}(\mathrm{Frob}_t \mid \mathcal{V}_\ell) \equiv A_p(t) \pmod p$ by the Hasse–Witt congruence. The Mellin sum $\mathscr{M}_{p,j} = \sum_t \tau_p(t)\, \omega_p(t)^{-j}$ is then realized on $R\Gamma_c(U_{\overline{\mathbb{F}}_p}, \mathcal{V}_\ell \otimes \mathcal{L}_{\omega_p^{-j}})$ with bounded rank and conductor. However, the event $b_j \equiv 0 \pmod p$ is the *same-characteristic* divisibility $\mathfrak{p}_p \mid \mathscr{M}_{p,j}$, a crystalline condition outside the scope of ℓ-adic equidistribution (Forey–Fresán–Kowalski [13]). The missing theorem is a *crystalline Mellin anti-concentration* estimate controlling $\#\{j : p \mid \mathscr{M}_{p,j}\}$.

6. **Jacobi counterexample: bounded complexity is not enough.** Fix an integer $d \ge 2$. For every prime $p \equiv 1 \pmod d$, the rank-1 Kummer sheaf $\mathcal{L}_\psi$ on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ attached to $\psi = \omega_p^{-(p-1)/d}$ (order d) has Mellin transforms $M_j = J(\omega_p^{-j}, \psi)$. The Gross–Koblitz/Stickelberger valuation $v_p(g(\omega^{-k})) = k/(p-1)$ and the Jacobi factorization $J(\alpha, \beta) = g(\alpha)g(\beta)/g(\alpha\beta)$ give $v_p(M_j) = 1$ whenever $j \ge (p-1)(d-1)/d$, yielding at least $(p-1)/d - 1$ characters with $p \mid M_j$. Thus a sheaf of *fixed* rank, conductor, and Weil bound $O(\sqrt{p})$ can have $\gg p$ bad tame characters. This shows that bounded cohomological complexity alone cannot control $Z(p)$; any proof of Hypothesis Z must use Apéry-specific arithmetic.

7. **Sufficient criteria for Hypothesis Z.** Two clean algebraic conditions, either of which would imply $Z(p) = O(1)$:

   (a) Bounded tame-Hasse polynomial. Suppose there exist an absolute integer D and, for each sufficiently large p, a nonzero Laurent polynomial $Q_p \in \mathbb{F}_p[X, X^{-1}]$ of degree at most D and units $u_{p,j} \in \mathbb{F}_p^\times$ such that $\mathscr{M}_{p,j} \equiv u_{p,j}\, Q_p(\omega_p^j) \pmod p$ for every nontrivial tame character. Then $Z(p) \le 2D$, since Q_p (after clearing negative exponents) has at most $2D$ roots.

   (b) Bounded nonordinary twists. Suppose for each p the set E_p of tame characters χ for which the twisted crystal $H^1_{\mathrm{rig},c}(U, \mathcal{V} \otimes \mathcal{K}_\chi)$ fails to have a unique unit-root eigenvalue satisfies $|E_p| \le C$. Then $Z(p) \le C + O(1)$, since a unique unit root forces $v_p(\mathscr{M}_{p,j}) = 0$.

   Criterion (a) is the algebraic version of Hypothesis Z; criterion (b) is its p-adic-slopes formulation. The Jacobi counterexample above shows that neither criterion holds for general bounded-complexity sheaves.

8. **Linear-complexity obstruction.** The Sym^2 factorization $H_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2$ with B_p squarefree of degree $\lfloor (p-1-2\varepsilon_p)/2 \rfloor$ implies that H_p has at most $\lfloor (p-1)/2 \rfloor + 2$ distinct zeros in $\mathbb{F}_p^\times$. Choosing a primitive root g and forming the canonical character interpolation $Q(X) = -\sum_{r=1}^{p-2} H_p(g^r)\, X^r$, whose values $Q(g^j) = b_j$ for $1 \le j \le p-2$, we find that Q has at least $(p-5)/2$ nonzero coefficients. Any Laurent polynomial of degree at most D representing the same function on the character group must have $2D + 1 \ge (p-5)/2$, giving $D \ge (p-7)/4$. Thus criterion (a) cannot be satisfied in its direct (unnormalized) form: a nontrivial Gauss/Jacobi normalization of the Mellin coefficients is necessary before bounded tame degree could emerge.

9. **Palindromic symmetry.** For $0 \le j \le p-1$ and $0 \le k \le p-1-j$, the identity $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod p$ holds (each of the k falling factors $p{-}1{-}j{-}i \equiv -(j{+}1{+}i)$). When $k \le j$, $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$; when $k > j$, Lucas's theorem gives $\binom{p-1-j+k}{k} \equiv 0$ (since $p{-}1{-}j{+}k \ge p$ and $\binom{k-j-1}{k} = 0$). Therefore each term of $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$ with $k > j$ vanishes, and for $k \le j$ the $(-1)^{2k}$ signs cancel, giving $b_{p-1-j} \equiv b_j \pmod p$. The extra terms of b_j when $j > (p{-}1)/2$ (i.e., $k > p{-}1{-}j$) likewise vanish: $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$ since $j{+}k{-}p < k$. Equivalently, $H_p(t)$ is palindromic: $t^{p-1} H_p(1/t) \equiv H_p(t)$. This explains the observed pairing: whenever $b_j \equiv 0$, also $b_{p-1-j} \equiv 0$, so zeros of b_j come in pairs $\{j,\, p-1-j\}$ (except at the self-paired center $j = (p-1)/2$, which vanishes only at non-ordinary primes). Proposition 58 proves the stronger statement that *every* solution of the Apéry recurrence is palindromic modulo p, in particular the companion solution a. The "pair count" $K(p) = Z(p)/2$ (or $(Z(p)-1)/2$ at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 36).

10. **Sym^2 squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [25]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [7] proved the mod-p squareness rigorously: for every prime $p \ge 5$, setting $\varepsilon_p = \frac{1}{2}(1 - \left(\frac{-6}{p}\right))$, there exists a unique (up to sign) polynomial $B_p \in \mathbb{F}_p[t]$ such that
$$H_p(t) \;=\; \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$
where $\Delta(t) = t^2 - 34t + 1$ is the conifold discriminant and B_p is squarefree and coprime to Δ. The proof uses the explicit identity $F(t(x)) = (1+x)\, h(x)^2$ between the Apéry and Franel generating functions, the p-Lucas property of both sequences, and a quadratic-descent argument through the covering $\mathbb{F}_p(x)/\mathbb{F}_p(t)$ where $t = x(1-8x)/(1+x)$. The deck-transformation character $(-6/p)$ determines whether $\varepsilon_p = 0$ or 1:
$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \qquad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$
Equivalently, $\varepsilon_p = 0$ iff $p \equiv 1, 5, 7, 11 \pmod{24}$. These two cases occur with equal density 1/2 by Dirichlet's theorem.

Splitting the mean $Z(p)$ by residue class modulo 8 (and modulo 24) yields no significant difference (mean ≈ 1.00 in each class, $n \ge 2000$ primes per class up to 10^5); the squareness structure does not directly influence $Z(p)$, consistent with $Z(p)$ counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure: b_j is (up to the Δ^{ε_p} correction) the j-th coefficient of B_p^2, i.e., $b_j \equiv \sum_k \beta_k\, \beta_{j-k} \pmod p$ where β_k denotes the coefficients of B_p. The vanishing $b_j \equiv 0$ thus requires cancellation across $\sim p/2$ convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree $(p-1)/2$ over $\mathbb{F}_p$, for $p \in \{101, 503, 1009, 5003\}$) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean ≈ 1, variance ≈ 1, total variation distance to Poisson(1) equal to 0.018). However, the observed $Z(p)$ distribution over 667 primes $p \le 5000$ is *not* Poisson(1) (TV = 0.43); instead it matches $2 \cdot$ Poisson(1/2) with TV = 0.023. The explanation is the palindromic pairing: since $b_{p-1-j} \equiv b_j$, zeros come in pairs $\{j,\, p{-}1{-}j\}$ and $Z(p)$ is almost always even. Denoting by $Z_{\mathrm{pair}}(p)$ the number of zero pairs, the data show $Z_{\mathrm{pair}} \sim$ Poisson(1/2), so $Z(p) = 2Z_{\mathrm{pair}}(p) \sim 2 \cdot$ Poisson(1/2). Thus squareness explains the $O(1)$ mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts $\max_{p \le N} Z(p) \sim 2 \log\log N$

via extreme-value theory, strongly suggesting that $Z(p)$ is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of B_p imposes a bounded *average* (Hypothesis $\bar{Z}$) remains the central open question.

The factorization does *not* improve the divisor-counting bound $D(P,Q) \le Q^2/\log P$. In the regime $P \ge Q^{4/3}$ where the divisor estimate is active, one has $q < p/2$, so the coefficient b_q mod p lies in the formally free triangular range of the square root: the map $(\beta_1,\dots,\beta_q) \mapsto (b_1,\dots,b_q)$ mod p is a triangular automorphism with Jacobian 2^q, hence the single equation $b_q \equiv 0$ imposes no algebraic constraint beyond fixing one coefficient of B_p. Thus the Sym^2 structure gives the correct *random model* (uniform square root $\Rightarrow$ Poisson(1/2)) but not the missing *distribution theorem* for the deterministic Apéry square root.

11. **Toric framework.** The Laurent polynomial $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$ on $(\mathbb{F}_p^\times)^4$ has Newton polytope $\Delta = \mathrm{conv}(\{0\} \cup \mathrm{Supp}(f))$, a 4-dimensional pyramid over the regular octahedron in $\mathbb{R}^3$. Its f-vector is $(7, 18, 20, 9)$, its normalized volume is $4! \cdot \mathrm{vol}(\Delta) = 8$, and it is Kouchnirenko nondegenerate for every prime $p \ge 5$: for each of the 27 nonempty faces σ not containing the origin, the toric critical system $\{x_i \partial f_\sigma/\partial x_i = 0\}$ has no solution in $(\mathbb{F}_p^\times)^4$. For the Adolphson–Sperber twisted Hodge polygon with w-Kummer twist $\chi_w = \omega^{-j}$ (character parameter $a = j/(p-1)$), the slopes are a, $a+1$, $a+2$, $a+3$ with multiplicities $1, 3, 3, 1$, confirming that the first slope equals a for every $a \in \{k/100 : 0 \le k \le 99\}$. By Newton-above-Hodge (Adolphson–Sperber [2]), the p-adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The $Z(p)$ bound thus reduces to proving ordinarity for all but $O(1)$ twist parameters j.
12. **Branch obstruction.** The $p-1$ Kummer twists χ^{-j} $(j = 0, \dots, p-2)$ are distinct characters of conductor p. In Iwasawa-theoretic coordinates $\chi^j = \omega^a \langle\cdot\rangle^s$ with $a = j \bmod (p-1)$, the horizontal family $\{j : 0 \le j < p-1\}$ hits *each* of the $p-1$ branches ω^a exactly once. Weierstrass preparation and bounded-λ rigidity control zeros *within one branch* (the vertical direction j, $j + (p-1)$, $j + (p-1)p, \dots$, where Dwork congruences live) but not *across* branches. Thus there is no p-adic analytic function of one variable whose zero count majorizes $Z(p)$. Any "rigidity" proof of Hypothesis $\bar{Z}$ would need a mechanism not supplied by existing Iwasawa–Dwork theory.
13. Any estimate $\log G_n = o(n)$—whether $O(\log n)$, $O(\sqrt{n})$, or $O(n^{0.13})$—yields $\gamma = 0$ in the exponent $G_n = e^{\gamma n + o(n)}$ and therefore does not improve the Rhin–Viola irrationality measure $\mu(\zeta(3)) \le 5.513$ [24], which depends only on the exponential growth rate $|b_n\zeta(3) - a_n| = e^{-(\log\alpha + o(1))n}$. An improvement would require an exponentially large common divisor, which is not predicted by any current model.
14. **Comparison with previous work.** Classical refinements of Apéry's method (Beukers [4], Rhin–Viola [24], Zudilin [27]) exhibit explicit common factors of the hypergeometric coefficients to improve denominator estimates and irrationality measures, but always uniformly for every index. They do not provide exceptional-set estimates for the *actual* reduced gcd G_n. A separate body of work, based on the Subspace Theorem, proves $\gcd(a^n - 1, b^n - 1) = e^{o(n)}$ pointwise for multiplicatively independent a, b (Bugeaud–Corvaja–Zannier [6]); these results apply to constant-coefficient linear recurrences via Binet representations, but the Apéry recurrence's polynomially varying coefficients preclude the multiplicative-independence hypothesis. To our knowledge, no previous work gives a quantitative power-saving exceptional-set estimate for gcd cancellation in a variable-coefficient holonomic recurrence.

15. **The remaining open problem.** The companion-block height bound (Proposition 59) shows that primes $p > \sqrt{n}$ with $D_q \equiv 0 \pmod{p}$ contribute $O(n^{2/3})$ to $\log G_n$ at *every* index. After removing this channel, the conjecture $G_n = e^{o(n)}$ for all n is equivalent to the *pointwise radical estimate*

$$\sum_{\substack{p>\sqrt{n} \\ p \mid b_{n \bmod p}}} \log p \;=\; o(n).$$

Theorem 64 proves this for all but $O_\varepsilon((\log N)^2)$ integers in $[1, N]$. However, the codegree method (Lemma 62) controls primes *shared* between pairs of integers, and does not bound the total weight of "private" primes at a single index. Closing the gap requires a one-point rainbow estimate: an argument that one fixed integer cannot simultaneously be bad at too many distinct primes. By the quotient reduction (Proposition 60), the estimate is only needed for the primes $p > n/(f(n) \log n)$ with f growing arbitrarily slowly—equivalently, uniformly over the quotient classes $q = \lfloor n/p \rfloor < f(n) \log n$—since prime counting alone handles the range $\sqrt{n} < p \le n/(f(n) \log n)$.

12 The second-moment approach

The codegree method of Section 8 exploits primes shared between pairs of integers to prove a density-one theorem. To close the remaining gap—proving $W(n) = o(n)$ for *every* n—we study the *second moment* of the bad-prime weight function.

12.1 The criterion

For a dyadic shell $I_N = (N, 2N]$, define

$$B(m) \;=\; \sum_{\substack{p>\sqrt{m} \\ p \mid b_{m \bmod p}}} \log p, \qquad M_2(N) \;=\; \sum_{m \in I_N} B(m)^2.$$

Proposition 72 (Second-moment criterion)**.** *If $M_2(N) \le C \cdot N^{2-\delta}$ for some $C > 0$ and $\delta > 0$, then $W(n) = o(n)$ for all n. More precisely, for every $n \in I_N$,*

$$W(n) \;\le\; B(n) \;\le\; \sqrt{M_2(N)} \;\le\; C^{1/2} \cdot N^{1-\delta/2}.$$

Proof. $B(n)^2$ is one nonneg. term of the sum $M_2(N)$, hence $B(n) \le \sqrt{M_2(N)}$. □

In particular, $M_2(N) = O(N \cdot (\log N)^A)$ for any fixed A gives $B(n) = O(\sqrt{N} \cdot (\log N)^{A/2}) = o(N)$.

Remark 73 (Shifted-gcd identity). The bad-prime weight admits a combinatorial reformulation. By the injectivity of $p \mapsto n \bmod p$ on primes $p > \sqrt{n}$ (two such primes giving the same residue would force their product $\le n$),

$$W(n) \;=\; \sum_{0<r<n} \log \operatorname{rad}_{>\max(\sqrt{n},\, r)} \gcd(n-r,\, b_r).$$

Each inner radical is 1 or a single prime $p = n - r$. For the top-half block $p > n/2$, the condition becomes "$n - r$ is prime and $(n - r) \mid b_r$."

Proposition 74 (Hard-core reduction)**.** *For each bad prime $p > \sqrt{n}$, write $n = q_p p + r_p$, $s_p = \min(r_p, p-1-r_p)$, $h_p = |p-1-2r_p|$. Let $1 \le Q, R, H < n/2$. Then the total weight of primes satisfying at least one of $q_p > Q$, $s_p \le R$, $h_p \le H$ is $O(n/Q + (R+H)\log n)$. In particular, with $Q = R = H = \sqrt{n/\log n}$, all "easy channels" contribute $O(\sqrt{n \log n})$; the remaining hard core consists of primes $p \gg \sqrt{n \log n}$ with zero positions far from both endpoints and the reflection centre.*

Proof. Large quotient ($q_p > Q$): then $p < n/Q$, so $\sum \log p \le \vartheta(n/Q) \ll n/Q$. Near endpoint ($s_p \le R$): by the injectivity of $p \mapsto n \bmod p$, at most two primes $> \sqrt{n}$ contribute for each value of s_p. Near centre ($h_p \le H$): the identity $(2q_p + 1)p = 2n + 1 \pm h_p$ forces p to divide a fixed integer of size $O(n)$, giving $O(1)$ primes per value of h_p. □

Remark 75 (The vertical main term is harmless). Write $W(n) = M(n) + \mathcal{E}(n)$ where $M(n) = \sum_{p>\sqrt{n}} Z(p)\,(\log p)/p$. The unconditional bound $Z(p) \le Cp^{2/3}$ gives $M(n) \ll \sum_{p \le n} p^{-1/3} \log p \ll n^{2/3}$. The entire problem is the oscillatory error $\mathcal{E}(n)$, which at each prime is $(\log p) \cdot (1_{n \bmod p \in \mathcal{Z}_p} - Z(p)/p)$.

12.2 Empirical verification

The following table, computed for all dyadic shells with $64 \le N \le 4096$ using the full zero-set data for all primes $p \le 20{,}000$, gives:

N	$M_2(N)$	M_2/N	$M_2/(N \log^2 N)$	$\max B(m)$
64	672	10.5	0.605	10.4
128	1,684	13.2	0.560	15.9
256	4,376	17.1	0.555	15.8
512	11,896	23.2	0.598	22.3
1024	28,632	28.0	0.583	21.7
2048	69,226	33.8	0.582	27.1
4096	166,671	40.7	0.587	32.1
8192	421,868	51.5	0.634	37.2

The ratio $M_2(N)/(N \log^2 N)$ stabilizes near 0.59, strongly suggesting

$$M_2(N) \;\sim\; C_2 \cdot N \cdot (\log N)^2, \qquad C_2 \approx 0.59. \tag{50}$$

This gives $B(n) \le \sqrt{0.59\,N}\,\log N \approx 0.77\,\sqrt{N}\,\log N = o(N)$.

12.3 CRT expansion

Expand M_2 dyadically. For distinct primes $p, q \in (P, 2P]$ with $P > \sqrt{N}$, write $\Omega_p = \{m \in I_N : m \bmod p \in \mathcal{Z}_p\}$. Then

$$M_2(N) \;=\; \underbrace{\sum_p (\log p)^2\,|\Omega_p|}_{\text{diagonal}} \;+\; \underbrace{\sum_{p \ne q} \log p\,\log q\,|\Omega_p \cap \Omega_q|}_{\text{off-diagonal}}.$$

Diagonal. $|\Omega_p| = Z(p) \cdot N/p + O(Z(p))$, so

$$\text{diag} \;=\; N \sum_p \frac{Z(p)\,(\log p)^2}{p} \;+\; O\Big(\sum_p Z(p)\,(\log p)^2\Big).$$

Under the Poisson model $\mathbb{E}[Z(p)] = 1$, the main term is $\sim N\cdot\sum_{P<p\le 2P}(\log p)^2/p \sim N\cdot(\log P)^2\cdot\log 2$. Summing dyadically $P = \sqrt{N}, 2\sqrt{N}, \ldots, N$ gives diagonal $\sim \frac{1}{2}\,N\,(\log N)^2$, matching $\approx 63\%$ of the empirical (50).

Off-diagonal. Since $p, q > \sqrt{N}$ we have $pq > N$, so each CRT class $r_p \pmod p$, $r_q \pmod q$ contains at most one element of I_N. The CRT main term of the off-diagonal is $\sim N\big(\sum Z(p)\log p/p\big)^2 \sim \frac{1}{4}\,N\,(\log N)^2$.

The predicted constant. Under the independent random model—each prime p hits a uniform m with probability $\sim 1/p$, independently across primes—the second moment of $B(m)$ at one integer is $\mathbb{E}[B(m)^2] = (\mathbb{E}[B])^2 + \mathrm{Var}(B)$. By weighted Mertens, $\mathbb{E}[B] = \sum_{\sqrt{m}<p\le m}(\log p)/p = \frac{1}{2}\log m + O(1)$, and

$$\mathrm{Var}(B) \;=\; \sum_{\sqrt{m}<p\le m} \frac{(\log p)^2}{p}\left(1-\frac{1}{p}\right) \;=\; \tfrac{3}{8}\,(\log m)^2 + O(\log m).$$

Therefore

$$\mathbb{E}[B(m)^2] \;=\; \big(\tfrac{1}{4}+\tfrac{3}{8}\big)(\log m)^2 + O(\log m) \;=\; \tfrac{5}{8}\,(\log m)^2 + O(\log m), \tag{51}$$

predicting $M_2(N) \sim \frac{5}{8}\,N\,(\log N)^2$. The diagonal constant is 3/8 (matching the observed $\mathrm{Diag}/N\log^2 N \approx 0.37$) and the off-diagonal contributes 1/4. The empirical ratio $M_2/(N\log^2 N) \approx 0.59$ is consistent with convergence to $5/8 = 0.625$, with lower-order boundary effects.

Centered variance and cross-prime covariance. The centered dispersion $V^\circ = M_2 - M_1^2/N$ decomposes as

$$V^\circ \;=\; \underbrace{\sum_p (\log p)^2\,|\Omega_p|\big(1-|\Omega_p|/N\big)}_{\le\,\mathrm{Diag}} \;+\; \underbrace{\sum_{p\ne q}\log p\,\log q\,\big(|\Omega_p\cap\Omega_q| - |\Omega_p|\,|\Omega_q|/N\big)}_{E_N^\circ}. \tag{52}$$

The first sum is at most the diagonal S (since $|\Omega_p|/N \ge 0$). The term E_N° is the centered cross-prime covariance: it vanishes exactly when the events $\{m \in \Omega_p\}$ are pairwise uncorrelated. The computed values are:

N	$M_2/N\log^2 N$	$\mathrm{Diag}/N\log^2 N$	V°/S	E_N°
256	0.556	0.359	1.014	+110
512	0.597	0.376	1.010	+215
1024	0.582	0.375	0.981	−152
2048	0.581	0.370	0.992	+7
4096	0.588	0.371	0.995	+41
8192	0.634	0.387	0.995	−188
16384	0.669	0.400	0.995	−922
32768	0.646	0.394	0.988	−14065

The ratio V°/S converges to 1 from below for $N \ge 1024$, with $V^\circ/S < 1$ at every scale tested beyond 512. The negative sign of E_N° for large N reflects genuine *anti-correlation* among the CRT-constrained incidence events: at $N = 1024$, over 96% of cross-prime covariance pairs $C_{p,q}$ are negative. This is consistent with a weak form of negative dependence arising from the partial-period CRT constraint. The magnitude $|E_N^\circ|$ is $O(\sqrt{N}\,(\log N)^2)$, so $|E_N^\circ|/S \to 0$ at rate $\sim 1/\sqrt{N}$.

Three-model comparison. To test whether the observed $V^\circ/S \approx 1$ is an Apéry-specific phenomenon or a generic consequence of the cardinality and reflection structure, we compare three models on each dyadic block I_N: (1) the actual Apéry zero-sets; (2) uniformly random reflected subsets of $\mathbb{F}_p^*$ with the same cardinalities $|Z_p|$; (3) random reflected subsets with cardinalities drawn from $2 \cdot \mathrm{Poisson}(1/2)$. Over 20 independent trials of each random model and $N \in \{256, 512, 1024, 2048\}$, all three models yield $V^\circ/S = 1.00 \pm 0.05$, with no statistically significant difference between the Apéry data and either random model. This rules out any detectable cross-prime alignment or repulsion specific to the Apéry sequence.

Remark 76 (Boundary correction). The naïve CRT main term $Z(p)Z(q)N/(pq)$ overestimates the random prediction for primes near $\sqrt{N}$ or N, because the admissible interval $J_{p,q} = J_p \cap J_q$ may be shorter than N. After replacing N by $L_{p,q} = |J_{p,q}|$ in the main term, the boundary-corrected CRT residual is positive and $O(\sqrt{N}\,(\log N)^2)$—the apparent negative sign in the uncorrected decomposition is entirely a boundary artifact.

12.4 The no-go model

Define the *adversarial zero-sets*: for one fixed $m_0 \in I_N$, set $\mathcal{Z}_p^* = \{m_0 \bmod p,\ p-1-(m_0 \bmod p)\}$. Then $|\mathcal{Z}_p^*| = 2$ (satisfying reflection), yet $m_0 \in \Omega_p^*$ for *every* prime $p > \sqrt{N}$, giving $B^*(m_0) = \sum_{p>\sqrt{N}} \log p = n - o(n)$. In particular:

- The adversarial model satisfies $Z^*(p) = 2$, $Z^*_{\mathrm{pair}}(p) = 1$, and all proved vertical bounds.
- Yet $M_2^*(N) \geq B^*(m_0)^2 \geq c\,n^2$, which is $\Omega(N^2)$, not $O(N)$.

Conclusion: *No proof using only vertical information (the sizes $Z(p)$, the reflection symmetry, nearest-neighbor exclusion, and gap-polynomial bounds) can establish $W(n) = o(n)$ for all n.* The missing theorem must use cross-prime *horizontal* information: the locations of the zeros $\mathcal{Z}_p$ across different primes p.

12.5 The Fourier form of the CRT error

Define the modular Fourier transform of $\mathcal{Z}_p$:

$$S_p(a) \;=\; \frac{1}{p} \sum_{r \in \mathcal{Z}_p} e\!\left(\frac{ar}{p}\right).$$

Then $S_p(0) = Z(p)/p$ and, by the palindromic pairing, $e(a(p-1)/(2p))\,S_p(a)$ is real-valued for every a. Fourier inversion gives

$$E_N(p,q) \;=\; \sum_{\substack{(a,b)\neq(0,0)\\ 0\leq a<p,\ 0\leq b<q}} S_p(a)\,S_q(b) \sum_{m\in I_N} e\!\left(\frac{am}{p} + \frac{bm}{q}\right).$$

The inner sum is a standard exponential sum $|K_N(\alpha)| \leq \min(N, 1/\|\alpha\|)$ where $\alpha = a/p + b/q$.

Applying the classical additive large sieve to bound $\sum_{p\neq q} |E_N(p,q)|$ gives $O(N^2/\log N)$ at the top prime scale $p \sim N$—one full logarithm larger than the threshold $o(N^2/\log^2 N)$ required for $M_2 = O(N(\log N)^A)$. Thus the classical large sieve does not close the gap.

12.6 The dispersion target

Define the centered dispersion

$$V^\circ(P,N) \;=\; \sum_{m\in I_N} \left| B_P(m) - \frac{S(P,N)}{N} \right|^2,$$

where $B_P(m) = \sum_{P<p\leq 2P} 1_{\Omega_p}(m)$ and $S(P,N) = \sum_p |\Omega_p|$.

Hypothesis 77 (AP–BDH dispersion)**.** $V^\circ(P,N) \ll N^{o(1)} \cdot S(P,N)$.

Proposition 78. *Hypothesis 77 together with the unconditional bound $Z(p) = O(p^{2/3})$ implies $W(n) = o(n)$ for all n.*

Proof. By hypothesis, for every $m \in I_N$,

$$B_P(m) \le \frac{S(P,N)}{N} + \sqrt{V^\circ(P,N)} \le \frac{S(P,N)}{N} + N^{o(1)}\sqrt{S(P,N)}.$$

By the zero-count bound, $S(P,N) \le (N/P+1) \cdot \sum_{P<p\le 2P} Z(p) \ll NP^{2/3}/\log P$. Therefore $B_P(m) \ll N^{o(1)} \cdot (NP^{2/3}/\log P)^{1/2}$. The weighted sum satisfies $W_P(m) \le B_P(m) \cdot \log(2P)$. Summing over $O(\log N)$ dyadic blocks gives

$$W(n) \ll N^{o(1)} \cdot N^{1/2} \cdot N^{1/3} = N^{5/6+o(1)} = o(N). \qquad \square$$

The dispersion hypothesis is a cross-prime equidistribution theorem stating that the Apéry zero-sets $\mathcal{Z}_p$ do not systematically align at one index across different primes. The no-go model of Section 12.4 shows that this hypothesis genuinely requires arithmetic input about the Apéry family, beyond what any vertical bound can provide.

Remark 79 (Covariance sign distribution). At $N = 1024$, more than 93% of the $\binom{113}{2} = 6{,}328$ cross-prime covariance terms $C_{p,q}$ are negative. This negativity has a clean structural explanation: for primes $p, q > \sqrt{N}$, the Poisson parameter $\lambda_{p,q} = A_pA_q/N < 1$, so $C_{p,q} < 0$ precisely when the joint count $J_{p,q}$ vanishes. In the sparse regime, $J_{p,q} = 0$ for $e^{-\lambda} \approx 96\%$ of pairs, while the rare pairs with $J_{p,q} = 1$ have $C_{p,q} = 1 - \lambda_{p,q} \approx 1$. Over a complete period $\pmod{pq}$, the average of $C_{p,q}$ is exactly zero: many small negatives are balanced by few large positives. Consequently, the high percentage of negative signs does *not* imply negative dependence and cannot be used directly. The aggregate off-diagonal $E^\circ = \sum_{p\ne q} w_pw_qC_{p,q}$ oscillates in sign: positive at $N = 1024, 4096, 8192$ and negative at $N = 2048$ (Table 1). The relevant theorem is the aggregate *sub-Poisson collision* estimate $|E^\circ| = o(S)$, which requires cancellation in the signed sum rather than a universal sign or a factored negative-dependence property. Neither the Dubhashi–Ranjan negative-lattice theorem (designed for balls-into-bins, not partial-period CRT) nor Janson's inequality (which requires nonnegativity of intersection parameters) applies to this partial-period setting.

Remark 80 (The dispersion route bypasses $\bar{Z}$). Hypothesis 77 does *not* require Hypothesis $\bar{Z}$. The adversarial singleton model $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1-(m_0 \bmod p)\}$ satisfies $\bar{Z}$ in its strongest form ($Z(p) = 2$ for all p), yet violates the dispersion hypothesis. Conversely, the dispersion hypothesis with $Z(p) = O(p^{2/3})$ already gives $W(n) = o(n)$, even without any average bound on $\sum Z(p)$. This makes the dispersion route *strictly stronger* than the $\bar{Z}$ route: it is both sufficient on its own and robust against arbitrary zero-count fluctuations. The three-model comparison confirms that $V^\circ/S \approx 1$ empirically, matching the prediction of the independent random model.

Remark 81 (Martingale decomposition). Order the active primes as $p_1 < \cdots < p_K$ and define the Doob martingale $M_k = \mathbb{E}[B \mid X_{p_1}, \ldots, X_{p_k}]$. The increments $D_k = M_k - M_{k-1}$ are orthogonal: $\mathrm{Var}(B) = \sum_k \mathbb{E}[D_k^2]$. However, $D_k = (X_{p_k} - \pi_k)\Delta_k$ where the "predictive cascade" $\Delta_k = w_{p_k} + \sum_{j>k} w_{p_j}(\mathbb{P}(X_{p_j}{=}1 \mid \mathcal{F}_{k-1}, X_{p_k}{=}1) - \mathbb{P}(X_{p_j}{=}1 \mid \mathcal{F}_{k-1}, X_{p_k}{=}0))$ contains all future primes. Under independence $\Delta_k = w_{p_k}$ and the variance equals the diagonal. For the Apéry measure on a partial CRT period, revealing that $m \bmod p_k$ falls in a zero can nearly identify m (when $A_{p_k} \approx 1$), making most subsequent indicators nearly deterministic. Controlling the cascade $\sum \mathbb{E}[\pi_k(1-\pi_k)\Delta_k^2] \le (1+o(1))\sum w_{p_k}^2 \alpha_{p_k}$ is equivalent to the dispersion inequality itself.

Remark 82 (Concentration and the all-n gap). The L^2-to-uniform upgrade (Lemma 84) gives $\max|B(m)-\mu| \le \sqrt{V^\circ}$. If $V^\circ \le C\,S$ for some constant C, then $\sqrt{V^\circ} = O(\sqrt{N}\,(\log N)^{3/2}) = o(N/\log N)$, so the all-$n$ theorem $W(n) = o(n)$ follows without any higher-moment, sub-Gaussian, or negative-association input (Corollary 85). Empirically, the actual concentration is much stronger: $\max B(m)/(N/\log N) \approx 0.07$ at $N = 4096$ (with $\max B \approx 36$ vs. $N/\log N \approx 492$), and the maximum $|B(m)-\mu|/\sigma$ stabilizes near 6–7 standard deviations, consistent with the Poisson prediction for the extremal order statistic.

Remark 83 (Block decomposition of V°). Decompose the prime sum $B(m) = \sum_j B_{P_j}(m)$ by dyadic prime blocks $P_j < p \le 2P_j$. Then $V^\circ = \sum_j V^\circ_{P_j} + 2\sum_{j<k}\mathrm{cross}(P_j, P_k)$. At $N = 4096$, the cross-block covariance is 0.03% of the full V°. This is because cross-block pairs (p,q) with $p \in (P, 2P]$, $q \in (Q, 2Q]$, $Q \ge 2P$ satisfy $pq \ge 2P^2 > 2N$, so the CRT approximation is even tighter for cross-block than within-block terms. Consequently, the dispersion hypothesis reduces to $V^\circ_P \ll N^{o(1)}\,S_P$ for each dyadic block *separately*—a within-scale CRT independence statement.

12.7 The covariance structure and the L^2-to-uniform upgrade

N	K	V°/S	E°	$\lvert E^\circ\rvert/S$	J=0	J=1
512	67	0.998	+130.6	0.015	95.6%	4.1%
1024	113	0.998	+215.6	0.010	96.6%	2.9%
2048	205	0.994	+96.2	0.002	98.1%	1.8%
4096	380	0.996	+120.0	0.001	98.9%	1.1%
8192	742	1.001	+1344	0.005	99.3%	0.6%

Table 1: Covariance structure of the off-diagonal sum $E^\circ = V^\circ - S_c$, computed over all $\binom{K}{2}$ pairs of active primes $p > \sqrt{N}$. J=0 and J=1 are the fractions of pairs (p,q) with joint occupancy $J_{p,q} = 0$ or 1. The Cauchy–Schwarz bound on $|E^\circ|$ exceeds the actual value by a factor of 3,000–6,000.

The following deterministic lemma is the bridge from the dispersion hypothesis to the all-index conclusion.

Lemma 84 (L^2-to-uniform upgrade)**.** *Let $F_N\colon I_N \to \mathbb{R}$ with $\mu_N = \frac{1}{N}\sum_{m\in I_N} F_N(m)$ and $\mathcal{V}_N = \sum_{m\in I_N}(F_N(m)-\mu_N)^2$. Then*

$$\max_{m\in I_N} |F_N(m)-\mu_N| \;\le\; \sqrt{\mathcal{V}_N}. \tag{53}$$

Proof. Each squared deviation $(F_N(m_0)-\mu_N)^2$ is a single summand of $\mathcal{V}_N$, hence at most $\mathcal{V}_N$. □

Corollary 85 (Dispersion implies the all-index Apéry bound)**.** *If there exists a constant C such that $V^\circ_w(N) \le C\,S_w(N)$ for all sufficiently large dyadic N, where*

$$V^\circ_w(N) = \sum_{m\in I_N}(B(m)-\mu)^2, \qquad S_w(N) = \sum_{p>\sqrt{N}}(\log p)^2 A_p,$$

then $W(n) = o(n)$ for all n, and consequently $G_n = e^{o(n)}$ for every positive integer n.

Proof. By Lemma 84 with $F_N = B$,

$$\max_{m\in I_N}|B(m)-\mu| \;\le\; \sqrt{V^\circ_w} \;\le\; \sqrt{C\,S_w}.$$

Since $S_w \le (\log 2N)^2 \sum A_p$ and $\sum A_p = O(N \log N)$ (from the bound $Z(p) \le Cp^{2/3}$ or even $\sum Z(p) \ll \pi(x)$), we have $S_w = O(N(\log N)^3)$. Thus $\sqrt{V_w^\circ} = O(\sqrt{N}\,(\log N)^{3/2})$. The mean satisfies $\mu = O((\log N)^2)$. Therefore

$$W(n) \;\le\; B(n) \;\le\; \mu + \sqrt{V_w^\circ} = O(\sqrt{N}\,(\log N)^{3/2}) = o(N/\log N),$$

which gives $W(n) = o(n)$ upon taking $N = n$. □

Table 1 and the extended computation of V°/S through $N = 32{,}768$ (Section 10) provide strong evidence for the bound $V^\circ \le C \cdot S$ with $C \le 1.01$. The ratio $|E^\circ|/S$ decreases from 0.015 at $N = 512$ to 0.001 at $N = 4096$; the brief increase at $N = 8192$ is consistent with fluctuations of $O(1/\sqrt{N})$. No concentration inequality, negative-association theorem, or fourth-moment bound is needed: the L^2-to-L^∞ trivial inequality together with $V^\circ = O(S)$ completes the proof.

12.8 The exact random CRT model

The independent fixed-margin CRT model gives a rigorous null prediction for the covariance structure. For each active prime p, retain the observed column size A_p and choose the incidence set Ω_p uniformly among all A_p-subsets of I_N, independently across primes. Put $D_p = A_p(1 - A_p/N)$ and $D_w = \sum_p w_p^2 D_p = S_{\text{diag}}$.

Proposition 86 (Random CRT second moment)**.** *Under the independent fixed-margin model,* $\mathbf{E}\, C_{p,q} = 0$ *and* $\operatorname{Var}(C_{p,q}) = D_p D_q/(N-1)$*. Moreover,* $\mathbf{E}(C_{p,q}C_{p,r}) = 0$ *for all triples of distinct primes. Consequently,*

$$\mathbf{E}\big[(E^\circ)^2\big] \;=\; \frac{2}{N-1}\Big(D_w^2 - \sum_p w_p^4 D_p^2\Big). \tag{54}$$

If no single prime column carries a positive proportion of D_w*, then*

$$\frac{\operatorname{sd}(E^\circ)}{S} \;\sim\; \sqrt{\frac{2}{N}}, \tag{55}$$

matching the empirical ratio $|E^\circ|/S$ *in Table 1 to within the expected fluctuations.*

Proof. Given Ω_p, the intersection count $J_{p,q}$ is hypergeometric with parameters (N, A_p, A_q), giving $\mathbf{E}\, C_{p,q} = 0$ and $\operatorname{Var}(C_{p,q}) = D_p D_q/(N-1)$. For triples, conditioning on Ω_p makes $C_{p,q}$ and $C_{p,r}$ conditionally independent with zero mean, so $\mathbf{E}(C_{p,q}C_{p,r}) = 0$. Expanding $(E^\circ)^2$ and using orthogonality gives (54). □

12.9 The operator norm and eigenvector localization

Define the normalized off-diagonal covariance matrix H with entries

$$H_{p,q} = \frac{C_{p,q}}{\sqrt{D_p D_q}} \quad (p \ne q), \qquad H_{p,p} = 0.$$

Then $V^\circ = S_{\text{diag}} + v^T H v$ where $v_p = w_p\sqrt{D_p}$ and $|v|^2 = D_w$. Consequently, $V^\circ \le (1 + \|H\|_{\text{op}})\, S_{\text{diag}}$.

The striking gap between $\|H\|_{\text{op}} \approx 1$ and the Rayleigh quotient ≈ 0.01 is explained by *eigenvector localization*. At every computed N, the top eigenvector concentrates on two or three primes with very sparse columns ($A_p \le 4$): these are large primes that share a common marked integer by CRT coincidence. At $N = 8192$ ($\lambda_{\max} = 1.21$), the cluster consists of $p = 14851, 7937, 7559$ with $A_p = 1, 2, 3$ respectively. The logarithmic weight vector $v_p = (\log p)\sqrt{D_p}$ assigns negligible mass to

N	K	$\|H\|_{\mathrm{op}}$	$\lambda_{\max}$	$\lambda_{\min}$	Rayleigh(v)
256	39	0.824	+0.824	−0.706	0.022
512	67	0.987	+0.987	−0.576	0.031
1024	113	1.088	+1.088	−0.707	0.021
2048	205	0.896	+0.896	−0.735	0.004
4096	380	0.904	+0.904	−0.734	0.002
8192	742	1.210	+1.210	−1.000	0.009

Table 2: Spectral data for the normalized covariance matrix H. The operator norm $\|H\|_{\mathrm{op}}$ appears bounded (fluctuating between 0.8 and 1.2, not growing with N). The Rayleigh quotient $v^T Hv/|v|^2 = E^\circ/D_w$ for the logarithmic weight vector v is much smaller and non-monotone with mean ≈ 0.01, explaining why $V^\circ \approx S$. At $N = 8192$, two eigenvalues equal -1 exactly, arising from a three-prime collision cluster with $A_p \in \{1, 2, 3\}$.

these sparse primes (combined contribution $< 0.5\%$ of $|v|^2$ at every N), so it is nearly orthogonal to the localized extreme eigenvectors. At $N = 4096$, only 3% of $|v|^2$ lies in the top 5 eigenvectors; the largest projection of v (10%) falls on an eigenvalue of magnitude 0.14, deep in the spectral bulk. At $N = 8192$, the projection drops to 0.5%.

This localization has a combinatorial origin: the extreme eigenvalues of H arise from k-fold CRT collisions—integers m marked by $k \geq 3$ distinct large primes. The corresponding eigenvectors span a $(k-1)$-dimensional space of eigenvalue -1 and one direction of eigenvalue $k-1$. The logarithmic weight vector, being smoothly spread across all K active primes, is structurally immune to these localized collision artifacts.

12.10 Palm decorrelation and the cancellation mechanism

The identity (52) decomposes $V^\circ = S_{\mathrm{diag}} + E^\circ$ where the off-diagonal $E^\circ = v^T Hv$. The Rayleigh quotient $R_v = v^T Hv/|v|^2 = E^\circ/D_w$ admits a transparent conditional-expectation interpretation. Define the *load excess*

$$\eta_p \;=\; \frac{(Hv)_p}{v_p} \;=\; \frac{\dfrac{1}{A_p}\displaystyle\sum_{m\in\Omega_p}\big(B_{-p}(m) - \mu_{-p}\big)}{w_p\,(1 - A_p/N)}, \tag{56}$$

where $B_{-p}(m) = B(m) - w_p X_p(m)$ is the total load excluding prime p, and $\mu_{-p} = \mu - w_p A_p/N$. The quantity η_p measures the average excess load from all *other* primes, conditional on $m \in \Omega_p$, normalized by the centered self-weight. Under exact independence, $\eta_p = 0$ for every p. The Rayleigh quotient is the weighted average

$$R_v \;=\; \sum_p \frac{v_p^2}{D_w}\,\eta_p. \tag{57}$$

Thus $V^\circ \approx S$ says: averaged with the natural variance weights, conditioning on one prime hitting does not bias the load from all other primes. This is the arithmetic content of the observed approximate orthogonality.

Remark 87 (Cancellation, not near-null). The small Rayleigh quotient does *not* mean that v is a near-null vector of H. Computation gives $\|Hv\|/|v| \approx 0.31$ at all tested sizes $N = 512, \ldots, 4096$, while $R_v = v^T Hv/|v|^2$ decreases from 0.031 to 0.002. The cancellation indicator $\|Hv\|/(|v|\,|R_v|)$ grows from 12 to 154: positive and negative spectral contributions to R_v individually exceed 0.13 but nearly cancel. This is the standard trace-zero mechanism: since $\operatorname{tr} H = \sum_j \lambda_j = 0$ and the projections $|\langle u, e_j\rangle|^2$ are approximately uniform ($u = v/|v|$ is delocalized), the identity $R_v = \sum_j \lambda_j(|\langle u, e_j\rangle|^2 - 1/K)$ produces cancellation to order $K^{-1/2}$. The extreme eigenvalues are localized (IPR of top eigenvector ≈ 0.25, effective dimension ≈ 4), so their projections deviate from $1/K$ but contribute little to the weighted sum.

12.11 Dyadic block compression

Partition the active primes into $L = O(\log N)$ dyadic blocks $\mathcal{B}_1, \ldots, \mathcal{B}_L$ according to $\lfloor \log_2 p \rfloor$. For each block, define the restricted weight $s_j = \|v|_{\mathcal{B}_j}\|$ and the unit share $a_j = s_j/|v|$. The *compressed covariance matrix* $\tilde{H} \in \mathbb{R}^{L\times L}$ is

$$\tilde{H}_{i,j} \;=\; \frac{(v^{(i)})^T H\, v^{(j)}}{s_i\, s_j}, \tag{58}$$

where $v^{(j)}$ is v restricted to $\mathcal{B}_j$. Then the Rayleigh quotient factors exactly as

$$R_v \;=\; a^T \tilde{H}\, a. \tag{59}$$

This reduces the $K \times K$ problem to an $L \times L$ one ($L \leq 8$) in the single direction of interest.

N	K	L	$\|H\|_{\mathrm{op}}$	$\|\tilde{H}\|_{\mathrm{op}}$	R_v	$\|Hv\|/\|v\|$
512	67	6	0.987	0.113	0.031	0.386
1024	113	6	1.088	0.111	0.021	0.302
2048	205	7	0.896	0.097	0.004	0.311
4096	380	7	0.904	0.040	0.002	0.313
8192	742	8	1.210	0.052	0.009	0.291

Table 3: Full operator norm $\|H\|_{\mathrm{op}}$ versus compressed operator norm $\|\tilde{H}\|_{\mathrm{op}}$. The full norm is bounded near 1 with a spike to 1.21 at $N = 8192$ (from the first singleton collision); the compressed norm is much smaller, ranging from 0.113 at $N = 512$ to 0.040–0.052 at $N = 4096$–8192. The column $\|Hv\|/|v|$ shows that v is not a near-null vector of H ($\|Hv\|/|v| \approx 0.3$ at all sizes); the small Rayleigh quotient arises from spectral cancellation.

The compressed norm $\|\tilde{H}\|_{\mathrm{op}}$ controls the Rayleigh quotient via $|R_v| \leq \|\tilde{H}\|_{\mathrm{op}}$, and is both numerically accessible (an $L \times L$ eigenvalue problem) and analytically tractable (each entry is a doubly-weighted average of covariances over a dyadic block pair).

The Marchenko–Pastur null model for K independent isotropic columns in $\mathbb{R}^N$ predicts $\|H\|_{\mathrm{op}} \approx 2\sqrt{K/(N-1)} + K/(N-1)$; with $K \sim N/\log N$, this gives $2/\sqrt{\log N} + 1/\log N$, which matches the observed range 0.8–1.2 to within 20–40% (the excess is consistent with sparse localized spikes from low-A_p columns). For the compressed matrix, the same model predicts $\|\tilde{H}\|_{\mathrm{op}} \sim \sqrt{L/N} \sim \sqrt{\log N/N} \to 0$, consistent with the observed decrease.

Remark 88 (Row leverage). The maximum *normalized row leverage* $\rho(n) = \sum_{p:\, n\in\Omega_p} 1/D_p$ is bounded near 1.5 at all tested sizes, dominated by one singleton column ($1/D_p \approx 1$) and one or two doublet columns ($1/D_p \approx 0.5$). No singleton collisions (two singletons at the same m) occur up to $N = 4096$. By the Gram identity $H + I = U^T U$ where U is the $N \times K$ normalized incidence matrix, $\|H\|_{\mathrm{op}} + 1 \leq \max_n \rho(n) \cdot K$ is a trivial upper bound. A tighter bound via $\max_n \rho(n)$ governs the localized spectral spikes; if $\rho(n) \leq C$ uniformly, then $\|H\|_{\mathrm{op}} \leq C - 1$.

12.12 Gram identity and block aggregates

The compression admits an exact Gram-matrix interpretation. Define the *normalized block aggregate*

$$Y_i(m) \;=\; \frac{1}{s_i}\sum_{p\in\mathcal{B}_i}(\log p)\left(X_p(m)-\frac{A_p}{N}\right), \qquad 1\le i\le L, \tag{60}$$

where each Y_i is a centered function on I_N with $\sum_m Y_i(m)=0$. Let Y be the $N\times L$ matrix with columns Y_i. Then

$$\boxed{I_L+\tilde H \;=\; Y^TY \;\succeq\; 0,} \tag{61}$$

or entrywise, $\tilde H_{i,j}=\langle Y_i,Y_j\rangle-\delta_{ij}$. This is the key structural identity: $I+\tilde H$ is the Gram matrix of L vectors in $\mathbb{R}^N$, so every eigenvalue of $\tilde H$ is at least -1. Only the positive edge requires an arithmetic bound.

The Rayleigh identity (59) now reads

$$R_v \;=\; a^T\tilde H\,a \;=\; \left\|\sum_i a_iY_i\right\|_2^2-1.$$

Thus $R_v=O(1)$ is equivalent to approximate orthonormality of the L aggregate functions Y_i.

Remark 89 (Low-to-high transfer). Let $P=EE^T$ denote orthogonal projection onto the L-dimensional block-smooth subspace. Then $PHu=E\tilde Ha$ and

$$\|Hu\|^2 \;=\; \|\tilde Ha\|^2+\|(I-P)HEa\|^2. \tag{62}$$

At $N=4096$, the data give $\|Hu\|\approx 0.31$ and $\|PHu\|\le\|\tilde H\|_{\mathrm{op}}\approx 0.04$, so $\|(I-P)Hu\|\ge 0.307$. Almost all of Hu lies in the within-block oscillatory subspace, not the block-smooth subspace. The covariance operator converts a smooth prime-scale profile into oscillations inside the blocks. This is a strictly sharper diagnosis than spectral cancellation alone.

12.13 Exact random CRT prediction

The independent *fixed-margin* CRT model (Section 12.8) yields exact second-moment formulas for the compressed matrix. Each normalized column satisfies the isotropy identity $\mathbf{E}[\phi_p\phi_p^T]=\frac{1}{N-1}P_0$, where $P_0=I-\frac{1}{N}\mathbf{1}\mathbf{1}^T$. For distinct primes, $\mathbf{E}[H_{p,q}^2]=\frac{1}{N-1}$.

Proposition 90 (Compressed second moments)**.** *In the independent fixed-margin model,* $\mathbf{E}[\tilde H_{i,j}]=0$ *for all* i,j*, and*

$$\mathbf{E}[\tilde H_{i,j}^2] \;=\; \frac{1}{N-1} \qquad (i\ne j), \tag{63}$$

$$\mathbf{E}[\tilde H_{i,i}^2] \;=\; \frac{2}{N-1}(1-\kappa_i), \qquad \kappa_i=\sum_{p\in\mathcal{B}_i}e_i(p)^4. \tag{64}$$

Consequently, the Frobenius norm satisfies

$$\mathbf{E}\|\tilde H\|_F^2 \;=\; \frac{L^2+L-2\sum_i\kappa_i}{N-1} \;\asymp\; \frac{L^2}{N}. \tag{65}$$

Proof. For $i\ne j$, disjoint block supports give $\tilde H_{i,j}=\sum_{p\in\mathcal{B}_i}\sum_{q\in\mathcal{B}_j}e_i(p)e_j(q)H_{p,q}$. Since different unordered prime pairs have uncorrelated entries (including those sharing one prime, by conditioning), only the diagonal terms survive: $\mathbf{E}[\tilde H_{i,j}^2]=\frac{1}{N-1}\sum_p e_i(p)^2\sum_q e_j(q)^2=\frac{1}{N-1}$. The diagonal entries involve $\sum_{p<q\in\mathcal{B}_i}e_i(p)^2e_i(q)^2$, giving the κ_i correction. $\square$

This proves $\|\tilde{H}\|_{\mathrm{op}} \to 0$ *in the random model* whenever $L = o(\sqrt{N})$. The GOE refinement predicts $\|\tilde{H}\|_{\mathrm{op}} \approx 2\sqrt{L/N}$, with the factor 2 being the asymptotic edge constant.

N	L	$\sqrt{L/N}$	$\|\tilde{H}\|_{\mathrm{op}}$	$\|\tilde{H}\|_{\mathrm{op}}/\sqrt{L/N}$
512	6	0.108	0.113	1.04
1024	6	0.077	0.111	1.45
2048	7	0.058	0.097	1.66
4096	7	0.041	0.040	0.97
8192	8	0.031	0.052	1.67

Table 4: The compressed operator norm tracks the random CRT prediction $\sqrt{L/N}$ with ratio ≈ 1–1.7.

Remark 91 (Random surrogate comparison). For each N, we generate 20 random palindromic zero sets of the same sizes $|Z_p|$, build the corresponding $\tilde{H}$, and compare $\|\tilde{H}\|_{\mathrm{op}}$. At $N = 2048$: actual $= 0.097$, random mean $= 0.097$, z-score $= -0.01$. At $N = 1024$: actual $= 0.111$, random mean $= 0.130$, z-score $= -0.91$. The actual Apéry matrix is statistically indistinguishable from its random palindromic surrogate.

12.14 AMTD: the precise remaining conjecture

The dispersion target (Hypothesis 77) is *not* a consequence of the Elliott–Halberstam conjecture, the generalized Elliott–Halberstam conjecture, or Barban–Davenport–Halberstam for fixed sequences. Those conjectures concern the discrepancy of a *fixed* arithmetic function in residue classes; here the tested residue function $1_{\mathcal{Z}_p}$ varies with the modulus. This is a fundamental structural distinction, not a technical gap.

We therefore propose:

Hypothesis 92 (Apéry Moving-Target Dispersion (AMTD))**.** *There exists an absolute constant C such that, for every dyadic interval $I_N = (N, 2N]$ and every dyadic prime block $P < p \le 2P$ with $P > \sqrt{N}$,*

$$V^\circ(P, N) \;\le\; C\, N^{o(1)}\, S(P, N).$$

The strongest structural sufficient condition is a *short-arc Fourier orthogonality* conjecture:

$$\int_{\mathbb{R}/\mathbb{Z}} \Bigg| \sum_{\substack{P<p\le 2P \\ 1\le a<p \\ x<a/p\le x+1/N}} \frac{(\log p)\, F_p(a)}{p} \Bigg|^2 dx \;\ll\; \frac{N^{o(1)}}{N} \sum_{P<p\le 2P} \sum_{a=1}^{p-1} \left| \frac{(\log p)\, F_p(a)}{p} \right|^2, \tag{66}$$

where $F_p(a) = \sum_{r\in\mathcal{Z}_p} e_p(ar)$. By Gallagher's lemma, (66) implies AMTD block by block.

The classical additive large sieve gives the right-hand side of (66) multiplied by $(1 + P^2/N)$. For $P > \sqrt{N}$, this factor exceeds 2 and grows to $O(N)$ at the top block $P \sim N$. The P^2 barrier is intrinsic to the generic large-sieve proof and cannot be removed by average-spacing refinements alone. A proof of (66) must therefore exploit the special structure of the Apéry Fourier coefficients $F_p(a)$—their palindromic phase constraint ($e_p(a/2)\, F_p(a) \in \mathbb{R}$), their connection to the Sym^2 K3 Hasse polynomial, or the arithmetic of the zero sets $\mathcal{Z}_p$ themselves.

Remark 93 (Partial large-sieve result for small primes). For dyadic blocks with $P \le \sqrt{N}$, the classical additive large sieve gives $V^\circ(P, N) \le (N + 4P^2) \sum_{P<p\le 2P} w_p^2 Z(p)/p \le 5\, S(P, N)$. Thus

AMTD holds *unconditionally* for primes below $\sqrt{N}$ with constant $C = 5$. The open case is $P > \sqrt{N}$, where the large-sieve factor $1 + 4P^2/N$ exceeds any fixed constant. At the top block $P \sim N$, this factor is $O(N)$, and the generic large sieve provides no useful bound.

Remark 94 (Linnik dispersion route). The most plausible analytic pathway is Linnik's dispersion method: expand the L^2-discrepancy, isolate the diagonal, and seek cancellation in the off-diagonal via reciprocity and bilinear trace-function estimates in the framework of Fouvry–Kowalski–Michel [11], extended by Kowalski–Michel–Sawin [12] and Drappeau [9]. This reduces the problem to a *bounded-complexity representation theorem*: either the Fourier coefficient $a \mapsto \hat{\mathcal{Z}}_p(a) = \sum_{r\in\mathcal{Z}_p} e_p(ar)$ or the phase $m \mapsto e_p(u\, b_m)$ must belong to a uniformly controlled trace-function family. The obstacle is that b_m mod p is computed from a nonautonomous recurrence whose complexity grows with p, and no fixed bounded-conductor ℓ-adic sheaf with trace function $A_p(t)$ uniformly in p is currently known. The closest available framework is Forey–Fresán–Kowalski [13], which studies arithmetic Fourier transforms of a fixed perverse sheaf over a fixed finite field; four mismatches remain (growing complexity, horizontal vs. extension-degree aspect, crystalline vs. ℓ-adic, and one- vs. two-characteristic dispersion). A polylogarithmic loss in (66) already suffices for the competition target.

Remark 95 (Palindromic Fourier factoring). For doublet primes in the top block, the exponential sum over hit positions is

$$S(\theta) = \sum_{p\in\mathcal{D}} \big[e(\theta\, m_1(p)) + e(\theta\, m_2(p))\big] = 2\sum_{p\in\mathcal{D}} e\Big(\theta\, \tfrac{3p-1}{2}\Big) \cos(\pi\theta\, h_p)\,,$$

where $h_p = m_2(p) - m_1(p) = p - 1 - 2r_p$ is the doublet gap. By Parseval, $\int_0^1 |S(\theta)|^2\, d\theta = T + F$. The diagonal AMTD requires this integral to be $O(T)$, equivalently that $S(\theta)$ has approximately square-root cancellation on the minor arcs.

The palindromic structure converts each doublet edge into a prime-indexed phase $e(3p\theta/2)$ modulated by the cosine $\cos(\pi\theta h_p)$. The gaps h_p are determined by the first zero r_p of b_n mod p. A proof of diagonal AMTD via this route would establish cancellation in $S(\theta)$ by exploiting the "pseudorandom" distribution of h_p as p varies.

Remark 96 (Hierarchy of targets). The compression produces a strict hierarchy of increasingly weak—and therefore increasingly tractable—conjectures:

(A) **Full operator bound:** $\|H\|_{\mathrm{op}} = O(1)$. This implies a Bessel inequality for all prime columns and is far stronger than needed.

(B) **Compressed Bessel bound:** $\|\tilde{H}\|_{\mathrm{op}} = O(1)$. This asks for approximate orthonormality of only $L = O(\log N)$ block-aggregate functions, and suffices for the all-index GCD estimate.

(C) **Scalar Rayleigh bound:** $a^T\tilde{H}a = O(1)$. This is the weakest statement that closes the all-index theorem, equivalent to $V^\circ \ll S$. It preserves all useful signs and remains a two-prime problem.

Each implication (A)⇒(B)⇒(C) is strict. The random CRT model predicts $\|\tilde{H}\|_{\mathrm{op}} \asymp \sqrt{L/N} \to 0$ and $|a^T\tilde{H}a| \asymp N^{-1/2}$, both of which are far stronger than (B) and (C).

Remark 97 (Diagonal AMTD via Cauchy–Schwarz). The all-index GCD theorem admits a simpler closing route that avoids cross-block estimates entirely. Define the uncentered block load $F_i(m) = \sum_{p\in\mathcal{B}_i} (\log p)\, g_p(m) = s_i\, Y_i(m)$, so that $V^\circ = \|\sum_i F_i\|_2^2$. If each dyadic block satisfies

$$\|F_i\|_2^2 \;\le\; C\, s_i^2 \qquad \text{(Diagonal AMTD)}, \tag{67}$$

equivalently $1+\tilde{H}_{ii} \le C$, then by Cauchy–Schwarz,

$$V^\circ \;\le\; \Big(\sum_i \|F_i\|_2\Big)^2 \;\le\; L\sum_i \|F_i\|_2^2 \;\le\; CL\,S.$$

Since $L = O(\log N) = N^{o(1)}$, this gives $V^\circ \le N^{o(1)}\,S$, which suffices for $\max_m |B(m)-\mu| \le \sqrt{V^\circ} = O\big(\sqrt{N}\,(\log N)^{3/2}\big) = o(N/\log N)$. The within-block estimate (67) is a *two-prime* signed dispersion problem—it preserves useful signs, does not require four-prime correlations, and avoids the full operator norm or Frobenius arguments. At the top block $P \sim N$, it reduces to a sparse translated-set collision problem: bounding the additive energy of $\{p + r : r \in \mathcal{Z}_p\}$ for primes p in a dyadic range.

Block-level data at $N = 4096$. The ratio $\|F_i\|_2^2/s_i^2$ for each dyadic block at $N = 4096$:

Block	K_i	$\|F_i\|_2^2/s_i^2$	$\tilde{H}_{ii}$
2^6	2	1.018	$+0.018$
2^7	13	0.998	-0.002
2^8	17	0.987	-0.013
2^9	30	0.988	-0.012
2^{10}	47	1.015	$+0.015$
2^{11}	94	1.016	$+0.016$
2^{12}	177	0.989	-0.011

Maximum ratio: 1.018 (well below any nontrivial constant). The Cauchy loss is $L \cdot \sum_i \|F_i\|_2^2/S = 7.00 = L$, confirming $V^\circ/S = 1.002$ while the Cauchy bound gives $V^\circ \le 7S$. At $N = 8192$ the maximum block ratio remains 1.029 (block 2^{10}) with $L = 8$ blocks. At $N = 32{,}768$ the collision count for top-block doublets is $F = 38$ with $T = 925$, giving $F/T = 0.041$ (decreasing from 0.055 at $N = 8192$), confirming the conjecture across all tested sizes.

Remark 98 (Translation variance). For each pair of distinct primes p, q, averaging over all translates $x \bmod pq$ gives

$$\frac{1}{pq}\sum_{x \bmod pq} |T_{p,q}(x;N)|^2 \;\le\; \frac{N\,Z(p)^2 Z(q)^2}{pq},$$

where $T_{p,q}(x;L) = \sum_{j=1}^{L} f_p(x+j)\,f_q(x+j)$. For bounded zero counts $Z(p) = O(1)$, this is already at the random scale for a *typical* translate. However, the GCD problem samples the single translate $x = N$ simultaneously for all prime pairs. Summing individual pair bounds absolutely costs $O(N^{3/2})$, which exceeds $S \asymp N(\log N)^2$. The P^2 large-sieve barrier arises because the Fourier support of f_p has $\sim p$ nonzero frequencies (not $Z(p)$), and palindromic symmetry improves only absolute constants.

12.15 Poisson variance formulation

The diagonal AMTD (67) admits a clean probabilistic reformulation. For a dyadic block $\mathcal{B}_i$, define the *column multiplicity function*

$$\lambda_i(m) = \#\{p \in \mathcal{B}_i : m \in \Omega_p\} = \sum_{p\in\mathcal{B}_i} X_p(m), \qquad m \in I_N.$$

Its empirical mean and variance over I_N are

$$\mu_i = \frac{1}{N}\sum_m \lambda_i(m) = \frac{1}{N}\sum_{p\in\mathcal{B}_i} A_p, \qquad V_i = \frac{1}{N}\sum_m (\lambda_i(m) - \mu_i)^2.$$

Expanding the square gives

$$N\,V_i = \sum_{p\in\mathcal{B}_i} D_p + \sum_{\substack{p,q\in\mathcal{B}_i \\ p\neq q}} C_{p,q}.$$

Since all primes in a dyadic block satisfy $\log p \asymp \log P$, one has $\|F_i\|_2^2 \asymp (\log P)^2\, N\, V_i$ and $s_i^2 \asymp (\log P)^2 \sum_p D_p$, so the diagonal AMTD is equivalent (up to asymptotically negligible corrections from the non-constant weight $\log p$) to:

$$V_i \;\leq\; C\,\mu_i \qquad \text{(sub-Poisson variance).} \tag{68}$$

If the indicators $X_p(m)$ were independent Bernoulli variables (with $\Pr[X_p = 1] = A_p/N$), each $\lambda_i(m)$ would be a sum of independent sparse Bernoullis and the variance-to-mean ratio would be ≤ 1. Condition (68) therefore says that the deterministic Apéry indicators behave as if they were *approximately independent* at the block level.

Table 5 records the empirical variance-to-mean ratio for the top dyadic block.

N	K_{top}	μ	V/μ	collisions	Poisson pred.
1024	47	0.086	1.073	14	7.6
2048	94	0.079	0.983	10	12.8
4096	177	0.079	0.989	22	25.6
8192	363	0.080	1.011	60	52.9

Table 5: Column multiplicity statistics for the top dyadic block. $\mu = \frac{1}{N}\sum_p A_p$ is the mean, V/μ is the variance-to-mean ratio (Poisson prediction: 1), "collisions" is $\sum_m \lambda(\lambda-1)$, and "Poisson pred." is T^2/N where $T = \sum_p A_p$.

Remark 99 (Collision count). The off-diagonal covariance sum has the exact identity

$$\sum_{p\neq q\in\mathcal{B}_i} C_{p,q} \;=\; \sum_m \lambda_i(m)\big(\lambda_i(m)-1\big) \;-\; \frac{(\sum_p A_p)^2 - \sum_p A_p^2}{N}.$$

The first term counts pairwise collisions (integers hit by at least two primes in the block). For the top block ($P \sim N$), each prime has $A_p \leq Z(p) \leq 3$ hits, so $\lambda_i(m) \leq 3$ for all m at $N = 8192$. The collision count fluctuates around the Poisson prediction T^2/N, confirming the near-independence of the indicators.

Remark 100 (Doublet dominance). Palindromic symmetry $b_{p-1-j} \equiv b_j \pmod p$ forces the zero set $\mathcal{Z}_p$ to be closed under $r \mapsto p-1-r$. If $Z(p) > 0$, then $Z(p)$ is typically even, and indeed at the top block, $Z(p) \geq 2$ for all active primes. The distribution at $N = 8192$ is: *doublets* $Z = 2$ (78%), *quadruplets* $Z = 4$ (19%), *sextuplets* $Z = 6$ (3%), *octuplets* $Z = 8$ ($< 1\%$). No singleton primes appear ($Z = 1$ requires $b_{(p-1)/2} \equiv 0 \pmod p$, which is extremely rare for large p).

For a doublet prime with $\mathcal{Z}_p = \{r_p, p-1-r_p\}$, the two hit positions are

$$m_1(p) = p + r_p, \qquad m_2(p) = 2p - 1 - r_p,$$

satisfying $m_1(p)+m_2(p)=3p-1$. A collision between two doublet primes p,q requires one of four types:

$$
\begin{aligned}
&\text{(A) } m_1(p)=m_1(q) &&\iff && r_q=r_p+(p-q),\\
&\text{(B) } m_1(p)=m_2(q) &&\iff && r_p+r_q=2q-1-p,\\
&\text{(C) } m_2(p)=m_1(q) &&\iff && r_p+r_q=2p-1-q,\\
&\text{(D) } m_2(p)=m_2(q) &&\iff && r_q=r_p+2(q-p).
\end{aligned}
$$

For $q>p$, type (B) is impossible: $m_1(p)=p+r_p<3p/2<3q/2<2q-1-r_q=m_2(q)$. The collision energy therefore decomposes into three, not four, pieces:

$$F=F_A+F_C+F_D, \tag{69}$$

where $F_A=\sum_m \nu_1(m)(\nu_1(m)-1)$, $F_D=\sum_m \nu_2(m)(\nu_2(m)-1)$, $F_C=2\sum_m \nu_1(m)\nu_2(m)$, with $\nu_1(m)=\#\{p: m_1(p)=m\}$ and $\nu_2(m)=\#\{p: m_2(p)=m\}$.

Two distinct doublet columns $\{m_1(p),m_2(p)\}$ and $\{m_1(q),m_2(q)\}$ can share at most one element: if both elements coincided, then $m_1(p)+m_2(p)=m_1(q)+m_2(q)$, i.e., $3p-1=3q-1$, forcing $p=q$. Thus $J_{p,q}:=|\Omega_p\cap\Omega_q|\le 1$ for distinct doublet primes in the top block. The doublet incidence system is a *simple graph* (no multi-edges), with the ordered collision count $F=\sum_m \lambda(m)(\lambda(m)-1)$. The exact variance identity is

$$V=T+F-T^2/N.$$

Since $T\asymp N/\log N$ and $T^2/N=o(T)$, the diagonal AMTD ($V\le CT$) is equivalent to the collision bound $F=O(T)$.

Collision decomposition.

N	K	T	T^2/N	F	F_A	F_C	F_D	F/T
1024	47	39	1.5	2	2	0	0	0.051
2048	93	70	2.4	4	2	2	0	0.057
4096	177	140	4.8	0	0	0	0	0.000
8192	363	293	10.5	16	10	4	2	0.055
16384	669	528	17.0	22	16	6	0	0.042
32768	1145	925	26.1	38	18	20	0	0.041

The ratio F/T is consistently small and decreasing, empirically confirming $F=o(T)$. Under the FPI prediction $F\approx 3K^2/N$, one expects $F/T\approx 3/(6.6\log N)$, giving 0.044 at $N=32{,}768$, in good agreement with the observed 0.041. At all sizes $\max_m \lambda(m)\le 3$.

Remark 101 (Two-prime divisor formulation). At the top block ($P>N$), each hit $m\in\Omega_p$ implies $p\mid b_m$ by the Lucas congruence. Therefore

$$\lambda_{\mathcal{B}}(m)=\omega_{(N,2N]}(b_m):=\#\{p \text{ prime}: N<p\le 2N,\ p\mid b_m\},$$

and the collision count becomes

$$F=\sum_{m\in I_N}\omega_{(N,2N]}(b_m)\big(\omega_{(N,2N]}(b_m)-1\big).$$

The diagonal AMTD is therefore equivalent to the following *two-large-prime divisor theorem*: on average over m, the Apéry number b_m has at most $O(1)$ prime divisors in the dyadic range $(N,2N]$.

This formulation reveals the arithmetic core: the collision bound $F = O(T)$ is an Apéry-specific Barban–Davenport–Halberstam theorem, asking for a level-of-distribution estimate at the second prime moment. A single-prime level-of-distribution (equivalently, $T = O(N/\log N)$) follows from the Poisson conjecture and is not in doubt; the missing step is the *bilinear* independence: the events $p \mid b_m$ and $q \mid b_m$ are approximately uncorrelated as p, q range over a dyadic block.

Vertical vs. horizontal projections. The incidence graph has primes on one side and integers on the other; an edge (p, n) records $p \mid b_n$. Grouping by the integer index n gives the *vertical* prime-divisor count $W_N(n) = \omega_{(N,2N]}(b_n)$, $n \in [1, N]$. Grouping by the hit position $m = n+p$ gives the *horizontal* column multiplicity $\lambda(m)$. These are different projections. By the Lucas congruence $b_{p+r} \equiv 5\, b_r \pmod{p}$ (for $r < p$), the horizontal multiplicity $\lambda(m) = \omega_{(N,m]}(b_m)$—it counts large prime divisors of b_m itself. Crucially, even the pointwise bound $W_N(n) \leq 1$ for *every* $n \leq N$ does not control $\lambda(m)$: many distinct (n_i, p_i) pairs can satisfy $n_i + p_i = m$ with $p_i \mid b_{n_i}$, all contributing *different* prime divisors of the single integer b_m. The collision bound $F = O(T)$ is therefore a genuinely horizontal, two-prime-divisor assertion about b_m.

Remark 102 (Why classical sieves do not close the argument). The collision bound $F = O(T)$ has the form of a two-dimensional upper-bound sieve (Selberg, Rosser–Iwaniec, Brun, GPY), but no classical sieve theorem deduces it from the one-prime data alone. After weight expansion, every such sieve introduces the counts $A_{pq}(N) = \#\{m : pq \mid b_m\}$ for products $pq \asymp N^2$. Since the observation interval has length only N, each individual expected count is $O(1/N)$ and only an *averaged* large-modulus discrepancy theorem can hold. The difficulty is intrinsic: even the strongest conceivable one-prime bound $A_p(N) \leq 1$ for every p does not control cross-prime alignment. An adversarial configuration with all zero sets concentrating at a single index m_0 satisfies $A_p = 1$ while $F \asymp (N/\log N)^2$. Thus the sieve machinery can *package* a two-prime distribution theorem as a collision bound, but it cannot *manufacture* that theorem from one-prime information. The minimum new input is the bilinear estimate

$$\sum_{\substack{p \neq q \\ p,q \in (N,2N]}} A_{pq}(N) \ll T + \frac{T^2}{N}, \tag{70}$$

which is precisely the block second-factorial-moment conjecture (diagonal AMTD, Hypothesis 92).

Proposition 103 (Conditional collision bound)**.** *Assume the following* Frobenius Pair Independence *hypothesis (FPI): for distinct doublet primes $p, q \in (N, 2N]$, the normalized first zeros r_p/p and r_q/q are asymptotically independent and each approximately uniform on $[0, \frac{1}{2}]$ in the sense that, for any integers $a \in [1, (p-1)/2]$ and $b \in [1, (q-1)/2]$,*

$$\frac{1}{K(K-1)} \#\big\{(p, q) \text{ distinct doublet primes in } (N, 2N] : r_p = a,\ r_q = b\big\} = \big(1 + o(1)\big)\,\frac{4}{pq}, \tag{71}$$

where K is the number of doublet primes. Then $F = o(T)$ and the all-index GCD theorem $G_n = e^{o(n)}$ holds for all n.

Proof. We bound each piece of the decomposition (69) separately.

Type A. The type-A collision count is

$$F_A = \#\big\{(p, q) \text{ distinct doublet primes} : r_q = r_p + (p - q)\big\}.$$

For a fixed ordered pair (p, q), the collision requires $r_q = r_p + (p-q)$. Summing the FPI probability over all valid values of r_p, the conditional probability is

$$\Pr[\text{type A} \mid p, q] \;=\; \sum_{r=1}^{(p-1)/2} \frac{2}{p} \cdot \frac{2}{q} \cdot \mathbf{1}\big[r + (p-q) \in [1, (q-1)/2]\big] \;\le\; \frac{4\,\min\big((p-1)/2,\, (3q-1)/2 - p\big)}{pq}.$$

For primes $p, q \in (N, 2N]$, this is at most $4/N$ when $|p - q| \le N/2$, and vanishes when $|p - q| > N$. Summing over all $K(K-1)$ ordered pairs:

$$F_A \;\le\; K(K-1) \cdot \frac{4}{N} \cdot (1 + o(1)) \;\le\; \frac{4K^2}{N} \cdot (1 + o(1)).$$

Since $K \asymp N/\log N$ and $T = 2K + O(K^{1/2})$, we obtain $F_A \le 4T^2/(4N) \cdot (1 + o(1)) = T^2/N \cdot (1 + o(1)) = o(T)$.

Type C. The collision condition $r_p + r_q = 2p - 1 - q$ constrains a sum of independent uniforms. Under FPI, $r_p + r_q$ has a trapezoidal distribution on $[2, (p + q - 2)/2]$ with maximum density $2/\min(p, q) \le 2/N$. The target $2p - 1 - q$ lies in the support when $q < 2p - 1$, which holds for $p > (q + 1)/2$. Thus $\Pr[\text{type C} \mid p, q] \le 2/N$, and $F_C \le 2K^2/N \cdot (1 + o(1)) = o(T)$.

Type D. The condition $r_q = r_p + 2(q - p)$ has the same structure as type A with gap $2(q - p)$ in place of $p - q$. By the same argument, $F_D \le 4K^2/N \cdot (1 + o(1)) = o(T)$.

Combining all three, $F = F_A + F_C + F_D = o(T)$. The diagonal AMTD $\|F_i\|_2^2 \le C\, s_i^2$ follows with $C = 1 + o(1)$ at each dyadic block, giving $V^\circ \le (1 + o(1))\, L\, S = O(\log N)\, S = N^{o(1)}\, S$. □

Remark 104 (Status of the FPI hypothesis)*.* The hypothesis (71) is a pair decorrelation statement for the zero positions across different primes. The marginal equidistribution of r_p/p is supported by the Kummer structure of the reduced generating series (Caruso–Fürnsinn–Vargas-Montoya–Zudilin [7]), which shows that the Galois group of $f_\alpha \bmod p$ is cyclic of order $(p-1)/\gcd(2, p-1)$. The rank-three Apéry Picard–Fuchs connection is the symmetric square of a rank-two connection, so its connected geometric monodromy is of $\mathrm{Sym}^2\mathrm{SL}_2 \simeq \mathrm{SO}_3$ type.

However, neither the cyclic Kummer group nor the SO_3 monodromy controls the zero *positions* r_p. To see why, observe that the zero-set $\mathcal{Z}_p = \{j < p : p \mid b_j\}$ is a *spectral* zero-set: writing $A_p(t) = \sum_{k=0}^{p-1} b_k t^k \in \mathbb{F}_p[t]$ and $f_p(u) = A_p(g^u)$ for a primitive root g, multiplicative orthogonality gives

$$\sum_{u=0}^{p-2} f_p(u)\, g^{-ju} = -b_j,$$

so $j \in \mathcal{Z}_p$ iff the j-th discrete Mellin coefficient of the Hasse-polynomial value vector vanishes. By contrast, the *roots* of $A_p(t)$ in $\mathbb{F}_p$ are evaluation zeros, controlled by the Kummer factorization $A_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2$. The Mellin coefficient index j and the discrete logarithm $\mathrm{ind}_g(x)$ of a root x live in dual spaces; no theorem identifies a coefficient-zero position with an evaluation-zero position.

The pair independence (71) is therefore a genuinely new hypothesis: it concerns the spectral zero process $\mathcal{A}_p(j) = \mathbf{1}_{\{b_j \equiv 0\}}$, not a Frobenius class function or a discrete logarithm. Neither Chebotarev, Sato–Tate, nor Katz equidistribution (which control traces and evaluation points) applies to this spectral-zero process.

A clean unconditional formulation replaces FPI by the *marked two-characteristic Hasse dispersion* conjecture:

$$\sum_{\substack{N<p,q\le 2N \\ p\ne q}} \sum_{m\in I_N} X_p(m)\, X_q(m) \;\ll\; T_N + \frac{T_N^2}{N}, \tag{72}$$

where $X_p(m) = \mathbf{1}_{\{p \le m,\, b_{m-p} \equiv 0 \pmod p\}}$. The left side equals the ordered collision count F_N^{ord}. Via the exponential-sum detection $\mathcal{A}_p(j) = p^{-1} \sum_{a \bmod p} e_p(a\, b_j)$, the conjecture (72) reduces to cancellation in the mixed-modulus sum

$$\sum_{p \neq q} \frac{1}{pq} \sideset{}{'}\sum_{\substack{a \bmod p \\ c \bmod q}} \sum_{m \in I_N} e_p(a\, b_{m-p})\, e_q(c\, b_{m-q}), \tag{73}$$

where $\sum'$ omits the term $a = c = 0$ (which produces the main term). The inner sum is over a nonautonomous recurrence orbit: $m \mapsto (b_{m-p}, b_{m-q})$. Standard Weil bounds do not apply.

A proof of (72) would require (1) a uniform crystalline Mellin object whose first Hasse coordinate tracks $b_j \bmod p$; (2) nondegeneracy of this coordinate on the monodromy torsor; (3) product monodromy for distinct Kummer twists; (4) rare-event anti-concentration at $1/p$ scale; and (5) a *horizontal* large sieve across different characteristics—a five-stage program beyond current technology but strongly supported by computation ($F/T \le 0.041$ at $N = 32{,}768$).

Proposition 105 (Profile decomposition)**.** *For each prime $p \in (N, 2N]$, let $A_p = |\Omega_p \cap I_N|$ and $L_p = 2N - p$. Define the smooth tail profile*

$$\rho(m) = \sum_{\substack{N < p \le 2N \\ p < m}} \frac{A_p}{L_p}, \qquad m \in I_N.$$

Then:

(i) $\sum_{m \in I_N} \rho(m) = T_N$;

(ii) $\|\rho\|_2^2 := \sum_{m \in I_N} \rho(m)^2 \le \rho_{\max} T_N$, *where* $\rho_{\max} := \max_{m \in I_N} \rho(m) \le \sum_p A_p/L_p$; *under the PNT profile approximation,* $\|\rho\|_2^2 = (c_2 + o(1))\, T_N^2/N$ *with* c_2 *as in* (75) *(note* $c_2 > 1$*: by Cauchy–Schwarz* $\|\rho\|_2^2 \ge T_N^2/N$ *always);*

(iii) $F_N \le 2\|\rho\|_2^2 + 2\|\lambda - \rho\|_2^2 - T_N$.

In particular, if $\|\rho\|_2^2 = O(T_N^2/N)$ (as the profile approximation and the data indicate), the residual dispersion estimate $\|\lambda - \rho\|_2^2 \ll T_N$ would imply $F_N = O(T_N)$, since $T_N^2/N = O(T_N)$.

Proof. Part (i): $\sum_m \rho(m) = \sum_p (A_p/L_p) \cdot L_p = \sum_p A_p = T_N$. Part (ii): $\|\rho\|_2^2 \le \rho_{\max} \sum_m \rho(m) = \rho_{\max} T_N$, and $\rho(m)$ is a partial sum of the positive terms A_p/L_p. For the profile asymptotics: by the PNT approximation $\rho(m) \approx (2/\log N) \log(N/(2N - m))$ and the integral $\int_0^1 (\log 1/t)^2\, dt = 2$, one obtains $\|\rho\|_2^2 \sim c_2\, T_N^2/N$. Numerically, $\|\rho\|_2^2/(T_N^2/N)$ stabilizes near 1.25 for $N = 512, \ldots, 32{,}768$ (Table 6), matching $c_2 = 1.262\ldots$ from (75). Part (iii) follows from $\|a + b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ applied to $\lambda = \rho + \eta$. $\square$

Remark 106 (Orthogonal decomposition). Setting $\eta = \lambda - \rho$, the identity $T + F = \|\lambda\|_2^2 = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + \|\eta\|_2^2$ gives

$$F = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + (\|\eta\|_2^2 - T). \tag{74}$$

Under the independent profile model (each $X_p(m)$ is an independent Bernoulli with mean $\pi_p(m) = A_p/L_p$ when $p < m$), the three terms in (74) have the following expected behavior:

- $\|\rho\|_2^2 = \Theta(T^2/N)$: the deterministic profile energy, tending to 0 relative to T.
- $\|\eta\|_2^2 \approx T$: the shot-noise variance of T independent Bernoulli hits.

- $\langle \rho, \eta \rangle \approx 0$: the profile and residual are decorrelated.

Hence $F \approx \|\rho\|_2^2 \approx T^2/N = o(T)$. Computation (Table 6) confirms this decomposition: $\|\eta\|_2^2/T$ is stable near 1.0 (matching palindromic random surrogates), while $F \approx \|\rho\|_2^2$ to within 1% at $N = 16{,}384$ and within 8% at $N = 32{,}768$. The cross-term $\langle \rho, \eta \rangle$ is $O(1)$ at every tested scale (e.g. -0.03 at $N = 16{,}384$ and 0.40 at $N = 32{,}768$, versus $T = 664$ and 1182). Moreover, the cross-term has a *deterministic* bound: $|\langle \rho, \eta \rangle| \le \rho_{\max} \cdot T$ where $\rho_{\max} = \max_m \rho(m) = \sum_p A_p/(2N - p)$. Since $\rho_{\max}/(T/N)$ stabilizes near 1.8 (Table 6), $|\langle \rho, \eta \rangle| \le 1.8\, T^2/N = o(T)$. The unconditional collision theorem $F = O(T)$ is therefore equivalent to the shot-noise estimate $\|\eta\|_2^2 = O(T)$, which is the residualized AMTD (Hypothesis 92).

Table 6: Profile decomposition diagnostics for the full hit measure μ_N. All norms over $I_N = (N, 2N]$. "Surr" denotes the mean of $\|\eta\|_2^2/T$ over 10 palindromic random surrogates.

N	T	F	$\|\rho\|_2^2$	$\|\rho\|_2^2/(T^2/N)$	$\|\eta\|_2^2/T$	Surr
512	21	2	1.1	1.28	1.05	0.98
1024	50	2	3.0	1.22	0.98	0.99
2048	86	4	4.5	1.26	0.99	1.01
4096	172	2	9.4	1.31	0.96	1.00
8192	354	20	19.6	1.28	1.00	1.00
16384	664	34	33.7	1.25	1.00	1.01
32768	1182	58	53.8	1.26	1.00	—
65536	2237	94	92.9	1.22	1.00	—
131072	4317	174	178.5	1.26	1.00	—
262144	7940	312	301.7	1.25	1.00	—

Remark 107 (Exact profile constant). The column $\|\rho\|_2^2/(T^2/N)$ in Table 6 converges to a constant c_2 as $N \to \infty$, with exact closed form

$$c_2 \;=\; \frac{2(\ln 2)^2 - 4\ln 2 + 2}{(2\ln 2 - 1)^2} \;=\; 1.261982\ldots \tag{75}$$

The derivation is direct. The triangular profile $\rho(m) = \sum_{N<p\le m} Z(p)/p$ for $m \in I_N$ has the continuous approximation $\rho(Nt) \approx (\ln t)/(\ln N)$ for $t \in [1,2]$, by Mertens' theorem. Then

$$T = \sum_m \rho(m) \;\approx\; \frac{N}{\ln N}\int_1^2 \ln t\, dt = \frac{N}{\ln N}\,(2\ln 2 - 1),$$
$$\|\rho\|_2^2 = \sum_m \rho(m)^2 \;\approx\; \frac{N}{(\ln N)^2}\int_1^2 (\ln t)^2\, dt = \frac{N}{(\ln N)^2}\,\big(2(\ln 2)^2 - 4\ln 2 + 2\big).$$

The ratio gives (75). The exact palindromic null expectation confirms the match: at $N = 1024$ the finite-N ratio is 1.209, at $N = 4096$ it is 1.281, at $N = 16{,}384$ it is 1.254, oscillating around c_2 with $O(1/\ln N)$ corrections from the prime number theorem. At all tested sizes, $E_{\text{null}}[F]$ agrees with the observed F to within the expected stochastic fluctuation.

Remark 108 (Concentration of F). Under the palindromic null (zero counts $Z(p)$ fixed, positions uniform among palindromic configurations), F concentrates around its mean. The dependency graph of collision indicators is the line graph of the complete graph on the active primes: two edge indicators share a dependency exactly when the corresponding prime pairs share one prime. This sparse structure gives

$$\mathrm{Var}(F_N) = \big(2 + O(1/\ln N)\big)\, E[F_N], \tag{76}$$

a rigorous Poisson-scale variance. *Proof sketch.* Write $F = 2U$ where $U = \sum_{p<q} I_{p,q}$ counts collision pairs. Edge indicators for disjoint prime pairs $\{p,q\} \cap \{r,s\} = \varnothing$ are independent. For a shared prime p, the conditional bound $\Pr(I_{p,q} = 1 \mid R_p = r) \le 2/h_q$ gives $\sum_{e<f} |\mathrm{Cov}(I_e, I_f)| \le 2\Delta\,\Lambda$ where $\Delta = 2\sum_p h_p^{-1} = O(1/\ln N)$ and $\Lambda = E[U]$. Combined with the Bernoulli variance $\sum \mathrm{Var}(I_e) = \Lambda(1 - O(1/N))$, this yields (76).

Since $E[F] = c_2\, T^2/N\,(1 + o(1)) \asymp N/(\ln N)^2 \to \infty$, Chebyshev gives $F/E[F] \to 1$ in L^2 and in probability, and $\Pr(F \ge C\,T) = O_C(\ln N/N)$ for every fixed $C > 0$.

The profile decomposition $F = \|\rho\|^2 + 2\langle \rho, \eta\rangle + \|\eta\|^2 - T$ separates deterministic and stochastic contributions. The deterministic cross-term satisfies $|\langle \rho, \eta\rangle| \le \rho_{\max}\, T$, with

$$\frac{\rho_{\max}}{T/N} \;\longrightarrow\; \frac{\ln 2}{2\ln 2 - 1} = 1.7943\ldots;$$

this bound is $O(T^2/N)$, the *same* order as $\|\rho\|^2$. The computationally observed $\langle \rho, \eta\rangle \approx 0$ is therefore genuine decorrelation evidence, not a consequence of the profile geometry.

If $\|\eta\|^2 = O(T)$, then all three terms are controlled: $\|\rho\|^2 = O(T^2/N)$, $|\langle \rho, \eta\rangle| = O(T^2/N)$, and $\|\eta\|^2 - T = O(T)$. Since $T^2/N = O(T)$ (because $T = o(N)$), this gives $F = O(T)$, exactly the BFM2 bound. The estimate $\|\eta\|^2 = O(T)$—that is, shot-noise-scale fluctuations of the actual zero positions around the triangular profile—is the sole gap.

Remark 109 (Prime-wise decomposition of $\|\eta\|^2$). For each prime $p \in (N, 2N]$, define the centred indicator

$$X_{m,p} \;=\; \mathbf{1}_{(m-p)\in Z_p} \;-\; \frac{Z(p)}{p}, \qquad m \in [p{+}1,\, 2N].$$

Then $\eta(m) = \sum_{N<p\le m} X_{m,p}$. Note that while distinct primes $p \ne q$ provide separate residue fields (and $b_j \bmod p$ is determined independently of $b_j \bmod q$ by CRT), this does not automatically make $X_{m,p}$ and $X_{m,q}$ independent for deterministic zero-sets—the zero positions within each Z_p are algebraically constrained. Under the independent fixed-cardinality residue model (Section 12.17), however, the columns are independent by construction.

Expanding the square:

$$\|\eta\|^2 \;=\; \underbrace{\sum_p \sum_m X_{m,p}^2}_{D} \;+\; \underbrace{\sum_{p\ne q} \sum_m X_{m,p}\, X_{m,q}}_{O}\,. \tag{77}$$

The diagonal $D = \sum_p D_p$ where $D_p = \sum_m X_{m,p}^2 = Z_p^{\mathrm{act}}(1-q_p)^2 + (L_p - Z_p^{\mathrm{act}})\, q_p^2$ with $q_p = Z(p)/p$, $Z_p^{\mathrm{act}} = \#\{j \in [1, L_p] \cap Z_p\}$, $L_p = 2N - p$. Expanding:

$$D \;=\; T \;-\; 2\sum_p Z_p^{\mathrm{act}}\, q_p \;+\; \sum_p L_p\, q_p^2. \tag{78}$$

Both correction terms are $O(1/\ln N)$: since $Z_p^{\mathrm{act}} \le Z(p)$ and $q_p = Z(p)/p \le Z(p)/N$, the second term is $\le 2\sum_p Z(p)^2/p \le (2/N)\sum_p Z(p)^2$, and the third is $\le N\sum_p Z(p)^2/p^2 \le (1/N)\sum_p Z(p)^2$. Under $\sum_p Z(p) = O(K)$ where $K \sim N/\ln N$, the Cauchy–Schwarz bound $\sum_p Z(p)^2 \le \big(\max Z(p)\big)\cdot \sum_p Z(p)$ and $\max Z(p) = O(1)$ (empirically) give both corrections $O(K/N) = O(1/\ln N)$. Thus $D = T + O(1/\ln N)$ unconditionally under $\sum_p Z(p)^2 = O(K)$. Under the null model, $E[O] = 0$.

Computation confirms this structure for the actual Apéry zero-sets: D matches T to $< 0.1\%$ for all tested N, and O/T fluctuates around zero at the level 10^{-3}–10^{-2} (see Table 7). Under the null model, $\mathrm{Var}(D) = O(T)$ (fluctuation of the truncated active mass Z_p^{act}) and $\mathrm{Var}(O) = O(T^2/N)$ (CRT averaging: each of the $\binom{K}{2}$ pair terms has overlap $\sim N$ within period $pq \sim N^2$). Together,

$$\frac{\mathrm{Var}(\|\eta\|^2)}{T^2} = O\left(\frac{1}{T} + \frac{1}{N}\right) \to 0,$$

proving $\|\eta\|^2/T \to 1$ in L^2 and in probability within the null model. The unconditional statement $\|\eta\|^2 = O(T)$ for the actual Apéry zero-sets is equivalent to a *two-prime dispersion* input: the zero positions of distinct primes are equidistributed modulo pq in the short interval $[1, N]$.

Table 7: Prime-wise decomposition $\|\eta\|^2 = D + O$.

N	T	D	O	O/T
512	21	20.94	1.14	0.054
1024	50	49.86	−0.83	−0.017
2048	86	85.90	−0.43	−0.005
4096	172	171.90	−6.76	−0.039
8192	354	353.90	1.22	0.003
16384	664	663.90	0.46	0.001
32768	1182	1181.91	3.52	0.003
65536	2237	2236.92	−4.16	−0.002
131072	4317	4316.92	−1.76	−0.0004
262144	7940	7939.93	11.92	0.0015

Note the structural identity $O = F - \|\rho\|^2 - 2\langle\rho, \eta\rangle$, so $O = o(T)$ is equivalent to $F = o(T)$ (since $\|\rho\|^2 = O(T^2/N) = o(T)$), and the prime-wise decomposition reduces the BFM2 bound to the assertion that the cross-prime interaction is asymptotically negligible.

Remark 110 (Higher factorial moments). The third factorial moment $F_3 = \sum_m \lambda(m)(\lambda(m)-1)(\lambda(m)-2)$ equals 6 at both $N = 16{,}384$ and $N = 32{,}768$ (a single row with $\lambda = 3$) but vanishes at $N = 65{,}536$ and $N = 131{,}072$ ($\max_m \lambda(m) = 2$ with 47 and 87 collision positions respectively), then returns to $F_3 = 6$ at $N = 262{,}144$ (a single row with $\lambda = 3$ among 154 collision positions). The palindromic model predicts $E[F_3] \approx c_3\, T^3/N^2$ where c_3 is a geometry-of-truncation constant. At $N = 262{,}144$ with $T = 7940$, $T^3/N^2 \approx 7.3$, so sporadic triple collisions at the single-digit level are consistent with the model. The fourth factorial moment $F_4 = 0$ at all tested sizes (no row has $\lambda \geq 4$).

12.16 Gap polynomial sparsity

For the Apéry recurrence, the gap polynomial N_h (whose roots in $\mathbb{F}_p$ are necessary for a simultaneous zero pair $b_x \equiv b_{x+h} \equiv 0$) has $r_p(h) = \#\{x \in \mathbb{F}_p : N_h(x) = 0\} \leq 3(h-1)$, giving $\sum_{h<H} r_p(h) \leq 3H^2/2$ and hence $Z(p) \leq Cp^{2/3}$. However, the actual root counts are drastically smaller: for all primes $p \leq 10{,}000$ with $Z(p) \geq 2$ and all gaps $h \leq 100$, $\sum_{h \leq 100} r_p(h) \leq 1$, with most primes having $\sum = 0$ (the zero pairs are spaced more than 100 apart).

Hypothesis 111 (Linear root sum). *$\sum_{1 \leq h < H} r_p(h) \ll H$ for every prime p and every $H \geq 1$.*

This would improve the zero-count to $Z(p) \ll p^{1/2}$, strengthening the exponent in Proposition 78 from $N^{5/6}$ to $N^{3/4}$. The heuristic justification is that N_h has degree $m_h = 3(h-1)/2$ in the primitive centred variable, and (conjecturally) $\mathrm{Gal}(N_h^{\mathrm{prim}}/\mathbb{Q}) = S_{m_h}$, so that by the Chebotarev density theorem $\mathbb{E}[r_p(h)] = 1$ for each h. For $h = 2, 3, 4, 5$, modular factorization certificates confirm $\mathrm{Gal} = S_{m_h}$ with $(m_2, m_3, m_4, m_5) = (1, 3, 4, 6)$, and the full hyperoctahedral groups $C_2 \wr S_{m_h}$ are verified via the single-flip lemma (the signed-group kernel is detected by a single pair flip in Frobenius factorization).

12.17 The exact random model

The palindromic identity $b_{p-1-j} \equiv b_j \pmod p$ forces the zero-set to consist of reflected pairs. Let V_p^+ denote the space of inversion-invariant functions $f : \mathbb{F}_p^\times \to \mathbb{F}_p$, i.e. $f(t) = f(t^{-1})$. The algebraic discrete Fourier transform $\hat{f}(j) = \sum_{t \in \mathbb{F}_p^\times} f(t)\, t^{-j}$ maps V_p^+ isomorphically onto the subspace of sequences satisfying $\hat{f}(j) = \hat{f}(-j)$.

Proposition 112 (Random inversion-invariant model). *Choose f uniformly from V_p^+. Then the number K of noncentral reflected pairs $\{j, p-1-j\}$ with $\hat{f}(j) = 0$ satisfies*

$$K \;\sim\; \mathrm{Binomial}\Big(\frac{p-3}{2},\, \frac{1}{p}\Big) \;\xrightarrow{p\to\infty}\; \mathrm{Poisson}(1/2).$$

Proof. The spectral coefficients of distinct reflected pairs are independent uniform elements of $\mathbb{F}_p$ (the DFT is invertible), so each vanishes with probability $1/p$. $\square$

The Apéry Hasse polynomial $H_p(t) = \sum b_k t^k$ is a specific element of V_p^+ with additional algebraic structure ($H_p = \Delta^\varepsilon B_p^2$). The conjecture that $Z(p) \sim 2\,\mathrm{Poisson}(1/2)$ is equivalent to asserting that H_p behaves, for the spectral-zero statistic, like a generic element of V_p^+.

The square structure can be incorporated directly. Let $V_p^- = \{g : \mathbb{F}_p^\times \to \mathbb{F}_p : g(t^{-1}) = -g(t)\}$ denote the anti-invariant subspace, with $\dim V_p^- = (p-3)/2$. For uniform $g \in V_p^-$, the function $f = g^2$ lies in V_p^+ and vanishes at the two fixed points $t = \pm 1$.

Proposition 113 (Random anti-invariant square model). *Choose g uniformly from V_p^- and put $f = g^2$. For any fixed distinct noncentral reflected pairs represented by $j_1, \ldots, j_r$,*

$$\Pr\big(\hat{f}(j_1) = \cdots = \hat{f}(j_r) = 0\big) \;=\; p^{-r} + O_r(p^{-c_r p})$$

for some $c_r > 0$. In particular, the pair-zero count $K^{\mathrm{sq}} \Rightarrow \mathrm{Poisson}(1/2)$.

Proof. Write $u_i = g(x_i)$ for orbit representatives. The spectral coefficient $\hat{f}(j) = \sum_i u_i^2 (x_i^j + x_i^{-j})$ is a diagonal quadratic form $Q_\lambda(u)$ in the independent uniform variables u_i. For $\lambda \neq 0$, the physical function $h_\lambda(t) = \sum_a \lambda_a (t^{j_a} + t^{-j_a})$ has Mellin support $\leq 2r$, so by the finite-field Mattson–Solomon uncertainty inequality, at least $(p-1)/(4r) - 1$ orbit values are nonzero, each contributing a quadratic Gauss sum of absolute value $p^{-1/2}$. The factorial-moment method then gives the result. $\square$

Multiplication by $\Delta(t)^\varepsilon$ affects only $O(1)$ orbits and does not change the limiting law. Thus the Sym^2 squareness of the Apéry Hasse polynomial is fully compatible with Poisson(1/2) in any random model where the square root B_p behaves generically.

Proposition 114 (Quadratic rank and both halves of the spectrum). *Let $B(t)$ be a uniformly random normalized reciprocal polynomial of degree $d = (p-1-2\varepsilon)/2$ over $\mathbb{F}_p$, with free variables $x_1, \ldots, x_m$ where $m = \lfloor d/2 \rfloor$. Put $H = \Delta^\varepsilon B^2$ and split the noncentral reflected-pair representatives into a* triangular *range $\mathcal{J}_{\mathrm{tri}}$ ($|\mathcal{J}_{\mathrm{tri}}| = m$) and a* nonlinear *range $\mathcal{J}_{\mathrm{nl}}$ ($|\mathcal{J}_{\mathrm{nl}}| = p/4 + O(1)$).*

(i) *In the triangular range, H_j is an affine-linear function of $(x_1, \ldots, x_m)$ with a unitriangular Jacobian. Hence $K_{\mathrm{tri}} = \#\{j \in \mathcal{J}_{\mathrm{tri}} : H_j = 0\} \Rightarrow \mathrm{Poisson}(1/4)$.*

(ii) *In the nonlinear range, each* H_j *is an affine quadratic form in* $(x_1, \ldots, x_m)$ *whose quadratic part has rank* $\geq m/2 - O(1) \gg p$. *By the Gauss-sum bound for affine quadratic forms of rank* R: $\Pr(F(x) = 0) = 1/p + O(p^{-R/2})$. *The pencil-rank estimate from Proposition 113 gives* $K_{\mathrm{nl}} \Rightarrow \mathrm{Poisson}(1/4)$.

(iii) *Without requiring independence between the two halves,*

$$\mathbf{E}\, Z(p) \;=\; 2\,\mathbf{E}\, K \;=\; 2\big(\mathbf{E}\, K_{\mathrm{tri}} + \mathbf{E}\, K_{\mathrm{nl}}\big) \;=\; 2\left(\frac{1}{4} + \frac{1}{4}\right) + o(1) \;=\; 1 + o(1).$$

If the mixed pencil-rank condition holds—i.e., every nonzero $\mathbb{F}_p$*-linear combination of* $H_{j_1}, \ldots, H_{j_s}$ *with at least one* $j_\nu \in \mathcal{J}_{\mathrm{nl}}$ *has quadratic rank* $\gg p/s$*—then the full pair-zero count* $K = K_{\mathrm{tri}} + K_{\mathrm{nl}} \Rightarrow \mathrm{Poisson}(1/2)$.

Remark 115 (Separation of random model from deterministic problem). Proposition 114 applies to the random reciprocal-polynomial model, not to the actual Apéry sequence. The actual B_p supplies one deterministic coefficient vector for each prime; converting the random-model statement into a theorem about $\sum_{p \leq x} Z(p)$ requires an arithmetic anti-concentration theorem asserting that these specific vectors avoid the coefficient quadrics with approximately random frequency. No such theorem is currently known.

12.18 Shift-correlation computation

Define the shift-correlation exponential sum

$$C_p(a) \;=\; \sum_{r=0}^{p-2} e_p(a\,(b_{r+1} - b_r))\,, \qquad a \neq 0.$$

A nontrivial bound $|C_p(a)| \ll p^{1-\eta}$ is the missing input for a Bombieri–Fouvry-type dispersion argument (cf. Section 12.4).

For all 666 primes $p \leq 5000$, we computed $\max_{a \neq 0} |C_p(a)|/\sqrt{p}$. The results:

- $\max_a |C_p(a)|/\sqrt{p}$ has mean 3.35 and maximum 5.07;
- $\max_a |C_p(a)|/\sqrt{p \log p}$ has mean 1.23 and maximum 1.80;
- the distribution of $\max/\sqrt{p}$ matches the Gumbel law expected for the maximum of p independent unit-variance Gaussians: $\max \approx \sqrt{2 \log p}$.

For shifts $s = 2, 3, 5$, the same behavior holds. The histogram of the differences $b_{r+1} - b_r \pmod p$ has std/mean ≈ 1 and the fraction of zero-bins is $\approx 1/e$, precisely the Poisson(1) model.

Conclusion: the Apéry sequence $b_r \bmod p$ is computationally indistinguishable from a random function $\{0, \ldots, p-1\} \to \mathbb{F}_p$ in its additive exponential-sum statistics. This provides strong computational evidence for the shift-correlation bound required by the dispersion approach.

12.19 Value-collision energy

A second pseudorandomness diagnostic is the *value-collision energy*

$$E(p) \;=\; \sum_{a \in \mathbb{F}_p} N_p(a)^2, \qquad N_p(a) = \#\{j \in \{0, \ldots, p-1\} : b_j \equiv a \pmod p\}.$$

By Cauchy–Schwarz, $Z(p)^2 = N_p(0)^2 \le E(p)$, so $E(p) \ll p$ would imply $Z(p) \ll \sqrt{p}$.

For a uniformly random function $\{0, \dots, p-1\} \to \mathbb{F}_p$, each $N_p(a) \sim \text{Poisson}(1)$ and $E(p)/p \to 2$. The palindromic symmetry $b_{p-1-j} \equiv b_j$ forces the half-sequence values to be doubled, giving $E(p)/p \to 3$: each half-position contributes independently to $N_p(a)$, and the doubling multiplies the second moment by 4 while the mean remains 1, yielding $\mathbb{E}[N_p(a)^2] = 4\big(\frac{1}{2} + \frac{1}{4}\big) = 3$.

We computed $E(p)/p$ for all 9590 primes $p \le 100{,}000$. The results confirm the palindromic prediction: $\text{mean}(E(p)/p) = 2.9995$, with 99% of primes in $[2.9, 3.1]$ and no dependence on $Z(p)$. The value-collision energy thus confirms that the Apéry sequence modulo p is anti-concentrated at the scale predicted by the palindromic random model.

Remark 116 (Second Fourier layer). By additive orthogonality,

$$E(p) = \frac{1}{p} \sum_{u \in \mathbb{F}_p} |S_p(u)|^2, \qquad S_p(u) = \sum_{j=0}^{p-2} e_p(u\, b_j).$$

A square-root bound $|S_p(u)| \ll \sqrt{p}$ for $u \neq 0$ would give $E(p) \ll p$, hence $Z(p) \ll \sqrt{p}$. This is a *second Fourier layer*: the first (multiplicative Mellin) transform produces the coefficients b_j; the second (additive) transform probes their residual distribution in $\mathbb{F}_p$. Ordinary monodromy of the K3 local system controls the first layer. It does not automatically construct a bounded-conductor object for the second layer. The modular form $f_4 = \eta(2z)^4\eta(4z)^4 \in S_4(\Gamma_0(8))$ controls only the central coefficient $b_{(p-1)/2} \equiv a_p(f_4) \pmod p$ via the quadratic Kummer twist; generic characters t^{-j} with $\text{ord}(\chi_{p,j}) \to \infty$ do not correspond to any fixed modular form. A residual anti-concentration theorem, proving the second-layer bound $E(p) \ll p$ or equivalently the additive exponential-sum bound above, would require genuinely new same-characteristic arithmetic geometry.

12.20 The high-moment criterion

The dispersion hypothesis (Hypothesis 77) is a second-moment statement localized in short windows. An alternative sufficient condition—global in the integer variable, but requiring a single fixed higher moment—is the following.

For a dyadic block of primes $(X, 2X]$ and an integer m, define

$$K_X(m) \;=\; \#\big\{p \in (X, 2X] :\; m \bmod p \in Z_p\big\}, \qquad \lambda_X \;=\; \sum_{X < p \le 2X} \frac{Z(p)}{p},$$

so that $\lambda_X \ll X^{2/3}/\log X$ by Proposition 9. Write $(K)_k = K(K-1)\cdots(K-k+1)$ for the falling factorial.

Hypothesis 117 (High-moment bound $(\text{HM})_k$)**.** *For one fixed integer $k \ge 3$, uniformly over dyadic X,*

$$\sum_{0 \le m < X^2} \big(K_X(m)\big)_k \;\ll\; X^{2+o(1)}\, \lambda_X^k.$$

Proposition 118 (The case $k = 2$ holds unconditionally)**.** $\displaystyle\sum_{0 \le m < X^2} \big(K_X(m)\big)_2 \;\le\; 5\, X^2 \lambda_X^2.$

Proof. Expanding the falling factorial over ordered pairs of distinct primes,

$$\sum_{m < X^2} \big(K_X(m)\big)_2 = \sum_{\substack{p,q \in (X,2X] \\ p \neq q}} \; \sum_{\substack{r \in Z_p \\ s \in Z_q}} \#\big\{m < X^2 : m \equiv r \ (p),\ m \equiv s \ (q)\big\}.$$

By the Chinese remainder theorem the inner count is at most $X^2/(pq)+1$, and since $pq \le 4X^2$ the additive 1 is at most $4X^2/(pq)$, so the count is at most $5X^2/(pq)$. Summing,

$$\sum_{m<X^2} \big(K_X(m)\big)_2 \le 5X^2 \sum_{\substack{p,q\in(X,2X]\\ p\neq q}} \frac{Z(p)Z(q)}{pq} \le 5X^2\lambda_X^2. \qquad \square$$

Remark 119 (Where the frontier begins). For $k \ge 3$, distinct primes have $p_1\cdots p_k > X^k > X^2$, so each residue tuple $(r_1,\ldots,r_k) \in Z_{p_1}\times\cdots\times Z_{p_k}$ has *at most one* CRT representative below X^2, and the trivial bound on the k-th moment is the total tuple count $\asymp (X\lambda_X)^k$, which exceeds the conjectured $X^2\lambda_X^k$ by the factor X^{k-2}. Hypothesis $(\mathrm{HM})_k$ thus asserts that only the expected proportion $\asymp X^2/(p_1\cdots p_k)$ of the actual Apéry zero tuples admit a representative in $[0, X^2)$ — a genuine cross-prime equidistribution statement, strictly beyond Proposition 118, yet far weaker than the Poisson conjecture (Hypothesis 44).

Remark 120 (Computational evidence). Using the actual Apéry zero sets Z_p for all primes $p \in (X, 2X]$, the moment ratios $R_k = \sum_{m<X^2}(K_X(m))_k \,/\, X^2\lambda_X^k$ are bounded and of order one at every tested scale:

X	$\#p$	λ_X	$\max K$	R_2	R_3	R_4
128	23	0.158	3	0.91	0.93	0.00
256	43	0.126	3	0.89	0.46	0.00
512	75	0.093	3	0.97	0.72	0.00
1024	137	0.082	4	0.97	0.83	0.50
2048	255	0.078	4	0.98	0.93	1.27

The $k=2$ ratio approaches 1 from below (the independence prediction), and R_3, R_4 fluctuate at order one—the expected count $X^2\lambda_X^4$ of 4-clusters is single-digit at these scales, so R_4 is dominated by extreme-value sampling noise. The observed $\max K = 4$ over $X^2 \approx 4\times 10^6$ integers matches the Poisson extreme-value prediction. Note $\lambda_X \approx 1/\log X$ here (the empirical Poisson mean), far below the worst-case input $X^{2/3}/\log X$ used in Theorem 121.

Theorem 121 ($(\mathrm{HM})_k$ implies a pointwise power saving)**.** *Assume Hypothesis 117 for one fixed $k \ge 3$. Then for* every *n,*

$$\log G_n \ll n^{2/3+2/k+o(1)}.$$

In particular $k=8$ gives $\log G_n \ll n^{11/12+o(1)}$, and any fixed $k>6$ proves the full conjecture $G_n = e^{o(n)}$ for all n.

Proof. Let $M_X = \max_{0\le m<X^2} K_X(m)$. If $M_X \ge 2k$, then $(M_X)_k \ge (M_X/2)^k$; taking the maximizing m inside the hypothesis,

$$(M_X/2)^k \;\le\; \sum_{m<X^2}\big(K_X(m)\big)_k \;\ll\; X^{2+o(1)}\lambda_X^k.$$

In either case ($M_X < 2k$ or not),

$$M_X \;\ll\; X^{2/k+o(1)}\lambda_X + 1 \;\ll\; \frac{X^{2/3+2/k+o(1)}}{\log X}.$$

Now fix n. The primes in $(\sqrt{n}, 2\sqrt{n}]$ contribute at most $\pi(2\sqrt{n})\log(2\sqrt{n}) = O(\sqrt{n})$ trivially. The remaining range $(2\sqrt{n}, n]$ is covered by dyadic blocks $(X, 2X]$ with $X = 2^j \le n$; every block meeting $(2\sqrt{n}, n]$ has $2X > 2\sqrt{n}$, hence $X > \sqrt{n}$ and therefore $n < X^2$. Hence $m = n$ is admissible in the maximum, and the block $(X, 2X]$ contains at most M_X lower-digit bad primes, each of weight $\log p \le \log 2X$. Therefore

$$\sum_{\substack{\sqrt{n}<p\le 2n\\ p\mid b_{n \bmod p}}} \log p \ll \sqrt{n} + \sum_{\substack{X=2^j\\ \sqrt{n}<X\le n}} X^{2/3+2/k+o(1)} \ll n^{2/3+2/k+o(1)},$$

the geometric sum being dominated by its top block. Adding the small-prime bound (Proposition 3), the companion-block bound (Proposition 59), and reducing to the lower-digit channel by Lemma 55(iii), the claim follows. For $k > 6$ the exponent satisfies $2/3 + 2/k < 1$, so $\log G_n = o(n)$. □

Remark 122. Among all sufficient conditions collected in this paper (Hypothesis Z, $\bar{Z}$, Poisson, AP–BDH, AMTD), Hypothesis $(\mathrm{HM})_8$ appears to be the weakest that still closes the full conjecture: it requires no pointwise information about any single prime, no short-window localization, and no variance identity—only one fixed factorial moment at the independence scale, compatible with the worst proved cardinality bound $Z(p) \ll p^{2/3}$.

The next two statements make the $(\mathrm{HM})_3$ frontier exact: a sharp reformulation as a positive pair–Palm excess, and an incidence model showing that *all* presently proved primewise and pairwise facts—including the fiber and energy bounds of Section 5.4 and even a uniform $O(1)$ row-codegree—are jointly insufficient for any power saving at $k = 3$. (In this block, Ω_p temporarily denotes the bad set $\{0 \le m < X^2 : m \bmod p \in \mathcal{Z}_p\}$.)

The third factorial moment: exact obstruction

Write

$$\mathcal{P}_X = \{p \text{ prime} : X < p \le 2X\}, \qquad M = X^2,$$

where $X \ge 2$ is integral, and put

$$\Omega_p = \{0 \le m < M : m \bmod p \in \mathcal{Z}_p\}, \qquad f_p(m) = \mathbf{1}_{\Omega_p}(m).$$

Thus $K_X(m) = \sum_{p\in\mathcal{P}_X} f_p(m)$. Set

$$A_p = |\Omega_p|, \qquad \rho_p = \frac{A_p}{M}, \qquad \mu = \sum_{p\in\mathcal{P}_X} \rho_p = \frac{1}{M}\sum_{m<M} K_X(m),$$

and retain

$$\lambda = \lambda_X = \sum_{p\in\mathcal{P}_X} \frac{Z(p)}{p}.$$

For ordered distinct primes $p, q \in \mathcal{P}_X$, define

$$\begin{aligned}
J_{p,q} &= |\Omega_p \cap \Omega_q|,\\
T_{p,q} &= \sum_{m\in\Omega_p\cap\Omega_q} \sum_{\ell\in\mathcal{P}_X\setminus\{p,q\}} f_\ell(m),\\
\lambda_{p,q} &= \sum_{\ell\in\mathcal{P}_X\setminus\{p,q\}} \frac{Z(\ell)}{\ell}, \qquad \mu_{p,q} = \sum_{\ell\in\mathcal{P}_X\setminus\{p,q\}} \rho_\ell.
\end{aligned}$$

The two aggregate positive pair–Palm excesses are

$$\mathscr{P}_3(X) = \sum_{p \neq q} \big(T_{p,q} - J_{p,q}\lambda_{p,q}\big)_+, \tag{79}$$

$$\mathscr{P}_3^{\mathrm{fin}}(X) = \sum_{p \neq q} \big(T_{p,q} - J_{p,q}\mu_{p,q}\big)_+. \tag{80}$$

The first uses the complete-period densities $Z(\ell)/\ell$; the second uses the exact marginals on $[0, M)$ and therefore absorbs every one-column boundary error.

Theorem 123 (Sharp pair–Palm characterization of $(\mathrm{HM})_3$)**.** *With all prime tuples ordered, one has the exact identities*

$$\sum_{p \neq q} J_{p,q} = \sum_{m<M} (K_X(m))_2, \tag{81}$$

$$\sum_{p \neq q} T_{p,q} = \sum_{m<M} (K_X(m))_3. \tag{82}$$

Moreover,

$$\mathscr{P}_3(X) \leq \sum_{m<M} (K_X(m))_3 \leq \mathscr{P}_3(X) + 5M\lambda^3, \tag{83}$$

$$\mathscr{P}_3^{\mathrm{fin}}(X) \leq \sum_{m<M} (K_X(m))_3 \leq \mathscr{P}_3^{\mathrm{fin}}(X) + 5(1 + 2/X)M\lambda^3. \tag{84}$$

Consequently $(\mathrm{HM})_3$ *is equivalent to either one of*

$$\mathscr{P}_3(X) \ll X^{2+o(1)}\lambda_X^3, \qquad \mathscr{P}_3^{\mathrm{fin}}(X) \ll X^{2+o(1)}\lambda_X^3.$$

More generally, for every fixed $\theta \geq 0$, the estimate in Goal G2 with exponent θ is equivalent to either positive-excess estimate with right-hand side $X^{2+\theta+o(1)}\lambda_X^3$.

There is also an exact centered formulation. Define

$$V_X = \sum_{m<M} (K_X(m) - \mu)^2, \qquad C_{3,X} = \sum_{m<M} (K_X(m) - \mu)^3$$

and the third factorial-moment defect

$$\mathfrak{D}_3(X) = (C_{3,X} - M\mu) + 3(\mu - 1)(V_X - M\mu). \tag{85}$$

Then

$$\sum_{m<M} (K_X(m))_3 = M\mu^3 + \mathfrak{D}_3(X). \tag{86}$$

Thus $(\mathrm{HM})_3$ *is equivalently the one-sided estimate*

$$\big(\mathfrak{D}_3(X)\big)_+ \ll X^{2+o(1)}\lambda_X^3.$$

The same equivalence holds at every G2 exponent $\theta \geq 0$.

Proof. If $p \neq q$ lie in $\mathcal{P}_X$, then $pq > M$. Hence every residue pair $(a \bmod p, b \bmod q)$ has at most one representative in $[0, M)$. This removes, rather than estimates, the usual CRT boundary term. At an integer with $K_X(m) = k$, there are exactly $(k)_2$ ordered choices of (p, q), and each has $k-2$ choices of a third prime. Its contribution to $\sum_{p \neq q} T_{p,q}$ is therefore $(k)_2(k-2) = (k)_3$. This proves (81)–(82); in particular, no factor 3! is missing.

For nonnegative A, B,

$$(A - B)_+ \leq A \leq B + (A - B)_+.$$

Apply this with $A = T_{p,q}$ and $B = J_{p,q}\lambda_{p,q}$, sum over ordered pairs, and use $\lambda_{p,q} \leq \lambda$ together with the proved second factorial moment:

$$\sum_{p \neq q} J_{p,q}\lambda_{p,q} \leq \lambda \sum_{m<M} (K_X(m))_2 \leq 5M\lambda^3.$$

This proves (83).

For each p and each residue in $\mathcal{Z}_p$, the number of its representatives below M differs from M/p by less than one. Therefore

$$|A_p - MZ(p)/p| \leq Z(p), \qquad |\mu - \lambda| \leq \frac{1}{M} \sum_p Z(p) \leq \frac{2\lambda}{X},$$

where the final inequality uses $p \leq 2X$. In particular $\mu \leq (1 + 2/X)\lambda$. Applying the same positive-part inequality with $B = J_{p,q}\mu_{p,q}$ gives

$$\sum_{p \neq q} J_{p,q}\mu_{p,q} \leq \mu \sum_{m<M} (K_X(m))_2 \leq 5(1 + 2/X)M\lambda^3,$$

which proves (84). Each equivalence follows in both directions from its sandwich. This is a characterization, not a new estimate hidden in a renamed hypothesis: the unproved arithmetic is exactly the positive conditional overload of third primes on the actual two-prime CRT loci.

Finally write $Y_m = K_X(m) - \mu$, so that $\sum_m Y_m = 0$. Expanding $K^3 - 3K^2 + 2K$ gives

$$\sum_{m<M} (K_X(m))_3 = C_{3,X} + 3(\mu - 1)V_X + M(\mu^3 - 3\mu^2 + 2\mu),$$

which rearranges to (86). Since the main term is nonnegative,

$$(\mathfrak{D}_3(X))_+ \leq \sum_{m<M} (K_X(m))_3 \leq M\mu^3 + (\mathfrak{D}_3(X))_+.$$

Equation (12.20) bounds $M\mu^3$ by a constant multiple of $M\lambda^3$, proving the last equivalences. When $\lambda = 0$ all zero sets, all pair loci, and all displayed quantities vanish. The same is true of the Palm sums when there is no pair locus, so no division by a possibly zero parameter has been suppressed. $\square$

Why the available primewise and pairwise tools do not imply G2

Proposition 124 (An anchored rainbow star with bounded codegrees)**.** *For every sufficiently large integral X there is a family $(\mathcal{Z}_p^*)_{p \in \mathcal{P}_X}$ with the following properties. Writing*

$$K_X^*(m) = \sum_{p \in \mathcal{P}_X} \mathbf{1}_{\{m \bmod p \in \mathcal{Z}_p^*\}}, \qquad \lambda_X^* = \sum_{p \in \mathcal{P}_X} \frac{|\mathcal{Z}_p^*|}{p},$$

$$\mu^* = X^{-2} \sum_{m<X^2} K_X^*(m),$$

one has:

1. *Every* $\mathcal{Z}_p^*$ *has size zero or two. Every nonempty set is reflection-invariant, has no consecutive elements, avoids* $0, 1, p-2, p-1$*, and satisfies*

$$|\mathcal{Z}_p^* \cap I| \leq 2|I|^{2/3}$$

for every nonempty interval $I \subseteq \{0, \ldots, p-1\}$*.*

2. *The universal CRT bound*

$$\sum_{m<X^2} (K_X^*(m))_2 \leq 5X^2(\lambda_X^*)^2$$

holds, and even the global centered second moment has the natural scale

$$\sum_{m<X^2} (K_X^*(m) - \mu^*)^2 \ll X^2 \lambda_X^*.$$

3. *If* $\mathcal{P}^*(m) = \{p : m \bmod p \in \mathcal{Z}_p^*\}$*, then for all distinct* $0 \leq m, n < X^2$*,*

$$|\mathcal{P}^*(m) \cap \mathcal{P}^*(n)| \leq 7.$$

Thus even a uniform $O(1)$ *row-codegree bound does not prevent the model.*

4. *Nevertheless,*

$$\lambda_X^* \asymp \frac{1}{\log X}, \qquad \sum_{m<X^2} (K_X^*(m))_3 \gg \frac{X^3}{(\log X)^3}.$$

Consequently the family violates every G2 estimate with fixed $\theta < 1$*.*

The family can additionally be extended to a reflection-invariant value map whose largest fiber is at most two and whose collision energy is at most $2p$*. Hence the numerical consequences of the bordered nonvanishing theorem—the* $O(p^{3/4})$ *fiber bound and* $O(p^{7/4})$ *energy bound—also do not rule it out.*

If one uses the gap-polynomial reflection identity $N_h(-x-h-1) = (-1)^{h-1}N_h(x)$*, the unique pair in every nonempty* $\mathcal{Z}_p^*$ *also satisfies its corresponding necessary gap certificate modulo* p*.*

Proof. Put $M = X^2$, choose $m_0 = \lfloor 3M/4 \rfloor$, and let $d = 2m_0 + 1$. Delete from $\mathcal{P}_X$ the primes dividing

$$E = (m_0 - 1)m_0(m_0 + 1)(m_0 + 2)d.$$

Only $O(1)$ primes are deleted: $|E| \ll X^{10}$ and every deleted prime exceeds X. For every remaining prime define

$$a_p = m_0 \bmod p, \qquad \mathcal{Z}_p^* = \{a_p, p - 1 - a_p\},$$

and put $\mathcal{Z}_p^* = \varnothing$ for deleted primes. The deletion makes the two displayed residues distinct and excludes the four endpoint residues. Their difference is a nonzero even integer, so they are not consecutive. Reflection is built in, and the interval estimate is immediate because an interval contains at most two marked residues.

Let L be the number of nonempty columns. The prime number theorem and Mertens' theorem give

$$L \sim \frac{X}{\log X}, \qquad \lambda_X^* = 2 \sum_{\substack{X<p\leq 2X \\ p\nmid E}} \frac{1}{p} \asymp \frac{1}{\log X}.$$

Every active prime marks m_0, so $K_X^*(m_0) = L$ and

$$\sum_{m<M} (K_X^*(m))_3 \geq (L)_3 \gg \frac{X^3}{(\log X)^3}.$$

Dividing by $X^{2+\theta+o(1)}(\lambda_X^*)^3$ leaves $X^{1-\theta-o(1)}$, which tends to infinity for $\theta < 1$. The second-factorial-moment estimate is universal and follows by the same CRT argument as Proposition 118. Also

$$\begin{aligned}\sum_m (K_X^*(m) - \mu^*)^2 &\leq \sum_m (K_X^*(m))^2 \\ &= \sum_m K_X^*(m) + \sum_m (K_X^*(m))_2 \\ &\ll M\lambda_X^* + M(\lambda_X^*)^2 \ll M\lambda_X^*.\end{aligned}$$

It remains to check the codegree. Fix $0 \leq m < n < M$ and put $h = n - m$. If an active prime marks both integers, their two residues are chosen from a_p and $-1 - a_p$. Equal choices give $p \mid h$; different choices give

$$h \equiv \pm(2a_p + 1) \equiv \pm d \pmod{p}.$$

Thus every common prime divides the nonzero integer

$$H = h(h-d)(h+d).$$

Here $0 < h < M < d$, and $|H| < 8M^3 = 8X^6$. If there are k common primes, their product exceeds X^k and divides $|H|$, whence

$$k \leq \frac{\log(8X^6)}{\log X} \leq 7$$

for sufficiently large X.

For the value-map assertion, partition the residues into the orbits of $r \mapsto p - 1 - r$. At an active prime, assign zero to the marked orbit and distinct nonzero values to all other orbits. At a deleted prime, assign distinct nonzero values to every orbit; this is possible because there are $(p+1)/2 \leq p - 1$ orbits. Thus the zero fiber is exactly $\mathcal{Z}_p^*$ in both cases. Every fiber has size at most two, and, since $t^2 \leq 2t$ for $t \in \{0, 1, 2\}$, the fiber energy is at most $2p$.

For completeness, the gap-polynomial reflection identity used here is symbolic. For $h \geq 2$, let $D_h(x)$ be the symmetric $(h-1) \times (h-1)$ tridiagonal matrix with diagonal entries $P(x+i)$, $1 \leq i < h$, and off-diagonal entries $(x+i+1)^3$, $1 \leq i < h-1$. Expansion in the last row and the defining recurrence for N_h give $\det D_h(x) = N_h(x)$. If R is the matrix that reverses the coordinate order, then $P(-v-1) = -P(v)$ gives, entry by entry,

$$D_h(-x-h-1) = -RD_h(x)R.$$

Consequently

$$N_h(-x-h-1) = (-1)^{h-1} N_h(x). \tag{87}$$

(For $h = 1$ this is immediate from $N_1 = 1$.) This proves the identity for every h, without numerical verification.

Finally let $u = \min(a_p, p-1-a_p)$ and $h_p = p - 1 - 2u$. The deleted endpoints ensure $1 \leq u < u + h_p < p$; moreover h_p is positive and even. Since

$$-u - h_p - 1 = u - p \equiv u \pmod{p},$$

the gap-polynomial reflection identity gives $N_{h_p}(u) \equiv -N_{h_p}(u) \pmod{p}$, hence $N_{h_p}(u) \equiv 0 \pmod{p}$ for odd p. $\square$

Remark 125 (Scope of the obstruction). The preceding family is an incidence model, not the actual zero family of the Apéry recurrence. It proves that reflection, no consecutive zeros, the interval $O(|I|^{2/3})$ bound, the universal CRT second moment, the primewise fiber/energy bounds, the gap-polynomial pair certificates, and even uniform $O(1)$ row codegree do not imply any G2 power saving. The missing arithmetic must constrain, across distinct primes, which of the structurally permitted reflection orbits is selected as the zero fiber. The pair–Palm quantity in Theorem 123 is the minimal aggregate positive overload that measures exactly this selection problem.

Computational findings

The direct CRT enumerator in `hm3_explore.py`, independently cross-checked against an m-scatter, gives the following ordered third factorial moments:

X	$\sum_{m<X^2}(K_X(m))_3$	$X^2\lambda_X^3$	R_3
256	60	131.989926	0.454580146
512	150	209.308570	0.716645284
1024	486	583.330960	0.833146247
2048	1824	1953.964920	0.933486564
4096	7218	7608.237134	0.948708600

At $X = 4096$, the 1203 canonical prime triples are supported on 1161 integers; 1147 have $K_X = 3$ and 14 have $K_X = 4$. The conditional third-prime extension rate per unordered pair hit is 0.073422, close to $\lambda_X = 0.0768283$. Quotient repetitions, reflection-orbit pileups, and residue correlations show no visible clustering. Full histograms and reproduction details are in `hm3_exploration.md`. These data are evidence for $(\mathrm{HM})_3$, not a proof.

Stall report and infrastructure audit

Goals G1 and G2 are not proved here, and no G3 hypothesis is claimed. In particular, a third-order Palm condition should not be advertised as "weaker than AP–BDH": the latter is an L^2 condition and the two statements are not ordered without extra hypotheses. The completed deliverable is G4, Theorem 123, together with the sharpness obstruction above.

Two issues in the supplied route are material to this conclusion. First, Lucas gives only

$$m \bmod p \in \mathcal{Z}_p \quad \Longrightarrow \quad p \mid b_m.$$

The converse can fail through the leading digit. For example, $b_2 = 73$; with $X = 64$, $p = 73$, and $m = 146 < X^2$, Lucas gives $b_{146} \equiv b_2 b_0 \equiv 0 \pmod{73}$, while $m \bmod 73 = 0 \notin \mathcal{Z}_{73}$. Hence $K_X(m) \le \omega_{(X,2X]}(b_m)$, but equality and the corresponding factorial-moment identity asserted in the specification are false.

Second, the written proof of Lemma 62 starts with the unavailable assertion $p > N/2 \ge h$ and then says that the two residues "differ by exactly h". Quotient wrap invalidates that sentence even under the displayed extra inequalities: for $p = 37$ one has $\mathcal{Z}_{37} = \{17, 19\}$, and $m = 93$, $n = 128$, $N = 70$ give $37 > N/2 \ge n - m$ while the residues are 19 and 17. The lemma may be repairable by a separate wrap argument using reflection, but the proof as written cannot be used as an established HM3 input. The anchored-star proposition shows that even a repaired, stronger $O(1)$ row-codegree estimate would still not control a one-row rainbow pileup.

Finally, Proposition 105 in the second-moment section states $\|\rho\|_2^2 \le T_N^2/N$, whereas Cauchy–Schwarz gives the reverse inequality and the proposition's own table reports a ratio near 1.25. No step above uses that proposition.

The trivial tuple estimate used for comparison should also be read as the uniform inequality

$$\sum_{m<X^2}(K_X(m))_3 \le \Big(\sum_{X<p\le 2X} Z(p)\Big)^3 \le 8X^3\lambda_X^3.$$

It need not be asymptotic to $(2X\lambda_X)^3$ when the zero mass is concentrated on fewer than three primes.

13 Conclusion

The principal unconditional result is the quantitative exceptional-set theorem: for every fixed $\varepsilon > 0$,

$$\#\{n \le X : \log G_n > \varepsilon\, n\} \ \ll_\varepsilon \ (\log X)^2.$$

Consequently $G_n = e^{o(n)}$ on a set of natural density one; the exceptional set has upper Banach density zero and finite harmonic weight. The arithmetic engine is the uniform interval estimate $Z(p) \ll p^{2/3}$, proved by the gap-polynomial and separated-block method.

The all-n theorem reduces to the block second-factorial-moment estimate

$$F_N \ \ll\ T_N + \frac{T_N^2}{N}, \tag{88}$$

which asserts that prime columns in the top block do not systematically align at one integer. The profile decomposition of F_N separates this into the deterministic profile energy $\|\rho\|^2 \sim c_2\,T^2/N$ (with c_2 given exactly by (75)), a cross-term $|\langle\rho,\eta\rangle| \le \rho_{\max}\,T = O(T^2/N)$, and the residual shot-noise term $\|\eta\|^2$. Since $T^2/N = O(T)$, the estimate (88) is equivalent to the shot-noise bound $\|\eta\|^2 = O(T)$, and computation through $N = 262{,}144$ gives $\|\eta\|^2/T \approx 1.0$.

This gap is genuine. All classical sieves (Selberg, Brun, Rosser–Iwaniec, GPY) are circular for (88): they expand into the same intersection counts $A_{pq}(N)$ that the estimate controls, and their quadratic-form optimization requires the two-prime level of distribution as input (Remark 102). On the geometric side, the K3 lisse sheaf $R^2 f_*\mathbb{Q}_\ell$ has bounded rank and conductor, but the event $p \mid b_j$ is a same-characteristic crystalline condition. The Jacobi counterexample shows that a rank-one sheaf with the same formal properties can have $\gg p$ bad tame characters. The gap-polynomial identities are monochromatic: they control two incidences of the *same* prime and do not constrain the rainbow intersections that determine $\|\eta\|^2$.

Two precise sufficient criteria for Hypothesis Z emerge: a bounded-degree tame Hasse polynomial (giving $Z(p) \le 2D$), or a unique-unit-root theorem for all but $O(1)$ tame twists. The fiberwise Hasse–Witt invariant $A_p(t) = \sum_{n<p} b_n\, t^n$ is explicitly known from Stienstra–Beukers [25], but it controls which *fibers* are nonordinary, not which *Mellin coefficients* vanish; the character-twisted Hasse computation is a fundamentally different problem on the global twisted cohomology $H^1_{\mathrm{rig},c}(U, \mathcal{V}_{\mathrm{tr}} \otimes \mathcal{K}_{\omega^{-j}})$. The raw interpolation $j \mapsto b_j$ has degree $p-2$ already for $p = 11$ and 13; moreover, the Sym^2 factorization forces at least $(p-5)/2$ nonzero Fourier coefficients, giving a rigorous lower bound $D \ge (p-7)/4$ on any Laurent polynomial of degree D representing the unnormalized Mellin transform. Bounded tame Hasse degree would require a nontrivial Gauss/Jacobi normalization of the first Frobenius block.

Computation through all 78,496 primes $p \le 10^6$ gives $Z(p) \le 12$ and mean ≈ 1; the value-collision energy ratio $E(p)/p \to 3$ (confirmed through $p = 100{,}000$) and the profile constant $c_2 \to 1.262$ (through $N = 262{,}144$) match the palindromic null at every tested scale. These observations support all conjectures but are not used as proofs.

The prime-wise decomposition (Remark 109) provides a structural reduction: $\|\eta\|^2 = D + O$ where $D = \sum_p D_p \approx T$ is an explicit diagonal and $O = \sum_{p\neq q} C_{pq}$ is the cross-prime interaction. Under the independent fixed-cardinality residue model (each Z_p drawn uniformly from $\binom{\mathbb{F}_p}{Z(p)}$, independently across primes), one obtains the exact variance formula

$$\operatorname{Var}(\|\eta\|^2) \;=\; \sum_p \operatorname{Var}(D_p) \;+\; \sum_{p\neq q} \operatorname{Var}(C_{pq}),$$

whose first sum is $O(T)$ (fluctuation of the active tail mass Z_p^{act}) and whose second is $O(T^2/N)$ (CRT averaging in intervals of length $\sim N$ within period $pq \sim N^2$). The null-model conclusion $\|\eta\|^2/T \to 1$ in L^2 is therefore unconditional within the model.

However, CRT does not imply independence for *deterministic* zero-sets: distinct residue fields provide separate coordinates, not a product measure. The adversarial construction $Z_p = \{2N \bmod p\}$ for all p places every shifted zero at $m = 2N$, giving $O = \Theta(T^2)$; this is a single explicit moving-target choice with $|Z_p| = 1$ for all p, so the two-prime dispersion statement is *false* for general $O(1)$-size zero-sets. For the Apéry sequence, this pattern is impossible: the fixed-column $Z_p = \{c - p \bmod p\}$ forces $p \mid b_c$ for all p, contradicting $b_c \in \mathbb{Z} \setminus \{0\}$. More generally, the palindromic structure (Z_p consisting of reflected pairs $\{j_p, p{-}1{-}j_p\}$) and the Galois-group structure of the gap polynomials prevent systematic alignment of zero positions across primes. Nonetheless, the quantitative control—that the residual alignment is $o(T)$, not merely $o(T^2)$—is a stronger statement, and cannot be established by any purely black-box argument: Bombieri–Vinogradov, the large sieve, and Barban–Davenport–Halberstam address single-modulus discrepancies, not cross-prime collisions, and the spectral bilinear framework of Duke–Friedlander–Iwaniec ("bilinear forms with Kloosterman fractions") requires a fixed numerator, which the moving-target zero-sets do not provide.

Transfer of the null-model theorem to the actual Apéry zeros is captured by the following precise conjecture.

Conjecture 126 (Two-prime dispersion)**.** *For $N \to \infty$ and primes $p, q \in (N, 2N]$ with $p \neq q$, define $C_{pq} = \sum_m X_{m,p}\, X_{m,q}$. Then*

$$\sum_{\substack{p,q\in(N,2N]\\ p\neq q}} |C_{pq}| \;=\; o(T_N).$$

This is equivalent to the BFM2 bound (88) and hence to the all-n conjecture $G_n = e^{o(n)}$.

Remark 127 (Anatomy-of-integers formulation)*.* For each $j \le N$, define $\omega(j) = \#\{p \in (N, 2N] : p \mid b_j\}$. Then $\sum_j \omega(j) = T + F$ (counting primes that divide some b_j in the top block), and

$$\sum_{p\neq q}\sum_{j\le N} \mathbf{1}[pq \mid b_j] \;=\; \sum_{j\le N} \binom{\omega(j)}{2}.$$

The two-prime dispersion is equivalent to $\sum_j \omega(j)(\omega(j){-}1) = o(T)$: at most $o(T)$ indices j are divisible by two or more primes from the same dyadic window. Since $\log b_N \sim N \log(17 + 12\sqrt{2}) \approx 3.53\, N$ while $\sum_{p\in(N,2N]} \log p \sim N$, each b_j has room in principle for every prime in the window as a factor. Pure size/counting arguments cannot exclude this conspiracy; the Poisson prediction relies on the equidistribution of the p-adic digits of b_j across independent residue fields.

Table 8 verifies the anatomy-of-integers formulation directly: at every tested scale, the distribution of $\omega(j)$ matches Poisson(λ) with $\lambda = T/(N{+}1)$ to within sampling noise. In particular, $\max_j \omega(j) \leq 3$ through $N = 131{,}072$, consistent with the Poisson prediction $N \cdot \lambda^3/6 \approx 5$ at this scale. The second factorial moment satisfies $\sum_j \omega(j)(\omega(j){-}1)/T \approx \lambda$, which is exactly the collision rate $T/N \approx 1/\ln N$ predicted by Mertens—the same rate appearing in the two-prime scaling law below.

Table 8: Distribution of $\omega(j) = \#\{p \in (N, 2N] : p \mid b_j\}$. $\lambda = T/(N{+}1)$ is the mean; the Poisson column gives $e^{-\lambda}\lambda^k/k!$.

N	T	λ	$\max\omega$	$\#\{\omega{\geq}2\}$	$\sum \omega(\omega{-}1)/T$	Poisson fit
8192	608	0.074	2	22	0.072	$< 0.1\%$
32768	2048	0.063	3	53	0.056	$< 0.1\%$
65536	3944	0.060	3	108	0.057	$< 0.1\%$
131072	7584	0.058	3	227	0.063	$< 0.1\%$
262144	14104	0.054	3	351	0.051	$< 0.1\%$

Computation through $N = 262{,}144$ gives $|O|/T < 2 \times 10^{-3}$ (Table 7), far stronger than the conjecture requires.

Table 9: Two-prime dispersion scaling. $\Sigma_{\text{abs}} = \sum_{p \neq q} |C_{pq}|$ measures the total unsigned cross-prime interaction. The ratio Σ_{abs}/T decreases as $\approx c/\ln N$ with $c \approx 1.08$.

N	T	Σ_{abs}	Σ_{abs}/T	$\ln N \cdot \Sigma_{\text{abs}}/T$	joint
8192	354	43.1	0.122	1.10	10
16384	664	75.3	0.113	1.10	17
32768	1182	123.2	0.104	1.08	29
65536	2237	214.1	0.096	1.06	47
131072	4317	391.1	0.091	1.07	87
262144	7940	682.3	0.086	1.07	156

Table 9 decomposes the cross-prime sum into its unsigned magnitude. The product $(\ln N) \cdot \Sigma_{\text{abs}}/T$ remains approximately constant at ≈ 1.08, giving the empirical scaling law

$$\frac{\sum_{p \neq q} |C_{pq}|}{T_N} \approx \frac{1.08}{\ln N}.$$

This is explained by a counting argument. Since $L \leq p$ for primes $p > N$, the map $n \mapsto n \bmod p$ restricted to $[1, L]$ is injective; CRT lifts at coprime moduli hit the interval at most once, giving the sharp pointwise bound $|C_{pq}| \leq \min(Z(p), Z(q))$. In practice, most pairs have no joint CRT hit (the "joint" column of Table 9 counts pairs with at least one), so $|C_{pq}| \approx Z(p)\,Z(q)\,N/(pq) = O(1/N)$ for the bulk. With $O(N^2/(\ln N)^2)$ pairs and $T = \Theta(N/\ln N)$, the sum-to-T ratio is $O(1/\ln N)$. The unsigned ratio $\Sigma_{\text{abs}}/T \sim 1/\ln N$ is therefore essentially *forced*—it measures the expected collision count $\sum_{p \neq q} E_{pq} \approx T^2/N$, which is $\Theta(T/\ln N)$ by Mertens. The genuine arithmetic content is in the *signed* sum: $|O|/T < 2 \times 10^{-3}$ says that the actual collision count $\sum_m r(m)(r(m){-}1)$ matches its Poisson prediction to within a few percent.

For non-CM motives over $\mathbb{Q}$, independence of Frobenius traces at distinct primes is a consequence of the generalized Sato–Tate conjecture (known for GL_2 by the work of Barnet-Lamb–Geraghty–Harris–Taylor, extended to the symmetric square L-function). However, the single-prime $Z(p)$ statistics are a *vertical* problem—equidistribution of Mellin-transform values as the character varies within one $\mathbb{F}_p$—for which Katz's finite-field Mellin equidistribution theory [15] is the right tool. The two-prime dispersion is a *horizontal* problem: correlating arithmetic events at distinct primes for a fixed motive, which is of Lang–Trotter type. No equidistribution theorem over a single finite field can see cross-prime structure, and no analog of the joint Sato–Tate theorem across two distinct primes exists in the literature. The two-prime dispersion asks for a compatible statement at the level of zero-counts $Z(p)$, which lives in the Kummer-twisted cohomology rather than the fiber-variable cohomology. The adversarial construction $\mathcal{Z}_p = \{2N \bmod p\}$ shows that any proof must use Apéry-specific structure (cf. Friedlander–Granville [10] for the general impossibility of target-agnostic equidistribution beyond the large-sieve range).

The strongest existing results for structured moving targets in sieve theory follow the Duke–Friedlander–Iwaniec template [8], which proves equidistribution of roots of a quadratic congruence to moduli beyond $\sqrt{x}$ using an integral formula over automorphic forms on GL_3. Their method requires the target set $\mathcal{Z}_p$ to arise as roots of a *fixed* polynomial with a global algebraic-variety structure—precisely the ingredient the Apéry recurrence lacks (the degree grows linearly with p, and no single variety parametrizes the zero set). The only result achieving the moving-target beyond-$\sqrt{x}$ barrier without a fixed polynomial degree is Maynard's breakthrough [17], which handles targets of size $O(p^{1/3})$ in arithmetic progressions using new estimates for trilinear Kloosterman sums—still requiring a multiplicative character structure absent from the Apéry setting. In summary, the two-prime dispersion for Apéry numbers falls in the gap between what sieve methods can reach (targets with global algebraic structure) and what the Apéry recurrence provides (a local, p-dependent set of size $O(1)$).

The Diophantine-approximation approach to GCDs of recurrence terms—the Subspace-Theorem framework of Bugeaud–Corvaja–Zannier [6], Levin [16], and Xiao [26]—requires the sequence to admit a finite exponential-polynomial representation (a constant-coefficient linear recurrence). The Apéry sequence satisfies a recurrence with polynomial coefficients and obeys $b_n \sim C\alpha^n n^{-3/2}$, where the fractional exponent $-3/2$ is incompatible with constant-coefficient recurrences. Moreover, even in the exponential-polynomial setting, same-sequence GCD bounds fail: $\gcd(2^m-1, 2^n-1) = 2^{\gcd(m,n)}-1$ is exponentially large when $m \mid n$. Therefore no existing Subspace-Theorem argument yields $G_n = e^{o(n)}$ for all n.

The most direct computational path is an explicit Hasse–Witt calculation for the rank-three transcendental crystal, determining the slope-zero multiplicity of the Kummer-twisted compactly supported cohomology as a function of the character parameter. The Galois-group analysis of reductions of Apéry-like series by Caruso–Fürnsinn–Vargas-Montoya–Zudilin [7] may provide algebraic input, though their results concern the fiber parameter and do not directly control the character-twisted Mellin structure or the two-prime dispersion.

On the single-prime side, the bordered-certificate analysis (Section 5.4) settles the value-distribution question at the 3/4 exponent: $\max_a N_p(a) = O(p^{3/4})$ and $E(p) = O(p^{7/4})$ unconditionally, with the certificate nonvanishing (Theorem 32) sharp at $k < p$. By the impossibility construction discussed in Section 12, no package of single-prime distributional facts—including these—can close the pointwise conjecture; their role is in the averaged theorems and as input to any future cross-prime argument.

Finally, the open problem now has a sharply localized form. The quotient reduction (Proposition 60) shows that prime counting alone disposes of every prime below $n/(f(n)\log n)$, so the conjecture is equivalent to controlling the lower-digit channel uniformly over the bounded quotient classes $q = \lfloor n/p \rfloor < f(n)\log n$; and among all sufficient conditions collected here, the single fixed factorial moment $(\mathrm{HM})_8$ (Hypothesis 117), proved unconditionally for $k = 2$ (Proposition 118), appears to be the weakest cross-prime statement that closes the full conjecture (Theorem 121).

References

[1] S. Ahlgren and K. Ono, *A Gaussian hypergeometric series evaluation and Apéry number congruences*, J. reine angew. Math. **518** (2000), 187–212.

[2] A. Adolphson and S. Sperber, *Twisted exponential sums and Newton polyhedra*, J. reine angew. Math. **443** (1993), 151–177.

[3] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13.

[4] F. Beukers, *Some congruences for the Apéry numbers*, J. Number Theory **21** (1985), 141–155.

[5] F. Beukers, *Another congruence for the Apéry numbers*, J. Number Theory **25** (1987), 201–210.

[6] Y. Bugeaud, P. Corvaja, and U. Zannier, *An upper bound for the G.C.D. of $a^n - 1$ and $b^n - 1$*, Math. Z. **243** (2003), 79–84.

[7] X. Caruso, F. Fürnsinn, D. Vargas-Montoya, and W. Zudilin, *Galois groups of Apéry-like series modulo primes*, arXiv preprint, 2026.

[8] W. Duke, J. B. Friedlander, and H. Iwaniec, *Equidistribution of roots of a quadratic congruence to prime moduli*, Ann. of Math. **141** (1995), 423–441.

[9] S. Drappeau, *Sums of Kloosterman sums in arithmetic progressions, and the error term in the dispersion method*, Proc. Lond. Math. Soc. **114** (2017), 684–732.

[10] J. Friedlander and A. Granville, *Limitations to the equi-distribution of primes I*, Ann. of Math. **129** (1989), 363–382.

[11] É. Fouvry, E. Kowalski, and P. Michel, *Algebraic trace functions over the primes*, Duke Math. J. **163** (2014), 1683–1736.

[12] E. Kowalski, P. Michel, and W. Sawin, *Bilinear forms with Kloosterman sums and applications*, Ann. of Math. **186** (2017), 413–500.

[13] A. Forey, J. Fresán, and E. Kowalski, *Arithmetic Fourier transforms over finite fields: generic vanishing, convolution, and equidistribution*, preprint, 2023.

[14] J. Greene, *Hypergeometric functions over finite fields*, Trans. Amer. Math. Soc. **301** (1987), 77–101.

[15] N. M. Katz, *Convolution and Equidistribution: Sato–Tate Theorems for Finite-Field Mellin Transforms*, Ann. of Math. Studies **180**, Princeton Univ. Press, 2012.

[16] A. Levin, *Greatest common divisors and Vojta's conjecture for blowups of algebraic tori*, Invent. Math. **215** (2019), 493–533.

[17] J. Maynard, *Primes in arithmetic progressions to large moduli III: uniform residue classes*, arXiv:2006.08250, 2020.

[18] I. Gessel, *Some congruences for Apéry numbers*, J. Number Theory **14** (1982), 362–368.

[19] S. Jin, W. Ma, and K. Ono, *A note on non-ordinary primes*, Proc. Amer. Math. Soc. **144** (2016), 4591–4597.

[20] A. Malik and A. Straub, *Divisibility properties of sporadic Apéry-like numbers*, Res. Number Theory **2** (2016), Art. 5.

[21] R. J. McIntosh, *A generalization of a congruential property of Lucas*, Amer. Math. Monthly **99** (1992), 231–238.

[22] Y. Mimura, *Congruence properties of Apéry numbers*, J. Number Theory **16** (1983), 138–146.

[23] E. Rowland and R. Yassawi, *Automatic congruences for diagonals of rational functions*, J. Théor. Nombres Bordeaux **27** (2015), 245–288.

[24] G. Rhin and C. Viola, *The group structure for* $\zeta(3)$, Acta Arith. **97** (2001), 269–293.

[25] J. Stienstra and F. Beukers, *On the Picard–Fuchs equation and the formal Brauer group of certain elliptic K3 surfaces*, Math. Ann. **271** (1985), 269–304.

[26] L. Xiao, *Greatest common divisors of integral points of numerically equivalent divisors*, Math. Z. **306** (2024), article 61.

[27] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), 251–291.